

WHICH CUBICAL SITES PRESENT SPACES? REFLECTION SUBGROUPS, CONNECTIONS, AND THE COMPARISON OF THE TYPE AND TEST MODEL STRUCTURES

RYOTA SHIBUSAWA

ABSTRACT. Every standard cube category — cartesian, with one or two connections, De Morgan, Kleene, Boolean — carries two canonical homotopy theories on its presheaves: the cubical-type model structure of cubical type theory, and Cisinski’s test structure, which presents spaces; a delocalization rigidity theorem shows that the type structure presents spaces if and only if the two coincide. This paper proves three things. *First*, on the De Morgan, Kleene, and Boolean sites the comparison is controlled, in every dimension, by reflection subgroups: an isotropy quotient $\square^n/H$ agrees in the two structures iff the normal closure of the reflections of H fixes a cell, and carries an explicit separating map iff that closure is mixed — settling these three families negatively at every dimension. *Second*, on the Dedekind site — the last open entry of the phase diagram — the natural finite witness fails: the *monotone dunce hat* cannot be contracted by any transfinite sequence of representable open-box fillings, yet is contracted by a three-line *shell* filling that every uniformly fibrant object admits; uniform fibrancy is strictly more connective than cellular box-filling. *Third*, the remaining Dedekind comparison is transformed into an explicit approximation problem and then driven deep into it: through Cisinski’s regularity theory, a cosimplicial resolution by cubical-simplex retracts whose two codegeneracies are the two connections, and a singular Quillen adjunction, the comparison is proved equivalent to a single statement — that uniform Kan complexes are approximated by their chain stratum. The singular complex of the resolution is shown to compute the *test* homotopy type of every presheaf, with no fibrancy or shape hypothesis, so the counit is always a test equivalence and the comparison is exactly a type-level upgrade; chain approximation is proved for every quotient by chain-cell identifications (the census included), for every cyclic atom, and over the square; a cone-descent theorem strictly contracts every split-pair fold quotient; and an adversarially audited assembly of these pieces reduces the entire comparison to two sharp combinatorial statements — the well-foundedness of the subatom order, and the intrinsic-chain-anchored asphericity of minimal atoms. Along the way every cyclic cell shape is shown to be regular, and the reversal separations reappear as failures of regularity. All finite claims are machine-verified, the core facts by independent implementations, in a version-pinned artifact.

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1. INTRODUCTION

Cubical type theories interpret dependent type theory in presheaves over a cube category, and the choice of cube category is the choice of an algebraic theory of the interval: faces and degeneracies only (the cartesian site of [1]), binary connections (the Dedekind site), a De Morgan involution (the site of CCHM type theory [6]), the Kleene inequality, or full Boolean complementation. Each site carries the *cubical-type model structure*, whose fibrations are the uniform Kan fibrations of the type-theoretic interpretation [6, 14, 20], and the *test model structure* of Cisinski [10], which presents the homotopy theory of spaces; the containment $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$ holds on all of them. Whether it is an equality is precisely whether cubical type theory *presents spaces* — an identification which is not a convention but a theorem, the delocalization rigidity of Proposition 2.2 below. The equality has been resolved case by case over the last years: negatively on the cartesian site, where the equivariant repair of [1] was needed; positively, in a recent breakthrough of Cavallo and Sattler, on the site with exactly *one* connection [8]; negatively on the De Morgan site, in a strong form — an explicit separating map, and an explicit finite presheaf contractible in the test structure but not in the type structure [21] — with a complete classification of the isotropy quotients $\square^n/H$, $H \leq B_n$, in dimensions $n \leq 3$ [11]. Two cases remained: the higher-dimensional and cross-site structure of the reversal separations, and the Dedekind site itself — the cube category with both connections, singled out by Buchholtz and Sattler in 2018 as the one standard site whose status was unknown, and shown in [8] to be inaccessible to every Reedy-elegance method.

This paper closes the first, and settles the second’s natural line of attack — negatively. Three theorems organize the results.

Theorem 1.1 (A: the phase diagram). *On every standard cube category — cartesian, one connection, Dedekind, De Morgan, Kleene, Boolean — the cubical-type model structure presents spaces if and only if $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$. Among these sites all but the Dedekind site are settled: the type structure presents spaces exactly on the one-connection site. On the Dedekind site the comparison remains open, and the natural finite witness fails (Theorem C).*

The new entries are the reversal sites at every dimension (Theorem B); the cartesian and one-connection entries are [1] and [8]. The equivalence of “presents spaces” with “ $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$ ” — so that the phase diagram answers both questions at once — is a delocalization rigidity theorem (Proposition 2.2): a *proper* left Bousfield localization of a model structure with the same cofibrations can never again present spaces. A second identification (Theorem 24.16) shows that, classically, the cubical-type structure is the

minimal Cisinski structure of its interval; the phase diagram therefore also decides on which cube sites Cisinski’s minimal interval structure presents spaces.

Theorem 1.2 (B: the criterion on every reversal site). *On the De Morgan, Kleene, and Boolean sites, for every n and every $H \leq B_n$, with N the normal closure of the reflections of H : if N fixes a cell then $\square^n/H \simeq B(H/N)$ in both structures, and an explicit test-not-type weak equivalence is attached to H iff N is mixed.*

Theorem 1.3 (C: the monotone dunce hat is no witness). *The monotone dunce hat $\square^2/((0, u) \sim (u, 0), (u, u) \sim (u, 1))$ is test-contractible and type-contractible: its identity is contracted in every fibrant object by a shell filler, although no transfinite sequence of representable open-box fillings contracts it. The Dedekind comparison remains open.*

For the remaining case the paper establishes the positive program: the comparison holds if and only if the type localizer is *regular* in the sense of Cisinski (Theorem 24.19), the question reduces to the cell shapes of the site, and regularity is proved here for all graph quotients — in particular all 1,533 census quotients — for all two-dimensional cell images, and for every atom whose vertex-fiber poset admits a monotone transversal (Theorems 24.25, 24.31 and 24.34); the five-chain atom, expected to be the first obstruction, is in fact isomorphic to the square (Proposition 24.32). A *triangulation resolution* then settles the whole contractible tract: the presheaf category carries a canonical cosimplicial resolution by cubical-simplex retracts, with the two connections as the two codegeneracies, and for every test-contractible shape the regularity question is identified with type-contractibility (Theorem 24.47) — so all cyclic atoms are regular (Corollary 24.48), and the whole comparison is equivalent to a single statement of cubical approximation: the counit of the singular adjunction of the triangulation is a type equivalence on every uniform Kan complex (Theorem 24.51). The approximation itself is then proved on a large tract: the singular complex computes the test homotopy type of *every* presheaf (Theorem C.13), so the counit is always a test equivalence and only the type-level upgrade remains; chain approximation holds unconditionally at every chain-cell quotient (Theorem 24.53), every cyclic atom (Theorem 24.55), and in ambient dimension two (Proposition 24.59); the cone-descent theorem (Theorem 24.60) strictly contracts every split-pair fold quotient; and an adversarially audited assembly (Remark 24.63) locates the entire remaining comparison in two combinatorial statements: the well-foundedness of the subatom order of the cube, and the intrinsic-chain-anchored asphericity of minimal atoms and their fold and stabilizer quotients.

Provenance of inputs. Figure 1 records what the three theorems consume. The published entries of the phase diagram are [1] (cartesian) and [8] (one connection). Every lemma imported from the unpublished [21] is restated and proved in site-general, self-contained form in Appendix A;

from the unpublished [11] the paper uses one covering argument, carried where it is applied by the covering theory of Part 2, and otherwise only classification data that the criterion revalidates machine-checkably (artifact, `separation.criterion.py`). Part 5 and the regularity program depend on neither, resting on the provenance calculus developed there and on the published [10]. The boundary of novelty is sharp: [21] establishes the explicit De Morgan separation in dimension two; every theorem of the present paper — the all-dimensional criterion across the three reversal sites, the shell contraction and the shell–tent calculus on the Dedekind site, and the regularity reduction — appears here for the first time. A byproduct of the reversal analysis is a bridge to the homotopy theory of finite group actions: the orbit space of the Milnor construction of a finite M -set is $\mathbb{F}_2$ -acyclic whenever a Sylow 2-subgroup of M has a fixed point (Proposition 17.1).

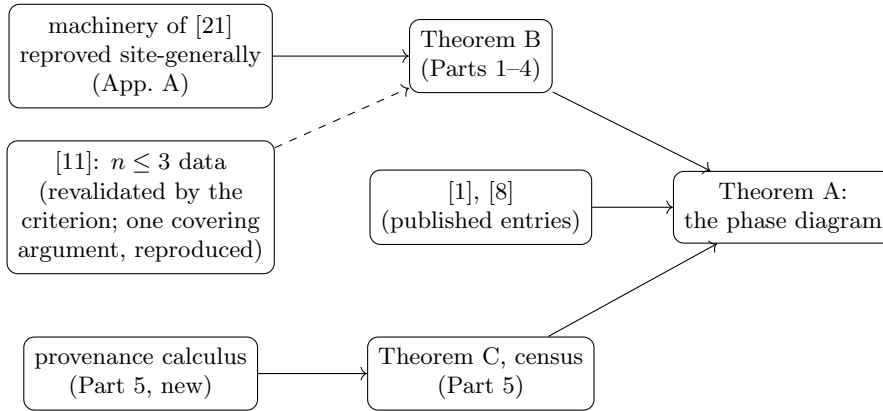


FIGURE 1. What the three theorems consume. Solid arrows are proof-level inputs; the dashed arrow is validation data. The unpublished [21] enters only through Appendix A, where its machinery is restated and proved for an arbitrary site of the paper.

Two mechanisms organize the results. On the reversal sites a single algebraic actor controls the entire interaction of the two homotopy theories with isotropy:

the reflection subgroup — the subgroup generated by the elements that fix a cell.

On the Dedekind site two mechanisms compete. The interaction of the diagonal with the connections — the same interaction that destroys Eilenberg–Zilber decompositions — is crystallized in a one-line monotonicity lemma (*a monotone map constant on the diagonal cylinder is constant*) and makes cellular box-filling rigid; and the *shell boxes* of uniform fibrancy, whose side data prescribe all fold tracks at once and which no representable-box calculus contains, dissolve that rigidity and contract the natural witness.

The architecture of the proofs is layered as follows.

Part I: reversal-site phase transfer. Which interval theories admit isotropy phenomena at all?

Theorem 1.4 (Phase diagram; Part 1). *(i) On every cubical site whose interval theory contains a connection and no reversal, every isotropy quotient $\square^n/H$ is contractible in both structures: a single equivariant connection contraction kills all witnesses. In particular the Dedekind comparison is structurally inaccessible to quotient witnesses; the congruence-quotient candidate of Part 5 is refuted there by the shell contraction, and the comparison remains open. (ii) On each of the De Morgan, Kleene, and Boolean sites, the explicit map-level and object-level separations of [21] hold verbatim: the separating median collage and the test-contractible, non-type-contractible subdivided cone exist on all three. The transfer is universal-algebraic: all term identities descend to subvarieties, and the only site-specific inputs are two parity lemmas whose witness points are unmixed.*

Part II: the landscape. On the three reversal sites the classification's conclusions coincide — but, provably, its proofs cannot.

Theorem 1.5 (Landscape; Part 2). *(i) (Rigidity discriminant) An invariant interior cell of a witness machine with twist δ is stage-zero slide-reachable to an end iff the character $\varepsilon \circ \delta$ vanishes, where ε is the strata-swap character; genuinely twisted cells survive every stage of the fibrant replacement, detected by a strict character projection to the interval quotient — test-side $\mathbb{Z}/2$ -covering theory obstructing type-side nullity. (ii) (Realizability divide) Twisted cells exist over a site iff the character is realizable by a twisted free-algebra element: always on the Boolean site (signed XOR), always in the relevant cases on the Kleene site (the Kleene inequality removes precisely the mixed points that carry the De Morgan parity obstructions), and essentially never on the De Morgan site — the site of cubical type theory is the exceptional site, the one where the certificate method happens to work. (iii) (Covering descent) Presheaf coverings are local systems on the category of elements and extend uniquely along the algebraic fibrant replacement; free quotients admit equivariant replacements, and a map of free quotients is a type weak equivalence iff its equivariant cover is. Separations therefore descend along towers, bypassing the broken certificates.*

Part III: the criterion on the Boolean site. The case analysis of [11] — four agreement families, three separation mechanisms — collapses into one statement.

Theorem 1.6 (Criterion; Part 3). *Let $H \leq B_n$ be mixed, on any reversal site, and let $N := \langle\langle \text{Refl}(H) \rangle\rangle$ be the normal closure of its reflections.*

- (1) *If N has a common fixed cell, then $\square^n/H \simeq B(H/N)$ in both structures.*
- (2) *If N is mixed, an explicit test weak equivalence that is not a type weak equivalence is attached to H : on the Boolean site for every n*

(Theorem 15.10, via the sieve-landing dichotomy and the dimension-reduction recursion), and on the De Morgan and Kleene sites for $n \leq 3$ by the landscape transfer of Part 2.

The engine of (2) is an exact characterization: the obstruction-certificate method succeeds for a group if and only if the group is generated by its reflections — because certificate consistency forces every twist to kill all vertex stabilizers, nontrivial vertex stabilizers consist of reflections, and every reflection fixes a vertex; N is reflection-generated by construction, and the separation descends along the free cover H/N by (iii) above.

In dimension ≤ 3 the criterion reproduces the 78/20 classification with zero mismatches — on the De Morgan site by direct comparison with [11], on the Kleene and Boolean sites via the landscape transfer; in dimension 4 the closed-form census decides all 30 reflection-generated mixed classes — 28 rigid, where the attached witness separates, and 2 tree-arrangement classes where the quotient reduces, in both structures at once, to the Klein quotient and to an order-4 mixed block. The agreement half of the criterion is itself a simplification discovery: the median retraction of [11], whose odd-arity self-dual majority term had seemed essential, is an avoidable device — retracting at the level of N , where no signs act, and only then passing to the free cover eliminates the equivariance problem the median was solving, and simultaneously handles degenerate groups (found by machine search in dimension 4) for which *every* one-step equivariant retraction provably fails.

Part IV: every reversal site. The Boolean criterion transfers to De Morgan and Kleene — proving Theorem B — by three tools: the De Morgan convexity interpolant is *self-dual*, an identity of free De Morgan algebras, so any two cells sharing exact deck data are one-step homotopic on every reversal site and the sieve-landing dichotomy applies verbatim; the theory quotients onto the Boolean site preserve deck data, ends, and stabilization, so rigidity projects; and stabilized cells factor algebraically through invariant loci, reducing to lower-dimensional isotropy quotients with no vertex-function analysis.

Part V: the Dedekind site. Theorem C has three layers. Test-contractibility is two homotopy pushouts of contractibles. Cellular rigidity is a provenance analysis of representable open-box fillings: the diagonal track of a fresh cylinder can never attach to earlier stages (the diagonal-track lemma), forcing an exact equation whose restriction to the identity end produces a constant edge where the monotone dunce hat has none. Type-contractibility is the shell contraction: an exact provenance calculus for the uniform replacement (a cartesian generating category and a complete based normal form) shows where fresh fillers can have entirely prescribed fold tracks, and an explicit six-prism shell over the constant cube then contracts the identity onto the strictly null endomorphism $(u \wedge v, u \vee v)$ in every fibrant object. Assembling Parts I–V yields Theorem A.

Methods and verification. The paper is written against a public, version-pinned artifact: every finite combinatorial claim — censuses, certificates, discriminants, towers, the 98-subgroup reproduction — is verified by exhaustive computation, with exact (parity- and satisfiability-based) decision procedures on the Boolean and Kleene sites. Several of the main theorems were found as empirical laws in this data — the rigidity discriminant, the realizability divide, the criterion itself — and then proved.

Related work. The comparison problem originates in the type-theoretic literature: the cartesian equivariant repair of Awodey, Cavallo, Coquand, Riehl and Sattler [1], the elimination of reversals from cubical type theories by Cavallo and Sattler [7], and the model-structure frameworks of [14, 20]. The explicit De Morgan separation and the three-dimensional classification are in the companion papers [21, 11]; the present paper subsumes and extends the archived series on the phase diagram and the landscape. The test-category side rests on [10, 18, 5]; free Kleene algebras on [3]. The slogan that reflection subgroups control the comparison appears to be new; its resonance with the theory of reflection arrangements is discussed in the outlook. The fundamental-group formula for isotropy quotients (Theorem 14.6) is a categorical relative of Armstrong’s theorem on quotients of simply-connected spaces [2]; the codiscrete objects $E(T)$ are Milnor constructions [17] and, equivariantly, classifying spaces for the family of subgroups with fixed points, so the arrangement dichotomy of the outlook sits beside Quillen’s poset conjecture [19] and the orbit-space theorems of Webb and Symonds [24, 22].

Part 1. The phase diagram

2. CUBICAL SITES FROM VARIETIES

2.1. The sites. Let $\mathcal{V}$ be a variety of algebras on the signature $\{\wedge, \vee, 0, 1\}$ or $\{\wedge, \vee, \neg, 0, 1\}$ containing the bounded distributive lattice axioms, and write $F_{\mathcal{V}}(n)$ for the free algebra on $x_1, \dots, x_n$. The *cube category* $\square_{\mathcal{V}}$ has objects $[n]$ for $n \geq 0$ and morphisms

$$\square_{\mathcal{V}}([m], [n]) = \text{Hom}_{\mathcal{V}}(F_{\mathcal{V}}(n), F_{\mathcal{V}}(m)) \cong F_{\mathcal{V}}(m)^n,$$

composition being substitution; presheaves on $\square_{\mathcal{V}}$ are $\mathcal{V}$ -*cubical sets*. We consider:

- the *Dedekind family*: $\mathcal{V} \supseteq$ bounded distributive lattices, no $\neg$ (this includes the Dedekind site itself, $F_{\mathcal{V}}(n)$ the free bounded distributive lattice, of cardinality the Dedekind numbers 3, 6, 20, 168, $\dots$);
- the *De Morgan site* $\square_{\text{DM}}$: $\mathcal{V} =$ De Morgan algebras (distributive lattices with an involution $\neg$ satisfying the De Morgan laws); $|F_{\text{DM}}(2)| = 168$;
- the *Kleene site* $\square_{\text{KL}}$: the subvariety of De Morgan algebras with $x \wedge \neg x \leq y \vee \neg y$; $|F_{\text{KL}}(n)| = 6, 84, 43918, \dots$ [3];

- the *Boolean site* $\square_{\text{BA}}$: the subvariety of Boolean algebras; $|F_{\text{BA}}(n)| = 2^{2^n}$;
- the *cartesian site* $\square_{\text{cart}}$: no lattice operations; morphisms are tuples of variables and constants.

All of these are strict test categories: the cube category with connections is a strict test category by Maltsiniotis [18], and the remaining cases follow from the criteria of [10]; see the systematic treatment of Buchholtz and Morehouse [5]. Hence each site carries the test model structure, with $\mathcal{W}_{\text{test}}$ created by the nerve of the category of elements $\text{el}(-)$, and each carries the cubical-type model structure with cofibrations the monomorphisms and fibrations the uniform Kan fibrations generated by the open-box inclusions $(m \hat{\otimes} \delta^\varepsilon)$ [14, 20]. The containment (U) is basic to everything that follows, so we prove it here in full.

Proposition 2.1 (The upper bound (U)). *On every site of the paper, $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$.*

Proof. The open boxes lie in $\mathcal{W}_{\text{test}}$. Representable presheaves are test-aspherical over any category — $\text{el}(a)$ has the terminal object id_a — so $\square^1 \rightarrow *$ lies in $\mathcal{W}_{\text{test}}$. The sites are *strict* test categories, i.e. totally aspherical test categories, and total asphericity is equivalent to the stability of $\mathcal{W}_{\text{test}}$ under finite products [10, 4.3.1–4.3.3]; hence the projection $X \times \square^1 \rightarrow X$, the product of id_X with $\square^1 \rightarrow *$, lies in $\mathcal{W}_{\text{test}}$ for every presheaf X . By two-out-of-three the end inclusions $X \times \{\varepsilon\} \rightarrow X \times \square^1$ lie in $\mathcal{W}_{\text{test}}$; being monomorphisms, they are test-trivial cofibrations. For a monomorphism $m: A \rightarrow B$ the open box $m \hat{\otimes} \delta^\varepsilon$ sits in the triangle

$$B \times \{\varepsilon\} \longrightarrow A \times \square^1 \cup_{A \times \{\varepsilon\}} B \times \{\varepsilon\} \xrightarrow{m \hat{\otimes} \delta^\varepsilon} B \times \square^1,$$

whose first map is a pushout of the test-trivial cofibration $A \times \{\varepsilon\} \rightarrow A \times \square^1$ along $A \times \{\varepsilon\} \rightarrow B \times \{\varepsilon\}$ — hence a test-trivial cofibration — and whose composite is the test-trivial cofibration $B \times \{\varepsilon\} \rightarrow B \times \square^1$. By two-out-of-three $m \hat{\otimes} \delta^\varepsilon \in \mathcal{W}_{\text{test}}$; it is a monomorphism (the Leibniz product of monomorphisms), hence a test-trivial cofibration.

Trivial cofibrations, trivial fibrations. Type-trivial cofibrations form the saturation of the open boxes under pushout, transfinite composition, and retract; test-trivial cofibrations are closed under these operations and contain the open boxes, so every type-trivial cofibration lies in $\mathcal{W}_{\text{test}}$. A type-trivial fibration has the right lifting property against all cofibrations — the monomorphisms, common to the two structures — and the maps with the right lifting property against all monomorphisms are exactly the test-trivial fibrations; so every type-trivial fibration lies in $\mathcal{W}_{\text{test}}$.

Conclusion. A type weak equivalence factors as a type-trivial cofibration followed by a type fibration, which by two-out-of-three is a type-trivial fibration. Both factors lie in $\mathcal{W}_{\text{test}}$, hence so does the map. $\square$

The two-out-of-three argument is that of [21, Prop. 2.5], here carried out in full at site-generalality.

2.2. Presenting spaces equals coinciding with the test structure.

Both model structures live on the same category with the same cofibrations, and (U) says $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$: the test structure is a left Bousfield localization of the type structure. The following rigidity of the homotopy theory of spaces shows that a *proper* such localization can never be homotopically invisible; it turns each separation theorem of this paper into a “does not present spaces” theorem, and conversely upgrades the agreement of [8] to a coincidence of weak equivalence classes. (We work classically; underlying ∞ -categories and derived adjunctions of Quillen adjunctions are as in [15, 16].)

Proposition 2.2 (Delocalization rigidity). *Let a bicomplete category carry two model structures with the same class of cofibrations and $\mathcal{W}_1 \subseteq \mathcal{W}_2$, and suppose the second presents spaces: its underlying ∞ -category is equivalent to the ∞ -category $\mathcal{S}$ of spaces. Then the first presents spaces if and only if $\mathcal{W}_1 = \mathcal{W}_2$.*

Proof. ($\Leftarrow$) is trivial. For ($\Rightarrow$): the identity is a Quillen adjunction from the first structure to the second — it preserves cofibrations, and trivial cofibrations of the first structure are trivial cofibrations of the second because $\mathcal{W}_1 \subseteq \mathcal{W}_2$. Let $L: \mathcal{C}_1 \rightleftarrows \mathcal{C}_2 : R$ be the derived adjunction of underlying ∞ -categories [16]. R is fully faithful: every fibrant object of the second structure is fibrant in the first (there are more trivial cofibrations to lift against in the second structure, so fewer fibrations), and the two structures have the same trivial fibrations (both are defined by lifting against the common cofibrations), hence the same cofibrant replacements; the derived counit $LRX \rightarrow X$ at a fibrant X is therefore represented by a cofibrant replacement $QX \rightarrow X$, a weak equivalence in both structures. So L is a reflective localization. R preserves the terminal object (it is a right adjoint), so $L(1) \simeq L(R(1)) \simeq 1$ is terminal. Now let $e: \mathcal{S} \simeq \mathcal{C}_1$ be an equivalence. The composite $L \circ e: \mathcal{S} \rightarrow \mathcal{S}$ preserves colimits and the point; since $\mathcal{S}$ is the free cocompletion of the point, a colimit-preserving endofunctor of $\mathcal{S}$ is determined by its value at the point [15, Thm. 5.1.5.6], so $L \circ e \simeq \text{id}_{\mathcal{S}}$ and L is an equivalence. Finally let $f \in \mathcal{W}_2$: the localization functor $\gamma_2 \simeq L \circ \gamma_1$ inverts f ; as L is an equivalence, γ_1 inverts f ; and the weak equivalences of a model structure are saturated — a map inverted in the homotopy category is a weak equivalence [19] — so $f \in \mathcal{W}_1$. $\square$

Corollary 2.3. *On every site of this paper the cubical-type model structure presents spaces if and only if $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$. In particular the theorem of Cavallo and Sattler [8] implies $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$ on the one-connection site, and each separation theorem of this paper implies that the corresponding cubical-type structure does not present spaces.*

Proof. Cofibrations coincide (the monomorphisms), (U) gives $\mathbf{W}_{\text{type}} \subseteq \mathbf{W}_{\text{test}}$, and the test structure presents spaces because the sites are test categories. Apply Proposition 2.2. $\square$

2.3. Symmetries and isotropy quotients. The invertible morphisms $[n] \rightarrow [n]$ of $\square_{\mathcal{V}}$ form the symmetric group Σ_n when $\neg \notin \mathcal{V}$ and the signed permutation (hyperoctahedral) group $B_n = \Sigma_n \ltimes (\mathbb{Z}/2)^n$ when $\neg \in \mathcal{V}$. For H a subgroup of this automorphism group, the *isotropy quotient* $\square^n/H$ is the levelwise orbit presheaf. These quotients are the natural finite testing ground for the comparison: they are where all known object-level phenomena live [23, 11, 21].

Definition 2.4. An element of B_n is a *reflection* if it fixes a nonempty cell of $\square^n$ — equivalently (Lemma 11.1), if every cycle of its signed cycle decomposition has even sign sum, equivalently if it fixes a vertex. A subgroup $H \leq B_n$ is *mixed* if it fixes no nonempty cell (for the group, fixing a cell and fixing a vertex again agree). We write $\text{SEP}(H)$ for the statement that an explicit map in $\mathbf{W}_{\text{test}} \setminus \mathbf{W}_{\text{type}}$ is attached to H — built naturally from $\square^n/H$ -data, as in the criterion — and call H an *agreement* class when $\square^n/H$ has equivalent homotopy types in the two structures via a common B -object.

2.4. Free-algebra representations. Computations use the standard function representations. Free De Morgan algebras are monotone $\{0, 1\}$ -valued functions on the literal cube $L_n = \{0, 1\}^{2n}$ (coordinates in pairs $(x_i, \neg x_i)$), with negation $(\neg\varphi)(p) = 1 - \varphi(\rho p)$, where ρ swaps and complements each pair. Free Kleene algebras are the monotone functions on the *unmixed* subposet $U_n \subseteq L_n$ — the points having no $(0, 0)$ -pair together with a $(1, 1)$ -pair — with the same formula for negation (ρ preserves U_n) [3]. Free Boolean algebras are all $\{0, 1\}$ -valued functions on $\{0, 1\}^n$, with negation the output complement. A point of L_n is called unmixed when it lies in U_n ; the transfer of the parity lemmas in Section 4 rests on the observation that the De Morgan proofs only ever evaluate at unmixed points.

3. CONNECTIONS KILL ISOTROPY WITNESSES

Theorem 3.1 (Connections kill isotropy witnesses). *Let $\square_{\mathcal{V}}$ be a cube category whose interval theory contains a connection ($\vee$ or $\wedge$) and no reversal, so that the automorphisms of $[n]$ in $\square_{\mathcal{V}}$ form the symmetric group Σ_n . Then for every n and every $H \leq \Sigma_n$ the isotropy quotient $\square^n/H$ admits a strict H -equivariant contraction, and is therefore contractible in both the type and the test structure. In particular the two structures agree on all isotropy quotients of $\square_{\mathcal{V}}$, and no analogue of the known object-level witnesses exists on such sites.*

Proof of Theorem 3.1. Suppose $\vee \in \mathcal{V}$ (the case of $\wedge$ is dual, using the 0-endpoint). Consider the cell

$$c = (x_1 \vee t, \dots, x_n \vee t) \in \square_{\mathcal{V}}([n+1], [n]),$$

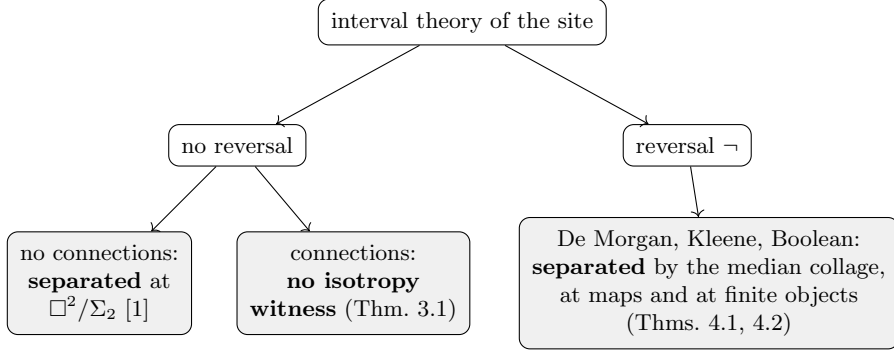


FIGURE 2. The phase diagram of cubical sites.

regarded as a map $c : \square^n \otimes \square^1 \rightarrow \square^n$ of representables, where t is the cylinder coordinate. Then: (i) $c|_{t=0} = \text{id}$ and $c|_{t=1}$ is the constant map at the vertex $(1, \dots, 1)$; (ii) c is Σ_n -equivariant: permuting the x_i commutes with the coordinatewise formula, and the vertex $(1, \dots, 1)$ is fixed; (iii) since quotients of presheaves are levelwise and the cylinder is formed in the x -coordinates only, c descends to a homotopy

$$\bar{c} : (\square^n/H) \times \square^1 \longrightarrow \square^n/H, \quad \bar{c}_0 = \text{id}, \quad \bar{c}_1 = \text{const},$$

for every $H \leq \Sigma_n$. A strict homotopy between the identity and a constant is a left homotopy in any model structure in which the projection $X \times \square^1 \rightarrow X$ is a weak equivalence, which holds in the type structure (the interval is contractible) and in the test structure (the site is a strict test category). Hence $\square^n/H \rightarrow 1$ is a weak equivalence in both structures. $\square$

Corollary 3.2. *On the Dedekind site the type and test structures agree on all isotropy quotients. In particular, whether the Dedekind site presents spaces cannot be decided by an isotropy witness.*

Remark 3.3. On the cartesian site the contraction of Theorem 3.1 is unavailable — there is no cell $x \vee t$ — and indeed the symmetric square $Q = \square^2/\Sigma_2$ is a witness: it is test-contractible, while its failure to be type-contractible is the observation of Coquand and Sattler that motivated the equivariant model structure of [1]. The phase boundary between the cartesian and Dedekind columns of Figure 2 is thus exactly the presence of a connection.

4. THE UNIFORM SEPARATION

Theorem 4.1 (Uniform map-level separation). *On each of the De Morgan, Kleene, and Boolean cubical sites, the median collage $\bar{\Phi} : J/K \rightarrow \square^2/K$ of [21] is a test weak equivalence that is not a type weak equivalence. Hence $W_{\text{type}} \subsetneq W_{\text{test}}$ on all three sites.*

Theorem 4.2 (Uniform object-level separation). *On each of the three sites of Theorem 4.1, the subdivided mapping cone C of [21] is an explicit finite presheaf that is test-contractible but not type-contractible: the two structures do not agree about which finite objects are aspherical.*

We recall the shape of the De Morgan separation [21]. $K = \{1, \text{sw}, \mathbf{n}, g\}$ is the Klein group acting on the square by the coordinate swap sw , the double reversal $\mathbf{n}(x, y) = (\neg x, \neg y)$, and $g = \text{nsw}$; $W := \square^2/K$. The join model J is the double mapping cylinder of the two strata circles, with a twisted K -action, and the median collage

$$\Phi = (F(d, a, t), F(d, \neg a, t)), \quad F(d, a, t) := (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a),$$

descends to $\bar{\Phi} : J/K \rightarrow W$. The main theorems of [21] state that $\bar{\Phi}$ is a test weak equivalence (both sides realize $B\mathbb{Z}/2 \star B\mathbb{Z}/2$) but not a type weak equivalence (the *nullity theorem*: every map from W into a fibrant replacement of J/K is null-homotopic), and that a subdivided mapping cone C of $\bar{\Phi}$ is test-contractible but not type-contractible. All constructions make sense verbatim on the Kleene and Boolean sites: the formulas for the actions, the join model, and F use only $\wedge, \vee, \neg, 0, 1$.

4.1. The audit.

Lemma 4.3 (Subvariety inheritance). *Every term identity of the De Morgan variety holds in the Kleene and Boolean varieties. In particular the self-duality $\neg F(d, a, t) = F(\neg d, \neg a, t)$, the flow identities $F(d, a, t \wedge s) = F(d, F(d, a, t), s)$ and $F(d, a, t \vee s) = F(F(d, a, t), a, s)$, the equivariance identities of the collage, and the sector formulas of the subdivided cone hold on all three sites.*

Proof. Kleene and Boolean algebras are subvarieties of De Morgan algebras: each is axiomatized by the De Morgan axioms plus further identities/inequalities. Identities valid in a variety hold in every subvariety. $\square$

Lemma 4.4 (No self-negated elements, all three sites). *On each site, no element of a free algebra equals its own negation. Consequently the double reversal $\mathbf{n}$ acts freely on the nonempty cells of the square, at every level.*

Proof. De Morgan and Kleene: the all-(1,0) point p_0 of the literal cube (each pair equal to (1,0)) satisfies $\rho p_0 = p_0$ and is unmixed, hence lies in U_n ; a self-negated φ would satisfy $\varphi(p_0) = 1 - \varphi(p_0)$. Boolean: negation is the output complement, so $\neg \varphi \neq \varphi$ pointwise. $\square$

Lemma 4.5 (First parity lemma, all three sites). *No φ in the free algebra on two generators satisfies $\varphi^{\text{sw}} = \neg \varphi$.*

Proof. De Morgan and Kleene: the point $q = (1, 0, 1, 0)$ satisfies $\text{sw } q = q$ and $\rho q = q$, and is unmixed, hence lies in U_2 ; at q the condition reads $\varphi(q) = 1 - \varphi(q)$. Boolean: at a diagonal input (a, a) the condition reads $\varphi(a, a) = 1 - \varphi(a, a)$. $\square$

Lemma 4.6 (Second parity lemma, all three sites). *No φ in the free algebra on two generators satisfies both $\varphi^{\text{sw}} = \varphi$ and $\varphi^n = \neg\varphi$.*

Proof. De Morgan and Kleene: the point $m = (1, 0, 0, 1)$ is unmixed and its bitwise complement equals $\text{sw } m$; symmetry gives $\varphi(\overline{m}) = \varphi(m)$ while the second condition gives $\varphi(\overline{m}) = 1 - \varphi(m)$. Boolean: at the input $(0, 1)$, the n -condition reads $\varphi(1, 0) = 1 - \varphi(0, 1)$ while symmetry gives $\varphi(1, 0) = \varphi(0, 1)$. $\square$

Proof of Theorems 4.1 and 4.2. We audit the proofs of the corresponding theorems of [21], replacing the De Morgan site by the Kleene or Boolean site throughout, and verify that every ingredient remains valid.

Site-uniform machinery. The following results of [21] are proved for an arbitrary presheaf category over a strict test cube category whose interval theory contains $\wedge, \vee, \neg$ and diagonals, and use no further structure of the free algebras; they are reproduced in self-contained form in Appendix A: the descent lemma for uniform fillers (Lemma A.1) and the algebraic fibrant replacement with its coequalizer presentation (Lemma A.2); the strictification of maps out of strict quotients (Lemma A.4); the equivariant filler slide (Lemma A.6; the slide formulas $\tau \wedge \neg z, \tau \vee z$ exist by Lemma 4.3); the upper bound (U), proved in full as Proposition 2.1; the equivariant descent dictionary (Lemma A.5) with the pasting and first-exit calculus (Lemma A.7); and the collage formula for nerves of sieve-cosieve decompositions (general category theory [4, 25]).

Term identities. Every algebraic identity invoked — self-duality and flow for F , the equivariance of Φ and of the cone slide, the face computations — holds by Lemma 4.3.

The test side. The stratification of the deck action on the cells of the square is site-uniform once Lemma 4.4 is in place: sw -fixed cells are the diagonal ones (φ, φ) , g -fixed cells the antidiagonal ones $(\varphi, \neg\varphi)$, and n acts freely; the strata are copies of the interval quotient $\square^1/\neg$, whose category of elements is $B\mathbb{Z}/2$ by the free-covering argument (freeness is again Lemma 4.4); the free parts are contractible covers with the generic cells as terminal objects (stabilizer-freeness of the generic cell: sw and g move it since $y \neq x$ and $\neg y \neq x$ in the free algebra on two generators, and n by Lemma 4.4). The collage computation then gives $\text{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ on each site, and the three-legs argument for $\text{el}(\bar{\Phi})$ being a weak equivalence proceeds verbatim: the collapse-locus closure and flow retraction use only the flow identities and Lemma 4.4; the middle layers use the terminal covers and free quotients. We have verified the stratification and the census ingredients by machine on both new sites (Section 26).

The type side. The nullity theorem’s stage induction is site-uniform; its base is the rigidity of invariant cylinder classes of $(J/K)([2])$, which is a formal consequence of Lemmas 4.5 and 4.6 exactly as in [21]: invariance under sw modulo the deck group forces componentwise sw -symmetry by

Lemma 4.5, and then invariance under n forces componentwise strict invariance by Lemma 4.6. The machine censuses confirm the rigidity exhaustively: on the Kleene site $(J/K)([2])$ has 96 invariant interior classes, every one with a strictly invariant representative; on the Boolean site, 8, likewise; and on neither site does any invariant class map to the identity class of W under the collage (no strict sections; on the Boolean site 64 non-invariant triples do hit the identity class — a genuinely Boolean phenomenon — but none is source-invariant). Non-contractibility of W in the type structure follows from (U) and $\text{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$ as in [21].

Assembly. The above is a complete dependency audit: together with Appendix A it constitutes a self-contained proof of the transfer, taking as input only the De Morgan base theorem of [21], whose proofs then run unchanged on each site: $\bar{\Phi}$ is a test weak equivalence but not a type weak equivalence (Theorem 4.1), and the subdivided mapping cone is test-contractible but not type-contractible (Theorem 4.2). $\square$

Remark 4.7 (What does *not* transfer). The transfer is not a triviality about quotients of proofs: the stage-zero *numerics* differ substantially across the sites (Table 1), and at least one qualitative phenomenon is new on the Boolean site — the existence of (non-invariant) collage preimages of the identity class, which on the De Morgan site is excluded outright. What transfers is the *mechanism*, because its site-specific content was already concentrated in Lemmas 4.5–4.6 by [21].

	De Morgan	Kleene	Boolean
$ F(2) $	168	84	16
classes of $W([2])$	7140	1806	72
strict endomorphism classes	72	38	12
invariant interior classes of $(J/K)([2])$	360	96	8
of which essential	0	0	0
collage-image = id triples	0	0	64
of which source-invariant	0	0	0

TABLE 1. Stage-zero censuses across the three reversal sites (machine-verified; De Morgan values from [21, 23]).

Part 2. The landscape

5. WITNESS MACHINES AND TWISTS

5.1. Witness machines, twists, and the swap character. For a mixed two-stratum subgroup $H \leq B_n$ (two reflections h_1, h_2 with disjoint fixed loci $\square^{m_1}, \square^{m_2}$), the *join model* J_H of [11] has coordinates $m_1 + m_2 + 1$ (the locus parameters and the join parameter t), a twisted H -action ν , and the median collage $\Phi = (F(\ell_1, \ell_2, t)\text{-components})$, $F(d, a, t) = (d \wedge \neg t) \vee (a \wedge$

$t) \vee (d \wedge a)$, descending to $\bar{\Phi}_H : J_H/H \rightarrow \square^n/H$. The *strata-swap character* $\varepsilon : H \rightarrow \mathbb{Z}/2$ takes value 1 on the elements exchanging the two loci; $\nu(h)$ acts on the t -coordinate by $t \mapsto \neg^{\varepsilon(h)} t$. An *invariant interior cell with twist* $\delta \in \text{Hom}(H, H)$ is a cell c of J_H (interior: t -component nonconstant) with $c \circ \sigma_h = \nu(\delta(h)) \cdot c$ for all h ; its class in (J_H/H) is invariant under the source substitutions, i.e., strict-map data for $\square^n/H \rightarrow J_H/H$ by strictification (Lemma A.4). An invariant cell is *slide-null* when an invariant homotopy (in one further cylinder variable w) joins its class to a class supported in an end (t -component constant).

6. THE REALIZABILITY DIVIDE

Lemma 6.1 (XOR character law). *In the free Boolean algebra, for $S \subseteq \{1, \dots, n\}$ let $e_S := \bigoplus_{i \in S} x_i$. Then for every signed permutation $h = (\sigma, s)$,*

$$e_S \circ \sigma_h = \neg^{\chi_S(h)} e_{h(S)}, \quad \chi_S(h) := \sum_{i \in S} s_i \bmod 2.$$

In particular, if H stabilizes S then e_S realizes the sign character $\chi_S|_H$.

Proof. $e_S \circ \sigma_h = \bigoplus_{i \in S} (x_{\sigma(i)} \oplus s_i) = e_{\sigma(S)} \oplus \sum_{i \in S} s_i$, and $\oplus 1$ is the output complement. (Verified exhaustively for $n \leq 3$.) $\square$

Theorem 6.2 (Unified realizability, all dimensions). *Let $H \leq B_n$, let $\psi : H \rightarrow \mathbb{Z}/2$ be a character, and let H_ψ denote the ψ -twisted action $h \mapsto \sigma_h \circ \rho^{\psi(h)}$ on the literal cube (ρ = swap-and-complement each pair; on the Boolean site ρ acts trivially on points). A ψ -twisted element ($T \circ \sigma_h = \neg^{\psi(h)} T$) of the free algebra on n generators exists over the site $\mathcal{V}$ if and only if ψ vanishes on the stabilizers, under H_ψ , of every middle point of the site:*

- (1) *Boolean: the vertices of $\{0, 1\}^n$;*
- (2) *Kleene: the proper middle points (every literal pair $(1, 0)$ or $(0, 1)$), which biject with the vertices;*
- (3) *De Morgan: the full middle level of the literal cube, including the mixed points.*

In particular Kleene realizability coincides with Boolean realizability, and the De Morgan site is exceptional exactly because the mixed middle points contribute additional twisted stabilizers.

Proof. Necessity (all sites): evaluating the twisted equation around a cycle of the H_ψ -action at a middle point p gives $T(p) = \psi(\text{cycle}) \oplus T(p)$, so ψ must vanish on $\text{Stab}_{H_\psi}(p)$; middle points are in the domain of the free algebra on each site (for Kleene, the proper points lie in the unmixed poset).

Sufficiency, Boolean: propagate values along H_ψ -orbits of vertices from arbitrary representatives; consistency at a stabilized point is the vanishing hypothesis, and every vertex function is a free Boolean element. When ψ is a stable-set sign character the solution is the signed XOR e_S .

Sufficiency, Kleene (the $\top$ -dominant extension): points of the unmixed poset U_n carry no $(0, 0)$ -pair together with a $(1, 1)$ -pair, so

$$T(p) := \begin{cases} 1 & \text{if } p \text{ has a } (1, 1)\text{-pair,} \\ 0 & \text{if } p \text{ has a } (0, 0)\text{-pair,} \\ t_\psi(p) & \text{if all pairs of } p \text{ are proper,} \end{cases}$$

is well defined, where t_ψ is the Boolean solution read on the proper values. T is monotone: proper points are pairwise incomparable in the literal order (raising any literal of a proper pair creates a $(1, 1)$ -pair), no point above a proper point has a $(0, 0)$ -pair, and the outer strata take the extreme values. T is ψ -twisted: substitutions preserve the three strata; ρ exchanges $(1, 1)$ - and $(0, 0)$ -pairs, so on the outer strata the equation reads $1 = \psi(h) \oplus 0$ exactly when $\psi(h) = 1$; on the proper stratum it is the Boolean equation. (For $n = 2$ the two solutions found by the machine are exactly T and $\neg T$.)

Sufficiency, De Morgan (the weight extension): given a parity-consistent assignment T_0 on the middle level, extend by $T(p) := 0$ for weight $< n$ and $T(p) := 1$ for weight $> n$. Substitutions preserve weight and ρ sends weight w to $2n - w$, so on the outer strata the twisted equation reads $0 = \psi(h) \oplus 1$ precisely when $\psi(h) = 1$, and on the middle level it is the consistency of T_0 . Monotonicity: points of equal weight are incomparable, and the strata are ordered $0 \leq T_0 \leq 1$. The closed form was verified against brute-force enumeration over all subgroups of B_2 and all their characters (25 cases, zero mismatches). $\square$

Remark 6.3. Theorem 6.2 inverts the naive expectation: adding equations to the variety (De Morgan $\rightarrow$ Kleene $\rightarrow$ Boolean) *increases* realizability, because the added equations destroy the carriers of the obstructions. The De Morgan site — the site of cubical type theory — is the exceptional site on which the certificate method works.

	De Morgan	Kleene	Boolean
$ F(2) $	168	84	16
essential invariant classes, $(J_{B_2}/B_2)([2])$	0	18	2
survivors: four-lines / B_3 / H_8 / H_{24}	0/6/5/7	9/19/5/7	9/19/5/7
sign characters realizable	no	yes	yes

TABLE 2. The realizability divide in the machine data (De Morgan values from [11]). “Survivors” are consistent nontrivial twists of the beyond-two-stratum certificate systems; on the Kleene side the counts are exact (monotone-realizability filter over U_3).

7. THE RIGIDITY DISCRIMINANT

Theorem 7.1 (Discriminant). *Let c be an invariant interior cell of J_H with twist δ , and $\psi := \varepsilon \circ \delta$.*

- (1) *If $\psi = 0$, then c is slide-null at stage zero: the slide $t \mapsto t \wedge \neg w$ in the cylinder variable, with the remaining coordinates extended w -constantly, is an invariant homotopy to the $t=0$ end.*
- (2) *If $\psi \neq 0$, then c is rigid at stage zero: no invariant homotopy joins its class to an end class.*

Proof. (1) The slide formulas are term operations available on all three sites; when $\psi = 0$ the deck action of the twist never flips t , so the slide is δ -equivariant, its $w=0$ face is c and its $w=1$ face has constant t -component. (2) Let $h \mapsto k_h$ assign to each generator a deck element witnessing the invariance of a homotopy c_4 with end faces. At the $w=1$ face the t -component is a constant ε_0 ; the twisted equation there reads $\varepsilon_0 = \psi'(h) \oplus \varepsilon_0$ for the homotopy's own character ψ' , forcing $\psi' = 0$. Hence the $w=0$ face has an *untwisted* t -component; but that face is a deck translate of c , whose t -component is ψ -twisted (negation preserves twistedness). For h with $\psi(h) = 1$ and any point p , twistedness gives $T(\sigma_h p) = \neg T(-)$ while untwistedness gives equality — a pointwise contradiction, so no such function, hence no such homotopy, exists. $\square$

Theorem 7.2 (Permanence). *Suppose $\varepsilon \neq 0$ and let c be as in Theorem 7.1(2), with $g_c : \square^n/H \rightarrow J_H/H$ the corresponding strict map. Then the class of g_c in $[\square^n/H, R(J_H/H)]$ differs from the class of every end map, at every stage of the fibrant replacement.*

Proof. The t -coordinate is a strict map $J_H \rightarrow \square^1$, equivariant over the ε -action of H on $\square^1$ by reversal; since $\varepsilon \neq 0$ it descends to

$$q : J_H/H \longrightarrow \mathbb{L} := \square^1/\neg,$$

and both ends of J_H land on the unique vertex of $\mathbb{L}$, so $q(\text{end maps})$ are constant. Suppose $[g_c] = [\text{end}]$; composing with $R(q)$ gives $[q \circ g_c] = [\text{const}]$ in $[\square^n/H, R(\mathbb{L})]$. A type cylinder homotopy is in particular a test homotopy (the unit of the replacement is a type-trivial cofibration, hence a test equivalence), and $\mathbb{L}$ is a $K(\mathbb{Z}/2, 1)$ in the test structure on each site [21], so

$$[\square^n/H, \mathbb{L}]_{\text{test}} \cong \text{Hom}(\pi_1 \text{el}(\square^n/H), \mathbb{Z}/2).$$

The map $q \circ g_c$ classifies the double cover with monodromy ψ : the twisted equation $T(\sigma_h p) = \psi(h) \oplus T(\dots)$ is precisely the statement that transporting around the loop of h in the free part $BH \subseteq \text{el}(\square^n/H)$ interchanges the sheets iff $\psi(h) = 1$. Since $\psi \neq 0$ the class is nontrivial, contradicting $[qg_c] = 0$. $\square$

Remark 7.3. Theorem 7.2 is a type-side obstruction detected by test-side covering theory. On the Boolean Klein machine the projection q factors strictly through the collage: the identity $F(d, a, t) \oplus F(d, \neg a, t) = t$ (verified

on all of $\text{FB}(2)^3$) gives $q = t^* \circ \bar{\Phi}$ with $t^* = [x \oplus y]$ the unique nontrivial strict map $\square^2/B_2 \rightarrow \mathbb{L}$.

8. COVERING DESCENT

Throughout this section G is a finite group acting on a presheaf X *freely on nonempty cells*: no nontrivial element fixes a nonempty cell at any level. Write $Y := X/G$ (levelwise orbits); $p : X \rightarrow Y$ is then a levelwise principal G -covering.

Lemma 8.1 (Coverings are local systems). *A presheaf map $E \rightarrow Z$ that is a levelwise principal G -covering is the same data as a functor $\text{el}(Z) \rightarrow G\text{-Tors}$, i.e., a $\pi_1(\text{el}(Z))$ -set with free transitive G -action. In particular isomorphism classes of G -coverings of Z are classified by $\text{Hom}(\pi_1 \text{el}(Z), G)$ up to conjugacy, and cylinder-homotopic maps pull coverings back to isomorphic coverings.*

Proof. A presheaf over Z is a presheaf on $\text{el}(Z)$; the levelwise torsor condition says each value is a G -torsor and each structure map a torsor map, which is a local system valued in G -torsors. The classification and homotopy invariance are the standard statements for local systems on a small category, via the nerve. $\square$

Lemma 8.2 (Simply connected box domains). *For every box inclusion $j = (S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, the category of elements $\text{el}(\text{dom}(j))$ is simply connected.*

Proof. The collapse $r : \text{dom}(j) \rightarrow \square^n \times \{\varepsilon\}$, given by $(s, t) \mapsto (s, \varepsilon)$ on the cylinder part and the identity on the body, is a strict left inverse of the inclusion, and the connection homotopy $(s, t, u) \mapsto (s, t \diamond u)$ (with $\diamond = \vee$ or $\wedge$ according to ε) is a strict deformation of the identity of $\text{dom}(j)$ onto $\iota \circ r$ — connections exist on all our sites. A strict homotopy equivalence induces an equivalence of nerves of categories of elements, and $\text{el}(\square^n)$ is contractible (terminal generic cell), so $\text{el}(\text{dom}(j))$ is contractible; in particular connected and simply connected. $\square$

Lemma 8.3 (Covering extension). *Let $Z \hookrightarrow Z'$ be a relative cell complex whose attachments are boxes of the uniform structure (e.g. the algebraic fibrant replacement $Z' = R(Z)$ of Lemma A.2). Then the restriction functor*

$$\text{Cov}_G(Z') \longrightarrow \text{Cov}_G(Z)$$

is an equivalence of groupoids. In particular every G -covering of Z extends to Z' , uniquely up to unique isomorphism relative to Z ; and the total object P of the extension of a free quotient $p : X \rightarrow Y$ along $Y \hookrightarrow R(Y)$ is a uniformly fibrant object containing X , with free G -action, $P/G \cong R(Y)$, and $X \hookrightarrow P$ a type-trivial cofibration — a G -equivariant fibrant replacement of X realizing the replacement of the quotient.

Proof. G -coverings of a presheaf are local systems valued in G -torsors on its category of elements (Lemma 8.1). For a single box attachment $Z' =$

$Z \cup_b \text{cod}(j)$, the category of elements is the pushout $\text{el}(Z') = \text{el}(Z) \cup_{\text{el}(\text{dom } j)} \text{el}(\text{cod } j)$, so

$$\text{Cov}_G(Z') \simeq \text{Cov}_G(Z) \times_{\text{Cov}_G(\text{dom } j)} \text{Cov}_G(\text{cod } j).$$

Now $\text{el}(\text{cod } j)$ is contractible ($\text{cod}(j) = \square^n \times \square^1$ is representable) and $\text{el}(\text{dom } j)$ is contractible as well (Lemma 8.2), so both covering groupoids are canonically equivalent to the groupoid of G -torsors — a connected groupoid with automorphism group G , not a contractible one — and the restriction $\text{Cov}_G(\text{cod } j) \rightarrow \text{Cov}_G(\text{dom } j)$ is an equivalence of groupoids (evaluation at any object of either category of elements identifies both with G -Tors, compatibly). Base change of an equivalence is an equivalence, so the projection $\text{Cov}_G(Z') \rightarrow \text{Cov}_G(Z)$ is an equivalence. Transfinite compositions and retracts of attachments pass to limits of groupoid equivalences.

For the second statement, let $P \rightarrow R(Y)$ be the essentially unique extension of $X \rightarrow Y$. Levelwise $P/G \cong R(Y)$ and the G -action is free away from nothing (torsor fibers). $X \hookrightarrow P$ is a transfinite composition of pushouts of boxes: over each downstairs attachment the covering restricts on $\text{dom}(j)$ to a trivialized system by Lemma 8.2, so its total object is G disjoint copies of $\text{dom}(j)$, and the extension attaches G disjoint lifted boxes. Fibrancy of P : fill a box of P by projecting to $R(Y)$, filling uniformly, and lifting the filler along the covering matching the given lift at the ε -end generator; the lift agrees with the given data on all of $\text{dom}(j)$ because two lifts of a map from a simply connected domain agreeing at one cell agree everywhere, and strict naturality is preserved since restriction preserves the ε -end base cell. $\square$

Theorem 8.4 (Covering descent). *Let G act freely (on nonempty cells) on X and X' , let $f : X \rightarrow X'$ be equivariant, and suppose the quotient map $\bar{f} : X/G \rightarrow X'/G$ is a type weak equivalence. Then f is a type weak equivalence. Contrapositively, non-equivalence descends: if $f \notin \mathbf{W}_{\text{type}}$ then $\bar{f} \notin \mathbf{W}_{\text{type}}$.*

Proof. Write $Y = X/G$, $Y' = X'/G$, and let P, P' be the equivariant replacements of Lemma 8.3. The functorial replacement $R(\bar{f}) : R(Y) \rightarrow R(Y')$ pulls the covering P' back to a covering of $R(Y)$ which restricts on Y to $\bar{f}^*(X' \rightarrow Y') \cong (X \rightarrow Y)$ (equivariance of f); by the uniqueness of Lemma 8.3 this pullback is P , giving a canonical equivariant $F : P \rightarrow P'$ over $R(\bar{f})$ with $F|_X = f$ up to the identification. Since $\bar{f} \in \mathbf{W}_{\text{type}}$ and all objects are cofibrant with $R(Y), R(Y')$ fibrant, $R(\bar{f})$ has a homotopy inverse $\bar{g}$ with cylinder homotopies $\bar{H} : \bar{g}R(\bar{f}) \simeq \text{id}$ and $\bar{K} : R(\bar{f})\bar{g} \simeq \text{id}$. By Lemma 8.1 the coverings are classified by characters of $\pi_1 \text{el}$, homotopy-invariantly; from $R(\bar{f})\bar{g} \simeq \text{id}$ we get $\bar{g}^*[P] = \bar{g}^*R(\bar{f})^*[P'] = [P']$, so $\bar{g}$ lifts to an equivariant $\tilde{g} : P' \rightarrow P$. Similarly the homotopies lift (a cylinder $Z \times \square^1$ is connected when Z is, so extensions of coverings along cylinders are again unique), each lift agreeing with the expected end maps up to a deck translation. Thus $\tilde{g}F \simeq \gamma_a$ and $F\tilde{g} \simeq \gamma_b$ for deck automorphisms γ_a, γ_b ; as these are isomorphisms, F has a left and a right homotopy inverse, hence

is a homotopy equivalence. Since $X \hookrightarrow P$ and $X' \hookrightarrow P'$ are type-trivial cofibrations and F extends f , two-out-of-three gives $f \in \mathcal{W}_{\text{type}}$. $\square$

Remark 8.5. The test-side counterpart is classical: el of a free quotient is the quotient of el , and free quotients of equivariant test equivalences are test equivalences (the Borel argument of [21]). Theorem 8.4 is its type-side partner, and the two together transfer an attached separation $(\bar{\Phi}_N \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}})$ from N to H along any tower.

9. SITE-INVARIANCE OF THE TEST SIDE

Lemma 9.1 (el-invariance). *For any two reversal sites and any $H \leq B_n$, the categories of elements $\text{el}(\square^n/H)$ (and those of the witness machines) are connected by weak equivalences determined by the common stratification data: the isotropy poset of H , the fixed-locus patterns, and the residual actions — all of which depend only on the signed-cycle combinatorics of H .*

Proof. We argue by induction on n ; all ingredients are those of Appendix A, checked to depend only on the group data.

The stratification is site-independent. A cell of $\square^n$ is stabilized by $h \in H$ iff its components satisfy the fixed-cell equations of h ; by the locus-pattern analysis (valid on each site because distinct signed generators are distinct in every free algebra, and no element is self-negated), the fixed locus of a subgroup $S \leq H$ is either empty or a subcube $\square^{d(S)}$ embedded by the pattern $x_i \mapsto \neg^{\varepsilon_i} m_{\text{cl}(i)}$, where the classes, signs, and dimension $d(S)$ are determined by the signed-cycle combinatorics of S alone. Hence the poset of isotropy types, the loci, their intersections (loci of joins), and the residual actions of normalizers are the same on all sites.

The collage is site-independent. Decompose $\text{el}(\square^n/H)$ by the sieve of stabilized cells and its free complement, as in [21]: (i) the free part has a contractible cover — the full subcategory of stabilizer-free cells of $\square^n$, with the generic cell terminal — on which H acts freely, so the piece is BH ; (ii) the sieve is the union of the strata, each stratum the quotient of a locus $\square^{d(S)}$ by its residual group — an isotropy quotient in dimension $d(S) < n$, equivalent across the sites by the inductive hypothesis (base $n = 0$: points); overlaps of strata are again loci, handled by the same induction over the isotropy poset; (iii) each comma piece is an arrow category fibered over its stratum with contractible fibers (coslices under the terminal generic cell), contractible free cover, and quotient $B(\text{Stab})$ by Quillen's Theorem A — all identified by the group data. The collage formula of [4, 25] assembles the pieces, and every piece and gluing map has been matched across the sites, so the nerves are connected by weak equivalences as claimed. The same induction applies verbatim to the witness machines, whose cells are built from cube cells, loci, and the combinatorial E_N -factor. $\square$

10. THE COLLAGE FORMULA

The decompositions used above assemble into a single closed formula, which both organizes the earlier special cases and makes the el-invariants of isotropy quotients exactly computable.

Theorem 10.1 (Collage formula). *Let $H \leq B_n$ and let $\Sigma \subseteq \square^n$ be the sieve of cells with nontrivial stabilizer. Then*

$$\mathrm{Nel}(\square^n/H) \simeq \mathrm{hocolim}\left(\mathrm{el}(\Sigma/H) \longleftarrow (\mathrm{el}\Sigma)_{hH} \longrightarrow BH\right),$$

where $(-)_hH$ is the Borel construction. Moreover $\mathrm{el}(\Sigma)$ is equivalent to the nerve of the incidence poset of the isotropy loci, so the middle term is the classifying space of a finite complex of groups whose vertex groups are the setwise stabilizers of the loci.

Proof. Decompose $\mathrm{el}(\square^n/H)$ along the closed sieve Σ/H and its open complement (the free cells): the nerve is the homotopy pushout of the sieve part, the cosieve part, and the comma category of arrows from the sieve into the cosieve. The sieve part is $\mathrm{el}(\Sigma/H)$. The cosieve part has the full subcategory of stabilizer-free cells of $\square^n$ as a cover, which is contractible (the generic cell is terminal) and carries a free H -action, so the piece is BH . The comma is $\mathrm{Arr}(\Sigma, F)/H$ with F the free part upstairs; projecting to the source, each fiber is a coslice with terminal object (the generic cell), so by Quillen's Theorem A the comma is equivalent to $\mathrm{el}(\Sigma)//H = (\mathrm{el}\Sigma)_{hH}$, the action on $\mathrm{el}(\Sigma)$ being honest and the F -component making it free. The legs are the collapse $(\mathrm{el}\Sigma)_{hH} \rightarrow BH$ and the natural projection $(\mathrm{el}\Sigma)_{hH} \rightarrow \mathrm{el}(\Sigma/H)$. For the addendum: each locus and each intersection of loci (the locus of the join of the stabilizers) is a subcube, hence has contractible el; by the nerve-of-cover lemma, $\mathrm{el}(\Sigma)$ is equivalent to the nerve of the incidence poset of the loci, and the Borel construction of a poset of contractible pieces with setwise stabilizers is the complex of groups stated. $\square$

$$\begin{array}{ccc} (\mathrm{el}\Sigma)_{hH} & \xrightarrow{\text{collapse}} & BH \\ \text{proj} \downarrow & & \downarrow \Gamma(\text{ho}) \\ \mathrm{el}(\Sigma/H) & \longrightarrow & \mathrm{Nel}(\square^n/H) \end{array}$$

FIGURE 3. The collage formula (Theorem 10.1): the nerve of an isotropy quotient is the homotopy pushout of the sieve quotient, the Borel construction of the reflection arrangement, and BH . When $\mathrm{el}\Sigma \simeq *$ the right leg is an equivalence and the sieve inclusion induces $\mathrm{el}(\Sigma/H) \simeq \mathrm{el}(\square^n/H)$ (Proposition 14.9).

Remark 10.2. The formula (Figure 3) reduces every $\text{el}(\square^n/H)$ — and, applied to the witness machines, every el appearing in this paper — to finite group cohomology plus finite poset data. All the $\mathbb{F}_2$ -cochain computations in Proposition B.3 below are exact evaluations of this formula (normalized bar complexes for the group pieces, iterated homotopy pushout cochain models for the gluings; artifact).

Part 3. The criterion

11. REFLECTIONS, VERTICES, AND AUTOMATIC HYPOTHESES

Two elementary facts about reflections anchor everything.

Lemma 11.1. *For a nontrivial element of B_n the following are equivalent: (a) it fixes a nonempty cell; (b) every cycle of its signed cycle decomposition has even sign sum; (c) it fixes a vertex. In particular the two descriptions of reflection in Definition 2.4 agree.*

Proof. (b) $\Rightarrow$ (c): on a cycle of even sign sum, propagate a vertex value around the cycle: the total twist is the sign sum, which is even, so a compatible assignment exists on each cycle. (c) $\Rightarrow$ (a): a vertex is a nonempty cell. (a) $\Rightarrow$ (b): a fixed cell forces, on each cycle, a fixed element of the free algebra whose propagation around the cycle twists by the sign sum; an odd twist would force the element to equal its own negation, impossible in every free algebra of the reversal varieties. $\square$

Lemma 11.2 (Odd order). *Every element of odd order in B_n is a reflection or the identity. Consequently, for every $H \leq B_n$ the quotient $H/\langle\langle \text{Refl}(H) \rangle\rangle$ is a 2-group; in particular, if it is nontrivial it has a $\mathbb{Z}/2$ -quotient.*

Proof. A signed cycle of length ℓ and sign s has order ℓ if $s = 0$ and 2ℓ if $s = 1$; an element of odd order therefore has all cycle signs zero, i.e., is a reflection or the identity. Hence all odd-order elements of H lie in $N := \langle\langle \text{Refl}(H) \rangle\rangle$. If the quotient H/N had an element of odd prime order p , a lift h would generate a cyclic group whose odd part maps onto it; the odd part consists of odd-order elements of H , hence lies in N — so the element was trivial. Thus H/N is a 2-group, and a nontrivial finite 2-group surjects onto $\mathbb{Z}/2$. $\square$

Lemma 11.3 (Automatic character triviality). *Let ψ be a character of H realizable as a twisted element of the free algebra on any of the three sites ($T \circ \sigma_h = \neg^{\psi(h)} T$). Then ψ vanishes on every reflection, hence on $\langle\langle \text{Refl}(H) \rangle\rangle$. In particular reflection-generated groups have no nontrivial realizable characters, on any reversal site.*

Proof. Evaluating the twisted equation at a vertex v fixed by h gives $T(v) = \psi(h) \oplus T(v)$ pointwise (Boolean; on the De Morgan and Kleene sites evaluate at the corresponding proper literal point, which is ρ -compatible), so $\psi(h) = 0$ for h in a vertex stabilizer. Apply Lemma 11.1(i). $\square$

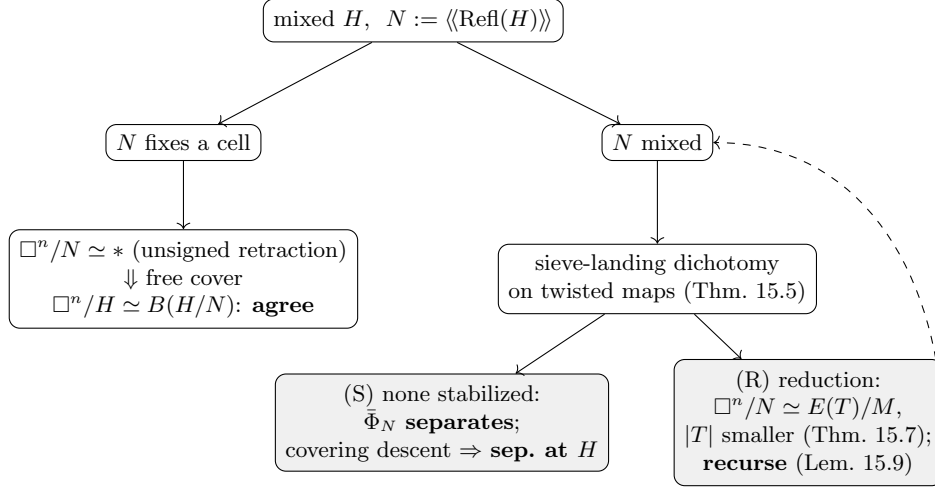


FIGURE 4. The criterion at every n (Theorem 15.10): the two halves, the dichotomy, and the well-founded recursion (dashed).

12. CERTIFICATES ARE EXACTLY REFLECTION GENERATION

We work with the exact certificate system of Part 2: for a mixed H with a reflection-free normal $V \trianglelefteq H$ and $G = H/V$, and a twist $\delta \in \text{Hom}(H, H)$, the *u-part* asks for an equivariant cell $c : \square^n \rightarrow \square^n$ with $c \circ \sigma_h = \delta(h) \circ c$, and the *e-part* for a G -valued vertex function E with $E(h \cdot v) = \bar{\delta}(h) \cdot E(v)$; on the Boolean site both systems are exact (parity-consistent assignments are cells), and on the Kleene site exactness holds after the monotone-realizability filter. H is *certified* when no nontrivial δ is consistent.

Theorem 12.1. *On the Boolean site — where the exact certificate system coincides with the actual-cell system, cells being arbitrary vertex functions — H is certified if and only if $H = \langle\langle \text{Refl}(H) \rangle\rangle$. On the Kleene and De Morgan sites the same equivalence holds for the abstract certificate system; the actual-cell systems refine it by the realizability filters of Theorem 6.2, which on the De Morgan site are strictly stronger.*

Proof. ($\Leftarrow$) Let δ be consistent. The u-part cell restricts to a vertex map $\hat{c}$ with $\hat{c}(h \cdot v) = \delta(h) \cdot \hat{c}(v)$, so $\delta(\text{Stab}_H(v)) \subseteq \text{Stab}_{B_n}(\hat{c}(v))$. The e-part, since G acts freely on itself, forces $\bar{\delta}(\text{Stab}_H(v)) = 1$, i.e. $\delta(\text{Stab}_H(v)) \subseteq V$. But V is reflection-free while nontrivial vertex stabilizers consist of reflections (Lemma 11.1(ii)), so $\delta(\text{Stab}_H(v)) = 1$ for every vertex v . By Lemma 11.1(i) every reflection lies in some vertex stabilizer, so δ kills $\text{Refl}(H)$, hence the normal closure ($\ker \delta$ is normal), hence H ; thus δ is trivial.

($\Rightarrow$) Suppose $N := \langle\langle \text{Refl}(H) \rangle\rangle \subsetneq H$. By Lemma 11.2 the quotient H/N is a nontrivial 2-group, so there is a character $\psi : H \rightarrow \mathbb{Z}/2$ vanishing on N and not on H . By Cauchy's theorem H contains an involution k ; the vertex

action of B_n is faithful, so k moves some vertex v_0 . Then $\delta(h) := k^{\psi(h)}$ is a well-defined homomorphism $H \rightarrow H$ (because $k^2 = 1$), nontrivial since $\psi \not\equiv 0$. Since ψ vanishes on all vertex stabilizers (they consist of reflections, which lie in N), there is a ψ -twisted vertex-level function $T : \{0, 1\}^n \rightarrow \mathbb{Z}/2$ with $T(h \cdot p) = \psi(h) \oplus T(p)$ (propagate along orbits; consistency at stabilized points is exactly the vanishing). The *multiplexer cell* $c(p) := k^{T(p)} \cdot v_0$ — encoded coordinatewise as a Boolean cell — satisfies $c(\sigma_h p) = \delta(h) \cdot c(p)$, and $E(h \cdot v_0) := \bar{\delta}(h)$ is well-defined on the orbit of v_0 (as δ kills the stabilizer) and extends freely elsewhere. Hence δ is a consistent nontrivial twist and H is not certified. $\square$

Remark 12.2. Theorem 12.1 explains the empirical tables of Part 2 at a glance (14 groups across $n = 2, 3, 4$, all matching), identifies the De Morgan certificates of [11] as instances (there the realizability filter is strictly stronger, but reflection-generation still implies certification by the same argument, and Lemma 11.3 discharges the character hypothesis), and turns “certified” from a computation into a group-theoretic property.

13. THE AGREEMENT HALF

Theorem 13.1 (Two-step covering agreement). *Let H be mixed and suppose $N = \langle\langle \text{Refl}(H) \rangle\rangle$ has a common fixed cell. Then, on each reversal site, $\square^n/N$ is contractible in both structures, H/N acts freely on its cells, and*

$$\square^n/H \simeq B(H/N)$$

in both structures. In particular $\square^n/H$ agrees.

Proof. Step 1. The common-fixed-cell condition for N is a consistent parity system on the coordinates, whose classes and signs define the fixed locus $L \cong \square^d$ with $d < n$ (a nontrivial reflection has a cycle of length ≥ 2 , so some class has size ≥ 2). N acts *trivially* on L : an element of N fixes every cell of L by definition of the common fixed pattern. Consequently the induced action of N on the classes carries no signs, and the consensus retraction

$$(R_t)_i = F(x_i, \neg^{\varepsilon_i} M_{\text{cl}(i)}, t), \quad M_k := \bigwedge_{j \in \text{class } k} \neg^{\varepsilon_j} x_j,$$

with F the median interpolation of [21], is N -equivariant — the only equivariance conditions are permutations *within* and *among* equal-size classes without signs, which the symmetric M satisfies — and no self-duality is required anywhere. R descends to the quotient and contracts $\square^n/N$ onto $L/N = L \cong \square^d$, strictly; hence $\square^n/N$ is contractible in both structures.

Step 2. H normalizes N , so acts on $\square^n/N$ through H/N . The action is free on nonempty cells: if $h \cdot c = \nu \cdot c$ for some $\nu \in N$ then $\nu^{-1}h$ fixes the cell c , hence is a reflection or the identity, hence lies in N , so $h \in N$. By the covering theory of Part 2 (equivariant replacement of the free quotient; on the test side, el of the free quotient is the Borel construction), the quotient

of a both-structures contractible object by a free action of the finite group H/N is a $B(H/N)$ in both structures — the argument of the free family of [11], applied verbatim with the contractible base $\square^n/N$ in place of $\square^n$. Finally $\square^n/H = (\square^n/N)/(H/N)$ since quotients are levelwise. $\square$

Remark 13.2. Theorem 13.1 subsumes the median-retraction family of [11]: for $H \leq \Sigma_n \times \langle z \rangle$ mixed, the reflections generate a subgroup fixing the diagonal-type locus, and the two-step argument applies with no parity condition on n . It also settles the five “edge” groups of dimension 4 found by the machine search — degenerate groups admitting *no* equivariant consensus pattern with odd negated classes — for which any one-step equivariant retraction provably fails: the two-step argument sidesteps the sign obstruction entirely, because all signs live in H/N and are absorbed by the covering step.

14. THE SEPARATION HALF: THE UNIFORM CYLINDER MACHINE

Let N be mixed and reflection-generated, on any reversal site. Write $\Sigma \subseteq \square^n$ for the *stabilized sieve*: the subpresheaf of cells with nontrivial stabilizer — the union of the fixed loci of the reflections of N , including all their intersections and nestings — and E_N for the 0-coskeletal codiscrete nerve of N ($E_N([m]) = N^{\text{vertices}([m])}$), contractible with N acting freely by translation.

Construction 14.1 (The cylinder machine). (The letter J is reused: J_N below is a new construction, unrelated to the join model of Part 2 except in spirit.)

$$J_N := (\square^n \times E_N) \cup_{\Sigma \times E_N \times \{0\}} (\Sigma \times E_N \times \square^1) / \sim, \quad (x, e, 1) \sim (x, e', 1),$$

the mapping cylinder of the projection $\Sigma \times E_N \rightarrow \Sigma$ glued into the body along its source. N acts diagonally; the action is free away from the collapsed end (on the body and the open cylinder because it is free on E_N ; the collapsed end is the bare Σ with its usual action). The projection $\Phi : J_N \rightarrow \square^n$ (forget E_N and the cylinder coordinate) is equivariant and descends to $\bar{\Phi}_N : J_N/N \rightarrow \square^n/N$.

$$\begin{array}{ccccc} \Sigma \times E_N \times \{0\} & \hookrightarrow & \square^n \times E_N & & \\ \downarrow & & \downarrow & & \\ (\Sigma \times E_N \times \square^1) / \sim & \longrightarrow & J_N & \xrightarrow{\Phi} & \square^n \end{array}$$

FIGURE 5. The uniform cylinder machine (Construction 14.1): the body $\square^n \times E_N$ (free, contractible), the cylinder over the stabilized sieve with collapsed end, and the equivariant projection Φ inducing $\bar{\Phi}_N : J_N/N \rightarrow \square^n/N$.

A single cylinder over the whole stabilized sieve (Figure 5) replaces the per-stratum horns of [11]: orbits, intersections, and nestings of loci require no case distinctions.

Proposition 14.2 (Test side). *$\text{el}(\bar{\Phi}_N)$ is a weak equivalence, for every mixed N .*

Proof. Decompose both sides by the stabilized/free dichotomy. On the target, the sieve is $\text{el}(\Sigma/N)$ and the cosieve is the free part, whose cover is contractible with terminal generic cell, so the cosieve piece is BN . On the source, the sieve consists of the cylinder cells; the cylinder direction is a strict deformation retraction onto the collapsed end, which is the *bare* Σ , so the sieve piece is again $\text{el}(\Sigma/N)$ and $\bar{\Phi}_N$ restricts to the *identity* on it. The source cosieve is $(\text{free cells}) \times_{E_N}$ (before the collapse), whose cover is contractible (product of a category with terminal object and the contractible E_N) with free N -action, so the piece is BN , and Φ induces the identity of N on fundamental groups. The comma pieces compare by the same contractible-cover argument (Quillen's Theorem A over the sieve, as in Part 2), and the collage formula finishes. $\square$

Theorem 14.3 (Type side: the trichotomy). *Suppose N is reflection-generated. Then every element of $[\square^n/N, R(J_N/N)]$ is homotopic to one of: (a) a constant; (b) a class factoring through the stabilized sieve $\Sigma/N \subseteq J_N/N$ (the collapsed end); or (c) an invariant-polynomial class — a *body class* $(c, \bar{e})$ whose cube component c is strictly invariant ($c \circ \sigma_h = c$ for all $h \in N$; equivalently, the components of c are N -invariant elements of the free algebra) and whose E_N -component is orbitwise constant.*

Proof. By the strictification and equivariant-descent dictionary (Lemmas A.4 and A.5), classes in $[\square^n/N, R(J_N/N)]$ are invariant cells of the algebraic replacement at level $[n]$, modulo chains of invariant homotopies one level up, and the replacement splits into *sector subreplacements* along the two closed sieves of J_N :

$$\bar{B} := \square^n \times E_N, \quad \bar{C} := (\Sigma \times E_N \times \square^1)/\sim, \quad \bar{B} \cap \bar{C} = \Sigma \times E_N \times \{0\},$$

the *body*, the *cylinder*, and their *interface*. Every cell of J_N is supported in $\bar{B}$ or in $\bar{C}$ (a cylinder cell has its τ -faces at the interface and at the collapsed end; no cell meets both the body beyond the interface and the end), and the sector bookkeeping for boxes, fillers and reindexings is exactly that of the sector calculus of Lemmas A.5 and A.7, whose proofs use only the coequalizer presentation of the replacement and the presence of diagonals.

The cylinder sector. The slide $(x, e, \tau) \mapsto (x, e, \tau \vee z)$ is a totally defined, N -equivariant strict homotopy of $\bar{C}$ onto the collapsed end, compatible with fillers by the uniformity relations (the equivariant filler slide, Lemma A.6). Hence every invariant class supported in the cylinder subreplacement — at every stage — is homotopic to a class supported in the collapsed end, i.e., factors through Σ/N : type (b).

The body sector. Here no slide is needed: the body is handled by covering theory. N acts freely on $\bar{B} = \square^n \times E_N$ (translation on E_N), and $\bar{B}$ is contractible in both structures (E_N is 0-coskeletal, $\square^n$ representable). By Lemma 8.3, the replacement of $\bar{B}/N$ is the free quotient P/N of an equivariant replacement $P \simeq *$, and by Lemma 8.1 together with the equivariant Whitehead argument for free actions, homotopy classes of maps $\square^n/N \rightarrow R(\bar{B}/N)$ are classified by their pullback coverings:

$$[\square^n/N, R(\bar{B}/N)] \cong \text{Cov}_N(\square^n/N)_{/\cong} \cong \text{Hom}(\pi_1 \text{el}(\square^n/N), N)/\text{conj}.$$

In particular every body-sector class, at every stage, is homotopic to a stage-zero body class. Stage-zero body classes are pairs (cube component with twist δ , E_N -component), and their consistency is precisely the exact certificate system: the cube component is the u-part and the E_N -component the e-part. By Theorem 12.1 only $\delta = 1$ occurs ($\delta = \text{id}$ -like twists are killed by the e-part, since vertex stabilizers are nontrivial and act freely on E_N); and $\delta = 1$ forces the cube component to be strictly invariant and the E_N -component to be orbitwise constant — type (c) — or the class is degenerate onto a point — type (a). Distinct orbitwise-constant E_N -labelings are joined by invariant E_N -edges (the codiscrete 1-cells, chosen orbitwise), so the class of type (c) depends only on the strictly invariant cube component c .

Mixed boxes and chains. A fresh filler of a box supported in one sector lies in that sector's subreplacement. A mixed box pastes through the interface by the excursion normalization of Lemmas A.5(2) and A.7 (their proofs use only the coequalizer relations and connections, both available): the chain through a mixed filler is replaced by a restricted step to an interface class followed by cylinder material. Interface classes are cylinder classes, hence of type (b). The first-exit induction (Lemma A.7), with the sector pair $(\bar{B}, \bar{C})$, therefore reduces every invariant chain to one ending in a class of type (a), (b), or (c). $\square$

Hypothesis 14.4 (Non-domination). *For every reflection-generated mixed N : (i) $\text{el}(\square^n/N)$ is not a homotopy retract of $\text{el}(\Sigma/N)$; and (ii) the identity of $\square^n/N$ is not homotopic, in the type structure, to any strictly invariant endomorphism class (the images of the invariant-polynomial classes under $\bar{\Phi}_N$).*

Remark 14.5. Part (ii) has a test-side necessary reduction: a homotopy $\text{id} \simeq \bar{\Phi}_N \circ f$ with f a body class factors the identity through $\bar{B}/N$, which is a BN -object by the covering classification above, so (ii) fails only if $\text{el}(\square^n/N)$ is a homotopy retract of BN — a third non-domination condition, decidable by el-data. Where rank comparisons do not settle it, (ii) is a stage-zero-visible type-side condition: on the Boolean site the strict endomorphism classes and the strict-homotopy component of the identity are finite, exactly computable sets, and for the Klein machine the component is $\{\text{id}\}$ while the identity is not strictly invariant; the artifact verifies (ii) at stage zero for the $n \leq 3$ bases.

Theorem 14.6 (Fundamental group of an isotropy quotient). *For every $H \leq B_n$, the group $\pi_1 \text{el}(\square^n/H)$ is a quotient of H in which the stabilizer of every cell dies. In particular, if H is generated by elements with nonempty fixed locus — e.g. if H is reflection-generated — then $\text{el}(\square^n/H)$ is simply connected.*

Proof. Cells of $\square^n/H$ are H -orbits of cells, and a morphism $[z] \rightarrow [x]$ of $\text{el}(\square^n/H)$ is a cube map u with $[x] \cdot u = [z]$. *Edges lift:* given a representative z' of $[z]$ and such a u , we have $x \cdot u = \sigma_h z'$ for some representative x and h , so $\sigma_{h^{-1}} x \cdot u = z'$; iterating along an edge path, every edge loop of $\text{el}(\square^n/H)$ at a base cell b lifts to an edge path in $\text{el}(\square^n)$ from a representative b' to $\sigma_h b'$ for a unique $h \in H$. Since $\text{el}(\square^n)$ is contractible (the generic cell is terminal), two lifts with the same h are homotopic, the concatenation of loops multiplies the h 's, and every h occurs (project any path from b' to $\sigma_h b'$). Hence $h \mapsto [\text{projected loop}]$ is a surjection $H \twoheadrightarrow \pi_1 \text{el}(\square^n/H)$. If h stabilizes some cell x , conjugate the basepoint to $[x]$: the lift of the corresponding loop can be taken constant at x (since $\sigma_h x = x$), so the loop is constant and h dies; a general element of the normal closure of the stabilizers dies because the kernel is normal. When H is generated by stabilizing elements the surjection kills a generating set, and reflections stabilize vertices (Lemma 11.1), giving the last claim. $\square$

Proposition 14.7 (Cover lifting). *Let N be reflection-generated and suppose the identity of $\square^n/N$ factors through Σ/N up to homotopy. Then $\text{el}(\square^n/N)$ is a homotopy retract of the universal cover of one component of $\text{el}(\Sigma/N)$. Consequently, if every component of $\text{el}(\Sigma/N)$ has an $\mathbb{F}_2$ -acyclic universal cover, then Hypothesis 14.4(i) follows from $\tilde{H}^*(\text{el}(\square^n/N); \mathbb{F}_2) \neq 0$ alone.*

Proof. A factorization $\text{id} \simeq q \circ g$ with $q: \Sigma/N \hookrightarrow \square^n/N$ induces $s := \text{el}(g)$ with $\text{el}(q) \circ s \simeq \text{id}$. The source is simply connected by Theorem 14.6, so s lifts along the universal covering $\pi: \tilde{E} \rightarrow E$ of the component E it hits: $\pi \circ \tilde{s} = s$. Then $(\text{el}(q) \circ \pi) \circ \tilde{s} \simeq \text{id}$, the claimed retraction; a retract of an $\mathbb{F}_2$ -acyclic space is $\mathbb{F}_2$ -acyclic, and the constant case is the empty-sieve instance of the same argument. $\square$

Remark 14.8. Theorem 14.6 is machine-confirmed: in the vertex Čech model, $H_1(\text{el}(\square^n/N); \mathbb{F}_2) = 0$ for every reflection-generated mixed class with $n \leq 4$ (1 + 3 + 30 classes; artifact), while $\tilde{H}_* \neq 0$ throughout (a nonzero H_3 in every case). For 24 of the 30 classes at $n = 4$ the inclusion-induced q_* already fails to be surjective on $\mathbb{F}_2$ -homology, which settles the factorization through the sieve there unconditionally.

Proposition 14.9 (Failure of non-domination at $n = 4$). *Part (i) of Hypothesis 14.4 is false in general. Among the six remaining classes at $n = 4$, three (of orders 8, 8, 24) have reflection-locus incidence posets whose order complexes are trees (5, 5, and 11 loci with no chains of length*

three; artifact), so $\text{el}(\Sigma)$ is contractible; by the collage formula (Theorem 10.1) the comma collapses onto BN and the sieve inclusion induces an equivalence $\text{el}(\Sigma/N) \xrightarrow{\simeq} \text{el}(\square^4/N)$. In particular $\text{el}(\square^4/N)$ is a homotopy retract of $\text{el}(\Sigma/N)$ there — consistently, the vertex Čech model gives $H_*(\text{el}(\Sigma/N)) \cong H_*(\text{el}(\square^4/N))$ with q_* an isomorphism in these three cases. The other three classes (orders 16, 32, 192) have incidence complexes with $\mathbb{F}_2$ -Betti numbers $(1, 1, 0)$, $(1, 5, 0)$, and $(1, 0, 15)$; the last explains the excess $H_2(\text{el}(\Sigma/N)) \neq 0$ over $H_2(\text{el}(\square^4/N)) = 0$ exactly; at this stage of the development part (i) was undecided for these three — the dichotomy of Section 15 below settles all of them.

Remark 14.10 (The governing invariant). Proposition 14.9 recasts the problem: what decides the sieve factorization at a reflection-generated mixed N is the homotopy type of the *incidence complex of the reflection arrangement* of N . At $n \leq 3$ every base has a noncontractible incidence complex (two points, four points, and a graph with $b_1 = 3$), which is why Hypothesis 14.4(i) is a theorem there (Proposition B.3); at $n = 4$ tree arrangements appear for the first time and the test side no longer separates the quotient from its sieve. Whether the identity factors through Σ/N up to *type* homotopy is then a separate, positive-decidable question, settled for two of the three tree classes by the following proposition.

Proposition 14.11 (Sieve factorization at tree arrangements). *On the Boolean site, for two of the three tree classes at $n = 4$ (the two of order 8) the identity of $\square^4/N$ factors through Σ/N up to type homotopy: there is an N -equivariant vertex map $r: \square^4 \rightarrow \square^4$ whose generic cell is stabilized (its image lies in a single plane stratum), and the pair (id, r) is a strict cylinder homotopy $(\square^4/N) \times \square^1 \rightarrow \square^4/N$ from the identity to $j \circ g$, where g is the class of r in $\text{Hom}(\square^4/N, \Sigma/N)$. Consequently the identity lifts along $\bar{\Phi}_N$ up to type homotopy, and $\bar{\Phi}_N$ is not a separating witness at these two classes. For the third tree class (order 24) no such one-step witness exists: an exhaustive search over all twisted-equivariance shapes $r \circ \sigma_h = \sigma_{c^{-1}hc} \circ r$ ($c \in N$) and all equivariant choices of values finds none (artifact).*

Proof. A one-step homotopy is a map $(\square^4/N) \times \square^1 \rightarrow \square^4/N$, i.e. a vertex function on $\{0, 1\}^5$ whose two slices share the same deck element for every $h \in N$; since the identity slice is a bijection its deck element is forced, so the second slice is exactly a c -twisted equivariant map for some $c \in N$, and it lands in the sieve iff its generic cell is stabilized. The cylinder $X \times \square^1 \rightarrow X$ is a genuine cylinder object in the type structure (every object is cofibrant, and connections contract $\square^1$), so a strict witness is an unconditional type homotopy; the search is exact because on the Boolean site cells are arbitrary vertex functions. The witnesses (and the exhaustion for the third class) are in the artifact; for one order-8 class there are 256 witnesses, for the other 32768. Lifting along $\bar{\Phi}_N$: the collapsed end includes Σ/N into J_N/N , and $\bar{\Phi}_N$ restricts to j there, so the end inclusion composed with g is the required lift. $\square$

Theorem 14.12 (Dimension reduction at tree arrangements). *For both order-8 tree classes, $\square^4/N$ is homotopy equivalent, simultaneously in both structures, to a lower-dimensional isotropy quotient by a mixed reflection-generated group: for the first class to $\square^2/K$ (the Klein quotient itself), for the second to $\square^3/R$ with R an order-4 mixed reflection-generated block. Consequently both classes are separation classes — the criterion’s law “SEP $\iff$ N mixed” survives at them — with the separating map obtained by transporting the lower-dimensional witness across the equivalence.*

Proof. In the witness census (artifact), every witness for the first class, and 8192 witnesses for the second, have image equal to the vertex set of a single stratum P whose setwise stabilizer is all of N , with $r|_P$ equal to an element of the residual group $R := \text{res}(P)$. Write $\iota: \square^d \rightarrow \square^4$ for the pattern embedding of P ($d = 2, 3$ respectively). Since the image of r lies in P and on the Boolean site vertex functions are maps, $r = \iota \circ \bar{r}$ with $\bar{r}: \square^4 \rightarrow \square^d$, and equivariance of r makes $\bar{r}$ intertwine the actions along $N \twoheadrightarrow R$; hence $\bar{r}$ and ι descend to $\bar{g}: \square^4/N \rightarrow \square^d/R$ and $\bar{j}: \square^d/R \rightarrow \square^4/N$. Then $\bar{g} \circ \bar{j} = [r|_P] = \text{id}$ strictly ($r|_P \in R$), and $\bar{j} \circ \bar{g} = [r] \simeq \text{id}$ by the witness cylinder — a homotopy valid in both structures. So $\square^4/N \simeq \square^d/R$ in both. The residual R is mixed (no fixed vertex on P) and generated by reflections (artifact census), so $\square^d/R$ is a separation class by the $n \leq 3$ criterion; a homotopy equivalence transports the attached map $\bar{\Phi}_R \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$ along it, using two-out-of-three in each structure. $\square$

Remark 14.13 (The recursion). Theorem 14.12 corrects the expectation raised by Proposition 14.11: the death of $\bar{\Phi}_N$ at tree arrangements signals not agreement but *recursion* — the quotient is equivalent to the isotropy quotient of an invariant stratum by its residual, and the criterion’s law survives with a refined witness dictionary: at a noncontractible arrangement, $\bar{\Phi}_N$ itself separates; at a tree with an equivariant stabilized retraction, dimension reduction hands the class to a lower-dimensional mixed reflection-generated residual and the law recurses. For the order-24 tree class — test-side reduced to its sieve, but with no one-step type-side retraction — the question is whether the identity factors through the sieve beyond one step; the dichotomy of Section 15 answers no (the class is rigid), completing the dictionary.

15. THE SIEVE-LANDING DICHOTOMY

The one-step analysis behind Proposition 14.11 globalizes into a complete decision procedure. Throughout this section the site is Boolean and N is reflection-generated mixed.

The codiscrete setting. The recursion of this section leaves the cube quotients and lands in a slightly larger class of objects, which we fix once. For a finite set T , let $E(T) := \text{cosk}_0(T)$ be the *codiscrete object* on T : its $[k]$ -cells are all functions $\{0, 1\}^k \rightarrow T$, so that $\square^n = E(\{0, 1\}^n)$ on the Boolean site.

An *instance* is a pair (T, M) with $M \leq \text{Sym}(T)$ fixed-point-free and generated by its fixed-point-having elements; the associated object is $E(T)/M$, and the cube quotients $\square^n/N$ of the criterion are the instances $(\{0, 1\}^n, N)$. Four standing facts make every argument below apply to instances verbatim: (i) maps from any presheaf into a codiscrete object are exactly vertex functions, so stage-zero maps $E(T)/M \rightarrow E(T')/M'$ are classes of vertex functions with the deck condition; (ii) $\text{el}(E(T))$ is contractible (the Čech nerve of T), $E(T) \times E_M$ is contractible and free, and the proofs of Theorem 14.6 and of the collage formula (Theorem 10.1) apply word for word — in particular $\text{el}(E(T)/M)$ is simply connected; (iii) the stabilized sieve consists of the cells valued in some $\text{Fix}(s)$, $s \neq 1$, whose loci and intersections are again codiscrete, hence contractible; (iv) the witness machine J_M over $E(T)$ and the sector machinery of Theorem 14.3 use only the coequalizer presentation of the replacement, connections and diagonals on the site, the contractibility and freeness of the body, and the fact that vertex stabilizers consist of fixed-point-having elements — all available for instances. The machinery is stated at instance generality below (Lemmas 15.1 and 15.3, Theorem 15.2), with the inductions organized so that no circularity arises: the classifying-space lemma is induction-free; visibility and the trichotomy at an instance of size $|T|$ use only instances of size $< |T|$ (the sieve pieces $E(\text{Fix}(s))$, $s \neq 1$, have strictly fewer vertices); and the criterion (Theorem 15.10) at size $|T|$ uses the trichotomy at size $|T|$ and the criterion at sizes $< |T|$ — a single simultaneous induction on $|T|$.

Lemma 15.1 (Classifying-space lemma). *Let X be a contractible object with a free G -action on the Boolean site and W/M any instance quotient. Then homotopy classes of maps into the replacement of the quotient are classified by coverings:*

$$[W/M, R(X/G)] \cong \text{Cov}_G(W/M)_{/\cong} \cong \text{Hom}(\pi_1 \text{el}(W/M), G)/\text{conj}.$$

Proof. By Lemma 8.3 the covering $X \rightarrow X/G$ extends to a G -equivariant fibrant replacement $P \supseteq X$ with $P \simeq *$, free G -action, and $P/G = R(X/G)$. A map $W/M \rightarrow P/G$ pulls the covering P back to a G -covering of W/M ; homotopic maps pull back isomorphic coverings (extend along the cylinder, Lemma 8.3). Conversely, a G -covering $Q \rightarrow W/M$ with Q 's components mapped to P G -equivariantly — possible, and unique up to equivariant homotopy, because P is contractible and fibrant and the action is free (equivariant obstruction theory reduces to ordinary obstruction theory on the free quotients) — descends to a map $W/M \rightarrow P/G$ with the given pullback. The two constructions are mutually inverse up to homotopy, and G -coverings of W/M are classified by $\text{Hom}(\pi_1 \text{el}(W/M), G)/\text{conj}$ by Lemma 8.1. $\square$

Theorem 15.2 (Codiscrete trichotomy). *For every instance (T, M) , the conclusion of Theorem 14.3 holds for the witness machine J_M over $E(T)$: every element of $[E(T)/M, R(J_M/M)]$ is homotopic to a constant, to a*

class factoring through the collapsed end Σ/M , or to an invariant-polynomial class.

Proof. Induction on $|T|$. The sector decomposition, the sector pasting of mixed boxes, and the equivariant slide of the cylinder sector onto the collapsed end use only the coequalizer presentation of the replacement and the connections and diagonals of the site, none of which sees the base object; they apply to $E(T)$ verbatim. The body sector is $E(T) \times E_M$ — contractible with free M -action — so its classes are classified by Lemma 15.1, reduce to stage zero, and are handled exactly as in Theorem 14.3 (the certificate analysis uses only that vertex stabilizers consist of fixed-point-having elements — tautological for an instance, where the stabilizer of a vertex $t \in T$ is by definition made of elements fixing t). The first-exit induction terminates through interface classes supported in the sieve, whose pieces are the codiscrete objects $E(\text{Fix}(s))$, $s \neq 1$, of strictly smaller vertex sets — covered by the inductive hypothesis. $\square$

For an instance — we write it as N acting on the vertex set for uniformity with the cube case — call $F: \{0, 1\}^n \rightarrow \{0, 1\}^n$ *c-twisted* ($c \in N$) if $F \circ \sigma_h = \sigma_{c^{-1}hc} \circ F$ for all $h \in N$, and *twisted* if it is c -twisted for some c ; write $D_F(h) := \{d \in N : F \circ \sigma_h = \sigma_d \circ F\}$ for the deck sets.

Lemma 15.3 (Stage-zero visibility). *On the Boolean site, for every instance — in particular for $\square^n/N$ — every class in $[E(T)/N, R(E(T)/N)]$ is represented by a strict cell (a vertex function with deck condition), and two strict classes are homotopic iff they are connected by a chain of strict cylinder homotopies $(E(T)/N) \times \square^1 \rightarrow E(T)/N$.*

Proof. Induction on $|T|$, simultaneous with Theorem 15.2. The sector and first-exit machinery applies to $E(T)/N$ itself (decomposed along Σ/N and its free complement) and, for homotopies, to the same decomposition of $(E(T)/N) \times \square^1$: every replacement-level class reduces stagewise to a stage-zero class — the body part by the covering classification (Lemma 15.1), the sieve part by the inductive hypothesis on the pieces $E(\text{Fix}(s))$ of strictly smaller vertex sets — and on the Boolean site stage-zero cells are exactly vertex functions, and stage-zero cylinders exactly pairs of slices with common deck elements. $\square$

Lemma 15.4 (Freedom equals stabilization). *Let F be twisted. Then some deck set $D_F(h)$ has more than one element iff F is a stabilized cell (iff its class lands in Σ/N). Moreover: one-step homotopies from the identity class are exactly the twisted classes; every twisted class is one-step homotopic to the identity; and if no twisted map is stabilized, a one-step homotopy from a twisted class leads only to twisted classes.*

Proof. $d_1, d_2 \in D_F(h)$ iff $\sigma_{d_1^{-1}d_2} \circ F = F$, i.e. iff F is stabilized by $d_1^{-1}d_2$; and a level- n cell lies in Σ iff it is stabilized. A one-step cylinder from the identity has slices $(\sigma_a, \sigma_b \circ r)$ sharing deck elements; since σ_a is a bijection

its deck element at h is forced to be aha^{-1} , which makes r $(a^{-1}b)$ -twisted; conversely, if F is c -twisted then the pair $(\sigma_a, \sigma_{ac}F)$ shares the deck elements aha^{-1} , so F is one-step homotopic to the identity. For the last claim: if all deck sets of all twisted maps are singletons, a shared-deck cylinder out of a translate of a twisted map forces the other slice to satisfy a full twisted equation (with the twist adjusted by the translations). $\square$

Theorem 15.5 (Sieve-landing dichotomy). *Exactly one of the following holds for each reflection-generated mixed N on the Boolean site.*

- (R) *Some twisted map is stabilized. Then the identity of $\square^n/N$ factors through Σ/N up to type homotopy by a single strict cylinder, and $\bar{\Phi}_N$ is not a separating witness (reduction).*
- (S) *No twisted map is stabilized. Then the strict homotopy component of the identity is exactly the set of twisted classes, which contains no constant class, no class landing in Σ/N , and no strictly invariant class. Consequently all three cases of the trichotomy (Theorem 14.3; Theorem 15.2 for general instances) are excluded for the identity and $\bar{\Phi}_N \in W_{\text{test}} \setminus W_{\text{type}}$ — no cohomological input is needed.*

Proof. (R) is Proposition 14.11’s argument. (S): by Lemma 15.4 the component of the identity under strict cylinder chains consists of twisted classes, and by Lemma 15.3 this computes the type homotopy component. A constant const_w is twisted only if $\sigma_{c^{-1}hc}$ fixes w for all h , i.e. only if N fixes a vertex — excluded for mixed N ; a class lands in Σ/N only if its cell is stabilized — excluded in case (S); a strictly invariant F that is also c -twisted satisfies $\sigma_{c^{-1}hc}F = F$ for all h , making it stabilized — excluded again. These are exactly the $\bar{\Phi}_N$ -images of the trichotomy’s cases (b), (c), and (a) respectively: in particular the constant case is excluded combinatorially, with no cohomological input. $\square$

Theorem 15.6 (The complete classification for $n \leq 4$). *Every reflection-generated mixed class with $n \leq 4$ ($1 + 3 + 30$ classes) is decided, in closed form, by the dichotomy: the two order-8 tree classes of Theorem 14.12 are of type (R) and are separation classes by dimension reduction; the other 32 are of type (S), so $\bar{\Phi}_N$ separates (the vertex Čech computation $\tilde{H}_3 \neq 0$ in every case is consistency data, not an input). Hence*

$$\text{SEP}(H) \iff \langle\langle \text{Ref}(H) \rangle\rangle \text{ is mixed}$$

holds for every $H \leq B_n$ with $n \leq 4$, unconditionally: Hypothesis 14.4 is eliminated. (Existence of a stabilized twisted map is decided orbitwise: a stabilized c -twisted map stabilized by s exists iff for every vertex orbit representative v some vertex is fixed by $c^{-1}\text{Stab}(v)c$ and by every N -conjugate of s ; artifact.)

Proof. The census is the artifact’s closed-form computation. For type (R), Theorem 14.12; for type (S), Theorem 15.5; for a general mixed H the

attached witness descends along the free H/N -quotient as in the criterion theorem below, and for N non-mixed the agreement half applies. $\square$

Theorem 15.7 (Structure of reductions). *In case (R), let F be a stabilized twisted map and $S \leq N$ its stabilizer. Then:*

- (1) S is normalized by N , and the image of F lies in $\text{Fix}(S)$, an N -invariant locus.
- (2) Some power F^k restricts to a bijection of $T := \text{im } F^k$, an N -invariant vertex subset of $\text{Fix}(S)$, and $\text{id} \simeq [F^k]$ by a single cylinder.
- (3) $\square^n/N$ is homotopy equivalent, simultaneously in both structures, to $E(T)/N$ — the quotient of the codiscrete object on T (levelwise all functions into T), a subobject of Σ/N (Figure 6).
- (4) N acts on T with no fixed point, through its image $M \leq \text{Sym}(T)$; in particular $|T| \geq 2$.

Proof. (1) From $F\sigma_h = \sigma_{c^{-1}hc}F$ we get $\sigma_{c^{-1}hc}^{-1}F = F\sigma_h^{-1}$, so for $s \in S$: $\sigma_{(c^{-1}hc)s(c^{-1}hc)^{-1}}F = \sigma_{c^{-1}hc}\sigma_s F\sigma_h^{-1} = \sigma_{c^{-1}hc}F\sigma_h^{-1} = F$; as $h \mapsto c^{-1}hc$ is a bijection of N , every N -conjugate of s stabilizes F . $\text{im } F \subseteq \text{Fix}(S)$ is the definition of stabilization, and $\sigma_h \text{Fix}(S) = \text{Fix}(hSh^{-1}) = \text{Fix}(S)$. (2) By induction F^k is c^k -twisted and S -stabilized; by finiteness some power restricts to a bijection of its image, and $\sigma_g T = F^k(\sigma_{c^k g c^{-k}} V) = T$; the homotopy is Lemma 15.4. (3) On the Boolean site cells are vertex functions, so the cells valued in T form the codiscrete subobject $E(T) \subseteq \square^n$, contained in Σ because the nontrivial S fixes T pointwise. F^k factors as $\iota \circ \bar{g}_0$, twisted equivariance descends to $\bar{g}: \square^n/N \rightarrow E(T)/N$ and $\bar{j}: E(T)/N \rightarrow \square^n/N$, and $\bar{j}\bar{g} = [F^k] \simeq \text{id}$ while $\bar{g}\bar{j} = [F^k|_{E(T)}]$ has strictly invertible power ($F^k|_T$ is a bijection of finite order), so both composites are isomorphisms in each homotopy category and $\bar{g}$ is an equivalence. (4) A fixed point of N on T would be an N -fixed vertex of $\square^n$, contradicting mixedness; $|T| = 1$ would be such a fixed point. $\square$

$$\begin{array}{ccccc}
 & & \bar{g}\bar{j}=[F^k|_T] \in \text{Aut} & & \\
 & & \downarrow & & \\
 \square^n/N & \xrightarrow{\bar{g}=[F^k]} & E(T)/N & \xrightarrow{\bar{j}} & \square^n/N \\
 & \searrow & & \nearrow & \\
 & & \text{id} \simeq \bar{j}\bar{g} & &
 \end{array}$$

FIGURE 6. Dimension reduction at a case-(R) class (Theorem 15.7): the stabilized twisted map F^k factors through the codiscrete object on its eventual image T , the composite on $E(T)/N$ is a strict finite-order automorphism, and the outer composite is a one-step cylinder homotopy — an equivalence in both structures at once.

Corollary 15.8 (The recursion is well-founded). *The dichotomy theory — twisted maps, deck sets, the sieve of stabilized cells, Lemmas 15.3 and 15.4, and Theorem 15.5 — uses only that cells on the Boolean site are vertex functions, so it applies verbatim to codiscrete quotients $E(T)/M$. By Theorem 15.7, each case-(R) step replaces (T, M) by a strictly smaller fixed-point-free instance, so the recursion terminates at a case-(S) leaf with $|T| \geq 2$, and the leaf’s separating map transports up the chain of equivalences by two-out-of-three. At $n \leq 4$ both reduction chains have length one and land on full loci — the Klein quotient and an order-4 mixed block. The leaf instances are again “reflection-generated mixed” in the $E(T)$ -sense by the following transport lemma, so the trichotomy machinery applies to them.*

Lemma 15.9 (Reflection transport). *In every reduction, the image M of N in $\text{Sym}(T)$ is generated by its fixed-point-having elements. More precisely: if N is generated by a conjugation-closed set R of elements each having a fixed point (for $N \leq B_n$, its reflections; Lemma 11.1), then every element of R fixes a point of T .*

Proof. Let $\tilde{\rho} \in R$ and set $\rho := c^k \tilde{\rho} c^{-k} \in R$ (conjugation-closed), with fixed point v . Twisted equivariance of F^k gives $F^k \circ \sigma_\rho = \sigma_{c^{-k} \rho c^k} \circ F^k = \sigma_{\tilde{\rho}} \circ F^k$, so $F^k(v) = F^k(\rho v) = \sigma_{\tilde{\rho}}(F^k(v))$: the element $\tilde{\rho}$ fixes the point $F^k(v) \in T$. Since $N = \langle R \rangle$, the image M is generated by the images of R , each with a fixed point; and the fixed-point-having generating set of a leaf is again conjugation-closed, so the lemma applies at every level of the recursion. $\square$

Theorem 15.10 (The criterion at every n). *On the Boolean site, for every finite set T and every $M \leq \text{Sym}(T)$ generated by its fixed-point-having elements:*

$$\text{SEP}(E(T)/M) \iff M \text{ has no fixed point on } T,$$

and consequently, for every n and every $H \leq B_n$,

$$\text{SEP}(H) \iff \langle\langle \text{Refl}(H) \rangle\rangle \text{ is mixed,}$$

with no hypotheses beyond the standing machinery (Lemma 15.3 and the trichotomy for codiscrete bases). Figure 4 displays the recursion.

Proof. Induction on $|T|$. If M fixes t_0 , the constant at t_0 is twisted (its deck elements can all be taken trivial), so the identity is one-step homotopic to a constant and $E(T)/M$ is contractible in both structures: agreement. If M is fixed-point-free, apply the dichotomy (Theorem 15.5): in case (S), $\bar{\Phi}_M$ separates; in case (R), Theorem 15.7 gives an equivalence, in both structures, to a strictly smaller instance $E(T')/M'$ which is fixed-point-free (Theorem 15.7(4)) and generated by fixed-point-having elements (Lemma 15.9); the inductive witness transports across the equivalence by two-out-of-three. For $H \leq B_n$: if $N := \langle\langle \text{Refl}(H) \rangle\rangle$ is non-mixed the agreement half applies; if mixed, N is generated by its reflections — fixed-point-having on the vertex set — so the first statement attaches a witness to $\square^n/N = E(\{0, 1\}^n)/N$

restricted along the faithful image, and it descends along the free H/N -quotient as in Theorem 16.1. $\square$

The conditional route that preceded the dichotomy — a separation theorem under Hypothesis 14.4 and its low-dimensional verification through exact $\mathbb{F}_2$ -cohomology evaluations of the collage formula — is superseded by Theorem 15.10; it is recorded, with its independently useful computations (Proposition B.3), in Appendix B.

16. THE CRITERION THEOREM

Theorem 16.1 (The criterion). *Let $H \leq B_n$ be mixed and $N := \langle\langle \text{Refl}(H) \rangle\rangle$. (1) On any reversal site, if N has a common fixed cell then $\square^n/H \simeq B(H/N)$ in both structures. (2) If N is mixed then an explicit test weak equivalence that is not a type weak equivalence is attached to H : on the Boolean site for every n , and on the De Morgan and Kleene sites for $n \leq 3$ — a restriction removed in Part 4 (Theorem 17.2).*

Proof. (1) is Theorem 13.1. (2): N is reflection-generated (the closure is generated by the H -conjugates of reflections of H , which are reflections) and mixed, so Theorem 15.10 attaches a separating map to N on the Boolean site; H/N contains no reflections, acts freely on $\square^n/N$ and on the witness objects, and the covering descent theorem (Theorem 8.4) descends the separation to H along the free quotient. For $n \leq 3$ the same conclusion transfers to the De Morgan and Kleene sites by Part 2, and the criterion reproduces the classification of [11] on all 98 subgroups with zero mismatches (Part 6). $\square$

17. OUTLOOK

The sieve-landing dichotomy (Theorem 15.5) replaces Hypothesis 14.4 entirely, needs no cohomological input, and — with the reflection-transport lemma — yields the criterion at every n (Theorem 15.10), with $n \leq 4$ decided in closed form (Theorem 15.6); the standing machinery is stated at codiscrete generality in Section 15, its proofs being the sector and first-exit inductions of Theorem 14.3 with the cube replaced by $E(T)$. What remains open beyond the comparison program itself is the converse of the Sylow cone below; the Dedekind side of the phase diagram is analyzed in Part 5, where its natural witness is refuted and the question remains open.

Independently of the criterion, the cohomology of the orbit spaces $E(T)/M$ — simply connected when M is generated by fixed-point-having elements (Theorem 14.6), computed by the collage formula from the fixed-point arrangement — exhibits a dichotomy of its own. One direction is a theorem:

Proposition 17.1 (Sylow cone). *If a Sylow 2-subgroup P of M has a fixed point $t_0 \in T$, then $E(T)/M$ is $\mathbb{F}_2$ -acyclic.*

Proof. The extra degeneracy $s_P(x_0, \dots, x_k) := (t_0, x_0, \dots, x_k)$ is P -equivariant (t_0 is P -fixed) and contracts the augmented chain complex $\tilde{C}_*(E(T); \mathbb{F}_2)$ over $\mathbb{F}_2 P$. The relative trace $s := \sum_{mP \in M/P} \sigma_m s_P \sigma_m^{-1}$ is well defined and M -equivariant, and $ds + sd = [M:P] \cdot \text{id} = \text{id}$ over $\mathbb{F}_2$; applying $(-)_M$ gives a contraction of $\tilde{C}_*(E(T)/M)$. $\square$

Hence nonvanishing forces the Sylow 2-subgroups to be fixed-point-free, and the computations (twelve cases, artifact) are consistent with the converse for the criterion-relevant class:

- (D) for M generated by its fixed-point-having elements, $\tilde{H}_*(E(T)/M; \mathbb{F}_2) \neq 0$ iff the Sylow 2-subgroups of M have no fixed point in T .

Outside that class the nonvanishing can be purely fundamental-group-driven ($\mathbb{Z}/4$ acting on a regular orbit plus a 2-orbit has $\tilde{H}_* = (1, 1, 1, \dots)$ from $\pi_1 = \mathbb{Z}/2$), while inside it, S_3 on three points is fixed-point-free with Sylow fixed points and $\mathbb{F}_2$ -acyclic orbit space (though not $\mathbb{F}_3$ -acyclic) — yet still a *separation* instance of the criterion, its twisted maps forced onto the full vertex set: the combinatorial dichotomy is strictly sharper than $\mathbb{F}_2$ -cohomology. Every mixed reflection-generated cube class with $n \leq 4$ contains even a fixed-point-free elementary abelian 2-subgroup (artifact), so the cube world lies on the conjecturally nonvanishing side of (D) throughout — consistently, $\tilde{H}_3 \neq 0$ is observed in every case — and (D) is the promised bridge from the model-structure landscape to the mod-2 homotopy theory of finite group actions on finite sets, with $E(T)$ the classifying space of the family of subgroups with fixed points. Beyond it, two directions. The Dedekind question — whether the connective site without reversal presents spaces — is untouchable by isotropy witnesses (Theorem 3.1); in Part 5 its natural congruence-quotient witness is refuted by the shell contraction, and the question remains open. And the criterion suggests a dictionary with reflection arrangements: the fixed loci of $\text{Refl}(H)$ form an arrangement of subcubes, and the two homotopy theories disagree exactly when the arrangement's symmetry cannot be disentangled from a fixed stratum; a Coxeter-theoretic form of the criterion is a natural sequel.

Part 4. Every reversal site

The criterion of Part 3 is a Boolean-site theorem. This part extends it to the De Morgan and Kleene sites — to every standard cubical site with reversal — at every dimension, proving Theorem 1.2.

Theorem 17.2 (The criterion on every reversal site). *On each of the De Morgan, Kleene, and Boolean sites, for every n and every $H \leq B_n$: $\text{SEP}(H) \iff \langle\langle \text{Refl}(H) \rangle\rangle$ is mixed; and when the closure has a common fixed cell, $\square^n/H \simeq B(H/\langle\langle \text{Refl}(H) \rangle\rangle)$ in both structures.*

Three tools drive the extension: an equivariant interpolation theorem making the sieve-landing dichotomy available verbatim on every reversal site; a projection principle along the theory quotients onto the Boolean site (Figure 7); and an algebraic reduction replacing the vertex-function analysis at reduction classes (Figure 8).

18. THE EQUIVARIANT INTERPOLANT

The interpolant χ below is the median interpolant F of Part 1 in the variables (u, v, t) ; we restate and reprove its properties in the exact form used in this part (the self-duality, implicit in Lemma 4.3, is here the central point).

Lemma 18.1 (Self-duality of the convexity interpolant). *In the free De Morgan algebra on u, v, t , the element*

$$\chi(u, v, t) := (u \wedge \neg t) \vee (t \wedge v) \vee (u \wedge v)$$

satisfies

$$\chi(u, v, 0) = u, \quad \chi(u, v, 1) = v, \quad \chi(\neg u, \neg v, t) = \neg \chi(u, v, t).$$

Consequently the same identities hold in every quotient variety, in particular in free Kleene and free Boolean algebras.

Proof. The first two are the defining computations of the convexity interpolant. For the third, De Morgan's laws give

$$\neg \chi(u, v, t) = (\neg u \vee t) \wedge (\neg t \vee \neg v) \wedge (\neg u \vee \neg v).$$

Distributing the first two factors: $(\neg u \vee t) \wedge (\neg t \vee \neg v) = (\neg u \wedge \neg t) \vee (\neg u \wedge \neg v) \vee (t \wedge \neg t) \vee (t \wedge \neg v)$. Meeting with the third factor and absorbing — using $t \wedge \neg t \leq t$, $\neg t$, so $t \neg t \neg u \leq \neg t \neg u$ and $t \neg t \neg v \leq t \neg v$, and $t \neg u \neg v \leq \neg u \neg v$ — leaves

$$(\neg u \wedge \neg t) \vee (t \wedge \neg v) \vee (\neg u \wedge \neg v) = \chi(\neg u, \neg v, t).$$

Identities of free De Morgan algebras hold in all quotients; the free Kleene and Boolean algebras are quotients of the free De Morgan algebra on the same generators. (The identities are also machine-verified on the literal-cube representation; artifact.) $\square$

Proposition 18.2 (Shared deck data implies one-step homotopy). *Let φ, φ' be level- n cells with $\varphi \circ \sigma_h = \sigma_{d(h)} \circ \varphi$ and $\varphi' \circ \sigma_h = \sigma_{d(h)} \circ \varphi'$ for all $h \in N$, with the same $d(h)$. Then the componentwise interpolant $G_i := \chi(\varphi_i, \varphi'_i, t)$ is a one-step homotopy between their classes, on every reversal site. Conversely, the slices of a one-step homotopy share deck data, and a one-step homotopy from the identity class forces its other slice to be a twisted cell.*

Proof. Writing $(\sigma_d\psi)_i = \neg^{\varepsilon_i}\psi_{\pi(i)}$, the substitution acts on the $\square^n$ -variables only, so

$$\begin{aligned} G_i \circ (\sigma_h \times \text{id}) &= \chi(\varphi_i\sigma_h, \varphi'_i\sigma_h, t) = \chi(\neg^{\varepsilon_i}\varphi_{\pi(i)}, \neg^{\varepsilon_i}\varphi'_{\pi(i)}, t) \\ &= \neg^{\varepsilon_i}\chi(\varphi_{\pi(i)}, \varphi'_{\pi(i)}, t) = (\sigma_{d(h)}G)_i, \end{aligned}$$

the third equality by Lemma 18.1 when $\varepsilon_i = 1$. The restrictions at $t = 0, 1$ are φ, φ' by the interpolation identities. For the converse, restricting the deck equality of a cylinder to the two ends gives the shared deck data; and if one slice is σ_a (a representative of the identity class), the deck element at h is forced to be aha^{-1} — automorphism cells are distinguished by their vertex maps — so the other slice satisfies a full twisted equation. $\square$

Corollary 18.3 (The dichotomy on every reversal site). *All the statements of the sieve-landing dichotomy of Part 3 (Section 15) hold verbatim on $\mathcal{V}$ with “twisted vertex map” replaced by “twisted cell”: deck freedom equals stabilization ($d_1, d_2 \in D_F(h)$ iff $\sigma_{d_1^{-1}d_2}F = F$); every twisted cell is one-step homotopic to the identity (translate one end by Proposition 18.2); if no twisted cell is stabilized, the strict homotopy component of the identity is exactly the set of twisted classes and contains no constant, no sieve-landing class, and no strictly invariant class.*

Proof. The proofs in Part 3 manipulate cell equalities only — never the Boolean description of cells — except for the existence of cylinders between shared-deck pairs, which is Proposition 18.2. Constants: a constant at w is twisted only if N fixes w , excluded for mixed N . Strict invariance plus twistedness forces stabilization, excluded in the rigid case. $\square$

19. THE PROJECTION LEMMA

Lemma 19.1 (Projection). *For $\mathcal{V} \in \{\text{DM}, \text{KL}\}$ the theory quotient induces surjections $\pi: F_{\mathcal{V}}(k) \twoheadrightarrow F_{\text{BA}}(k)$ commuting with substitution, hence a site functor $\square_{\mathcal{V}} \rightarrow \square_{\text{BA}}$ that is the identity on objects. On cells and cylinders, π preserves: the σ -actions; deck equalities (with the same deck elements); restrictions to cylinder ends; identity, constant, stabilized, and strictly invariant cells. Consequently every strict chain of one-step homotopies on $\mathcal{V}$ projects to a strict chain on the Boolean site connecting the projected classes.*

Proof. The signed permutations and the structural maps of the sites are given by terms in $\wedge, \vee, \neg, 0, 1$, and π is a homomorphism of algebras commuting with substitution in each variable; all listed conditions are equalities between cells built from such terms, and π of an equality is an equality. A one-step cylinder is a cell with deck and end conditions, so its image is one. $\square$

Proposition 19.2 (Rigidity transfers). *If no twisted Boolean vertex map for N is stabilized, then no twisted $\mathcal{V}$ -cell for N is stabilized.*

$$\begin{array}{ccc}
\text{strict chains on } \mathcal{V} : & \text{id} \rightsquigarrow^{G^{(1)}} F_1 \rightsquigarrow^{G^{(2)}} \cdots \rightsquigarrow \xi & \downarrow \pi \\
\text{strict chains on BA :} & \text{id} \rightsquigarrow_{\pi G^{(1)}} \pi F_1 \rightsquigarrow_{\pi G^{(2)}} \cdots \rightsquigarrow \pi \xi &
\end{array}$$

FIGURE 7. The projection lemma: chains of one-step homotopies project along the theory quotient, preserving deck data, ends, constants, stabilization, and strict invariance.

Proof. A stabilized c -twisted $\mathcal{V}$ -cell projects, by Lemma 19.1, to a stabilized c -twisted Boolean cell, i.e. a stabilized twisted vertex map. $\square$

20. THE ALGEBRAIC REDUCTION

Theorem 20.1 (Reduction on $\mathcal{V}$). *Let $N \leq B_n$ be mixed and reflection-generated, and let F be a stabilized c -twisted $\mathcal{V}$ -cell with stabilizer $S := \{s : \sigma_s F = F\} \neq 1$. Then:*

- (1) *S is normalized by N , and $\text{Fix}(S)$ is an N -invariant locus, a subcube $\square^d$ with $d < n$ embedded by a pattern $\iota : x_i \mapsto \neg^{\varepsilon_i} m_{\text{cl}(i)}$;*
- (2) *the stabilization equalities force the pattern relations on the components of F , so $F = \iota \circ \bar{F}$ for a unique level- n cell $\bar{F}$ of $\square^d$, algebraically;*
- (3) *$\bar{F}$ and ι descend to $\bar{g} : \square^n/N \rightarrow \square^d/\text{res}$ and $\bar{j} : \square^d/\text{res} \rightarrow \square^n/N$, where res is the image of N in $\text{Aut}(\square^d)$, and both composites are twisted cells, hence one-step homotopic to the respective identities: $\square^n/N \simeq \square^d/\text{res}$ simultaneously in both structures;*
- (4) *$\text{res} \leq B_d$ is mixed and generated by its reflections.*

Proof. (1) For $s \in S$ and $h \in N$, twisted equivariance gives $\sigma_{(c^{-1}hc)s(c^{-1}hc)^{-1}} F = \sigma_{c^{-1}hc} \sigma_s F \sigma_h^{-1} = F$, and $h \mapsto c^{-1}hc$ is a bijection of N ; invariance of the locus follows, and the locus-pattern analysis of Part 2 (Lemma 9.1) identifies $\text{Fix}(S)$ with a pattern-embedded subcube, of dimension $d < n$ since $S \neq 1$ acts nontrivially. (2) For each $s \in S$, $\sigma_s F = F$ reads componentwise as $\neg^{\varepsilon_i} F_{\pi_s(i)} = F_i$: exactly the statement that the components of F are constant up to the pattern signs along the cycles of S , i.e. satisfy the defining relations of the image of ι . Choosing one representative component per class defines $\bar{F}$ with $F = \iota \circ \bar{F}$; uniqueness because the components of ι are distinct literals, so ι is a monomorphism on cells. (3) From $\iota \circ \bar{F} \circ \sigma_h = F \sigma_h = \sigma_{c^{-1}hc} F = \iota \circ \text{res}(c^{-1}hc) \circ \bar{F}$ and monomorphy, $\bar{F}$ intertwines along $N \rightarrow \text{res}$, and $\sigma_h \circ \iota = \iota \circ \text{res}(h)$; both descend. The composite $\bar{j}\bar{g} = [F]$ is twisted by hypothesis; the composite $\bar{g}\bar{j} = [\bar{F} \circ \iota]$ satisfies $(\bar{F}\iota) \circ \text{res}(h) = \bar{F}\sigma_h\iota = \text{res}(c^{-1}hc)(\bar{F}\iota)$, so it is $\text{res}(c)$ -twisted. Every twisted cell is one-step homotopic to the identity (Corollary 18.3), and the χ -cylinders are strict cells, hence homotopies in

the type and the test structure simultaneously. (4) A fixed vertex of res on $\square^d$ is an N -fixed vertex of $\square^n$ inside the locus — excluded by mixedness. Reflection transport: for a reflection $\tilde{\rho}$ of N , the conjugate $\rho := c\tilde{\rho}c^{-1}$ is a reflection with a fixed vertex v ; the vertex map of F sends v into $\text{Fix}(S)$ with $F(v) = F(\rho v) = \sigma_{\tilde{\rho}}(F(v))$, so $\text{res}(\tilde{\rho})$ fixes the $\square^d$ -coordinates of $F(v)$. Vertex-fixing signed permutations are reflections, and N is generated by its reflections, so res is generated by reflections. $\square$

$$\begin{array}{ccccc}
 & & \bar{g}\bar{j}=[\bar{F}\iota] \text{ twisted } \simeq \text{id} & & \\
 & & \downarrow & & \\
 \square^n/N & \xrightarrow{\bar{g}=[\bar{F}]} & \square^d/\text{res} & \xrightarrow{\bar{j}=[\iota]} & \square^n/N \\
 & \searrow & & \nearrow & \\
 & & \text{id} \simeq \bar{j}\bar{g}=[F] & &
 \end{array}$$

FIGURE 8. The algebraic reduction (Theorem 20.1): both composites are twisted cells, hence χ -one-step homotopic to identities in both structures — no power trick, no invertibility.

21. THE MAIN THEOREM

We use from Parts 2–3, now at $\mathcal{V}$ -generality: the trichotomy for the witness machine J_N/N (constant / through the stabilized sieve / strictly invariant), whose sector and first-exit proofs use only the coequalizer presentation of the algebraic replacement, connections and diagonals, the contractible free body, and the covering classification, all present on every reversal site (indeed the machinery of [21] was built on the De Morgan site); the stage-zero visibility of Proposition 21.1; the test-side weak equivalence of $\bar{\Phi}_N$; covering descent along free quotients; and the agreement theorem. Because Lemma 15.3 is stated on the Boolean site, we record its site-general form explicitly.

Proposition 21.1 (Stage-zero visibility on reversal sites). *Let $\mathcal{V} \in \{\text{DM}, \text{KL}, \text{BA}\}$. For every instance (T, N) — in particular for $\square^n/N$ — every class in $[E(T)/N, R(E(T)/N)]$ is represented by a strict cell: a cell of $E(T)/N$ itself, i.e. the N -class of a T -valued vertex function $\{0, 1\}^k \rightarrow T$ subject to the deck condition. (On the cube instances $T = \{0, 1\}^n$ over the Boolean site these are exactly the tuples of free-algebra elements; over a general site or a general finite T the codiscrete cells are arbitrary vertex functions, and no free-algebra representation is implied.) Two strict classes are homotopic in the cubical-type structure iff they are connected by a finite chain of strict cylinder homotopies $(E(T)/N) \times \square^1 \rightarrow E(T)/N$.*

Proof. Induction on $|T|$, simultaneous with the trichotomy at $\mathcal{V}$ -generality, exactly as in Lemma 15.3, of which this is the site-general form. That proof manipulates the replacement only through: the coequalizer presentation and its filler calculus (Lemmas A.2 and A.5), the equivariant slide (Lemma A.6), the first-exit induction (Lemma A.7), the contractibility of the free body, and the covering classification (Lemma 15.1) — each stated and proved in Appendix A for an arbitrary site of the paper, with the site entering only through its interval theory (connections and diagonals), all present on DM, KL, and BA. The only Boolean-specific step of Lemma 15.3 is the identification of stage-zero data: on the Boolean site stage-zero cells are vertex functions, while on a general reversal site a stage-zero cell of $R(E(T)/N)$ is a cell of $E(T)/N$ itself — a $\mathcal{V}$ -cell tuple with deck condition — which is the definition of a strict $\mathcal{V}$ -cell; likewise stage-zero cylinders are pairs of slices sharing deck elements. The induction closes as before: the sieve pieces $E(\text{Fix}(s))$, $s \neq 1$, have strictly smaller vertex sets. $\square$

Remark 21.2. The shell phenomenon of Part 5 (Theorem 24.5) does not threaten the theorems of this Part: over group quotients, shell data are forced to be strict cylinders, and fresh invariant fillers never extend the strict homotopy relation (Proposition 24.67, Remark 24.68).

Proof of Theorem 17.2. (Figure 9 displays the recursion.) The agreement half and the descent along the free H/N -quotient are as in Part 3, on every site. For the separation half let N be mixed and reflection-generated; induct on n . If some twisted $\mathcal{V}$ -cell is stabilized, Theorem 20.1 produces an equivalence, in both structures, with $\square^d/\text{res}$, $d < n$, res mixed reflection-generated; the inductive witness transports across the equivalence by two-out-of-three. If no twisted $\mathcal{V}$ -cell is stabilized, then by Corollary 18.3 the strict component of the identity consists of twisted classes and excludes constants, sieve-landing classes, and strictly invariant classes; by stage-zero visibility this computes the type homotopy relation, and by the trichotomy every lift candidate along $\bar{\Phi}_N$ would put the identity in one of the three excluded classes. Hence the identity does not lift and $\bar{\Phi}_N \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$. (The recursion terminates: d strictly decreases, and no mixed reflection-generated subgroup exists at $n \leq 1$.) $\square$

Remark 21.3. Combined with Proposition 19.2, the *shape* of the recursion also transfers: a class that is rigid on the Boolean site is rigid on $\mathcal{V}$. A Boolean reduction class may a priori be rigid on $\mathcal{V}$ (if its witness system is not $\mathcal{V}$ -realizable) — the theorem is indifferent, since both branches yield separation — but at $n = 4$ this does not happen: see the next section.

Part 5. The Dedekind site: the monotone dunce hat and the shell contraction

The sites of Parts II–IV all have reversal. The remaining standard sites have none: the cartesian site (settled by [1]), the one-connection site (settled

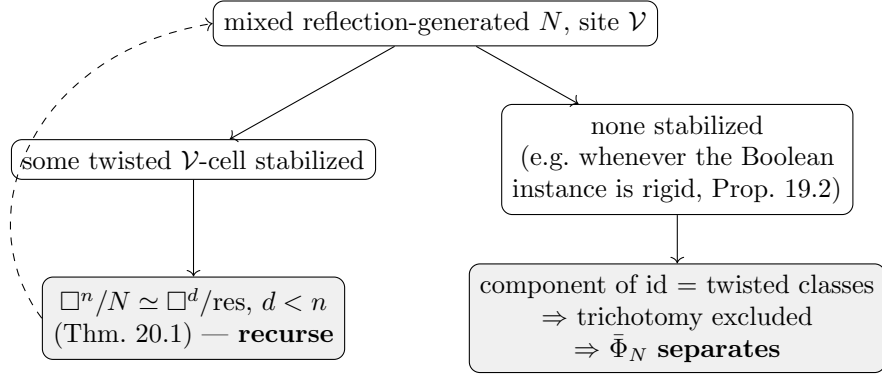


FIGURE 9. The recursion of Theorem 17.2 on an arbitrary reversal site.

by [8]), and the Dedekind site $\square_{\wedge\vee}$ — the Lawvere theory of bounded distributive lattices, whose comparison was the last open case of the Buchholtz–Sattler classification — and, after this Part, still the last open case. We settle the fate of its natural finite witness: the monotone dunce hat is test-contractible and, correcting the separation claimed in the frozen precursor of this Part in the artifact series [23], *type*-contractible. The contraction is found, and could only be found, through an exact provenance calculus for the uniform fibrant replacement, developed below and of independent use: cellular box-filling genuinely cannot contract the identity (Theorem 23.2), while the *shell boxes* of uniform fibrancy do (Theorem 24.5). This proves Theorem 1.3. The shell mechanism is then developed into a full calculus — cone shell moves (Theorem 24.7) and their compositions through fresh ends, the *tents* (Theorem 24.8) — and run as a census over all 1,533 small quotients of the square: *every* $\mathbb{F}_2$ -acyclic quotient is type-contractible (Theorem 24.11), the last resister, the monotone triple fold Q^* , falling to an explicit two-shell tent (Theorem 24.10). The expected direction of the still-open comparison thereby reverses (Remark 24.12), and the Part closes with the positive program: the criteria and the identification with the minimal Cisinski structure, the regularity reduction (Theorem 24.19), the shape-by-shape resolution of Section 24.5, and the triangulation resolution of Section 24.6, which settles the contractible tract — every atom is regular — and leaves the two tasks of Remark 24.66.

Definition 21.4 (The monotone dunce hat). $W_{\wedge\vee} := \square^2 / ((0, u) \sim (u, 0), (u, u) \sim (u, 1))$ (Figure 10).

Theorem 21.5. $W_{\wedge\vee}$ is test-contractible and type-contractible; it is not a separating witness. Moreover no transfinite composition of pushouts of representable open-box inclusions contracts its identity class (Theorem 23.2): the contraction genuinely requires a shell box.

The heart of the type-side rigidity is a one-line monotonicity lemma:

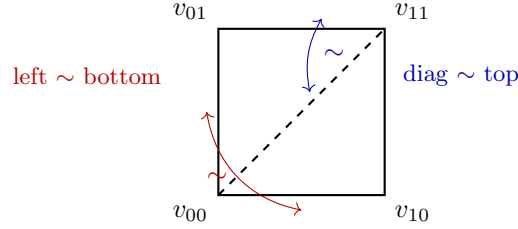


FIGURE 10. The monotone dunce hat $W_{\wedge\vee}$: the swap fold (left $\sim$ bottom) blocks the $\vee$ -connection contraction, the diagonal fold (diagonal $\sim$ top) blocks the $\wedge$ -connection contraction, and the diagonal-track lemma shows no representable open-box filling reassembles one — while the shell filler of Theorem 24.5 does. Vertices v_{00}, v_{01}, v_{10} merge.

Lemma 21.6 (Diagonal-track lemma). *A monotone function $\varphi: \{0, 1\}^3 \rightarrow \{0, 1\}$ constant on the diagonal cylinder $\{(u, u, t)\}$ is constant.*

Proof. The cylinder contains both the bottom vertex $(0, 0, 0)$ and the top vertex $(1, 1, 1)$; a monotone function with value e at the top is $\leq e$ everywhere, and with value e at the bottom is $\geq e$ everywhere. $\square$

No other track has this property (the middle variable is constant on the top cylinder but not constant) — this asymmetry is why the diagonal fold of $W_{\wedge\vee}$ blocks every cellular contraction, while each single-fold quotient is strictly contractible (artifact). The swap fold blocks the $\vee$ -connection contraction of the square, the diagonal fold blocks the $\wedge$ -connection contraction, and the lemma shows no higher-dimensional *representable-box* filling can reassemble a contraction. The shell boxes of Section 24.2 evade the lemma: their side data prescribe all fold tracks at once, so no track is fresh.

22. CONVENTIONS

$\square_{\wedge\vee}$ is the site of the Dedekind family of Part 1 with both connections and no reversal: $\square_{\wedge\vee}([m], [n]) = D(m)^n$ with $D(m)$ the free bounded distributive lattice on m generators (monotone maps $\{0, 1\}^m \rightarrow \{0, 1\}$). As before, $[a] \times [b] = [a + b]$, cofibrations are the monomorphisms, and fibrant objects are the uniform Kan complexes ([20, 9]). Two constructions enter the type-side analysis and must be distinguished. The *algebraic replacement* R of Lemma A.2 supplies the trivial cofibration $W \rightarrow R(W)$ into a fibrant object; it is *not* a free composition of open-box attachments — fibrancy on this site requires fillers for boxes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$ with S an arbitrary subpresheaf, strictly natural in substitution (Lemma A.1), and the naturality is implemented by Garner’s coequalizer identifications. A *cellular filling extension*, by contrast, is an honest transfinite composition of pushouts of representable open-box inclusions; such extensions are type-trivial cofibrations but need not be fibrant. Theorem 23.2 below is a statement about

cellular extensions; the exact provenance calculus of R itself, and the shell contraction it uncovers, are the subject of Section 24. We use three lemmas, the first about cellular extensions.

Lemma 22.1 (Provenance). *Let $X = Y_0 \subseteq Y_1 \subseteq \cdots \subseteq Y$ be a transfinite composition of pushouts of open-box inclusions $\square_\beta \hookrightarrow \square^{m_\beta}$ along attaching maps $\square_\beta \rightarrow Y_{<\beta}$. Then every cell of Y is either a cell of X (old) or arises, for a unique stage β and a unique cell c of $\square^{m_\beta}$ not lying in $\square_\beta$, as the image of c (fresh, with provenance (β, c)); no two distinct fresh cells and no fresh and old cell are equal, and a fresh cell's restriction along a cube map is computed by restricting c : it stays fresh if the restriction avoids $\square_\beta$, and attaches to the earlier cell given by the attaching map otherwise.*

Proof. Pushouts along monomorphisms of presheaves are computed levelwise by disjoint union of the complement, and monomorphisms are stable under pushout and transfinite composition. $\square$

Lemma 22.2 (Maps and cylinders from a quotient; the congruence case of Lemma A.4). *Let $W = \square^2/(A_1 \sim B_1, A_2 \sim B_2)$ be the coequalizer of two pairs of parallel cells $A_i, B_i \in \square^2([1])$, and Y any presheaf. Then maps $W \rightarrow Y$ are exactly the cells $h \in Y([2])$ with $h \cdot A_i = h \cdot B_i$ ($i = 1, 2$), and maps $W \times \square^1 \rightarrow Y$ are exactly the cells $H \in Y([3])$ with $H \cdot (A_i \times \text{id}) = H \cdot (B_i \times \text{id})$, where $\cdot$ denotes restriction along the evident cube maps. The two ends of such a cylinder are its restrictions at $t = 0, 1$.*

Proof. The universal property of the coequalizer, using $\square^2 = y[2]$, $\square^2 \times \square^1 = y[3]$, and that colimits of presheaves are computed levelwise. $\square$

Lemma 22.3 (Homotopy). *Two maps $f, g: W \rightarrow R(W)$ are homotopic in the cubical-type structure if and only if they are connected by a finite chain of cylinders $W \times \square^1 \rightarrow R(W)$ as in Lemma 22.2. The identity class of W is the class of the composite $\iota_{\wedge\vee}: W \rightarrow R(W)$ of the identity with the replacement unit, and W is type-contractible iff $\iota_{\wedge\vee}$ is homotopic to a constant map.*

Proof. All objects are cofibrant (cofibrations are the monomorphisms), $R(W)$ is fibrant, and $W \times \square^1$ is a cylinder object: the end inclusions are monomorphisms and the projection is a weak equivalence because the interval, having connections, is contractible. Left homotopy over a cylinder object between a cofibrant object and a fibrant one is an equivalence relation computed by finite chains. $\square$

23. THE IDENTITY IS ISOLATED

Fix $W = W_{\wedge\vee}$, with the two generating pairs $A_1 = (0, u)$, $B_1 = (u, 0)$ (swap fold) and $A_2 = (u, u)$, $B_2 = (u, 1)$ (diagonal fold), and let $W \subseteq Y$ be an *arbitrary* cellular filling extension: a transfinite composition of pushouts of open-box inclusions $\square \hookrightarrow \square^m$, of boxes of arbitrary dimensions, as in Lemma 22.1. We write $\iota_{\wedge\vee}$ also for the image of the generic cell of W in Y .

For a cylinder $H \in Y([3])$ write $d_L := H \cdot (A_2 \times \text{id})$, $d_R := H \cdot (B_2 \times \text{id})$ for its diagonal and top tracks.

Two finite facts about W itself enter the proof; both are verified exactly, by two independent implementations (Section 26).

Lemma 23.1 (Base facts). *(i) The class of the generic cell of W at level 2 is a singleton: no other cell of $\square^2$ is congruent to (x, y) . (ii) Among the cylinders $H \in W([3])$ (old cells), the only one with an end in the identity class is the constant cylinder. (iii) The class of the left edge $[(0, v)]$ of W is not the class of a constant edge.*

Proof. (i) The congruence is generated by the cells $A_i \cdot u \sim B_i \cdot u$; each generator has a component which is constant or a substitution instance of a single variable in a way the generic cell is not, and the transitive closure preserves this. (ii, iii) Finite enumeration over $D(3)^2$ and $D(1)^2$; artifact, two implementations. $\square$

Theorem 23.2 (Cellular isolation). *In every cellular filling extension Y of $W_{\wedge\vee}$: every cylinder $H \in Y([3])$ satisfying the two conditions of Lemma 22.2 with one end in the class of $\iota_{\wedge\vee}$ has both ends in the class of $\iota_{\wedge\vee}$; indeed the only such cylinders are constant in the sense that both ends equal $\iota_{\wedge\vee}$. Consequently no transfinite sequence of open-box fillings connects the generic class of $W_{\wedge\vee}$ to a constant.*

Proof. If H is old (a cell of W), Lemma 23.1(ii) applies. So let H be fresh with provenance (β, c) , c a level-3 cell of $\square^m$ avoiding $\square_\beta$, and suppose the end $e_0 := c \cdot (t=\varepsilon)$ has class $\iota_{\wedge\vee}$.

The diagonal track is fresh. The components of $d_L = c \cdot (A_2 \times \text{id})$ are the monotone functions $c_a(u, u, t)$. If d_L attached, some component would be constant at a present face's value; by Lemma 21.6 that component of c is globally constant, so c itself lies in that present face of $\square_\beta$ — contradicting freshness.

Hence d_R is fresh and $d_L = d_R$ exactly. A fresh cell never equals an old one (Lemma 22.1), so the condition $d_L = d_R$ in Y forces d_R fresh of the same stage and the equality of the underlying cube cells: $c \cdot (A_2 \times \text{id}) = c \cdot (B_2 \times \text{id})$ in $\square^m$.

The end attaches, exactly onto the generic cell. If e_0 were fresh its class would be a fresh singleton, not $\iota_{\wedge\vee}$. So e_0 lies in $\square_\beta$: writing f_0 for a present face containing it, $e_0 = f_0 \circ \rho$ for a unique level-2 cell ρ of $\square^{m-1}$, and its attachment is $A \circ \rho$ where $A \in Y_{<\beta}([m-1])$ is the attaching cell of the face. By Lemma 23.1(i) the class $\iota_{\wedge\vee}$ is a singleton, so $A \circ \rho = \iota_{\wedge\vee}$ exactly.

The contradiction. Restrict the exact equation $c \cdot (A_2 \times \text{id}) = c \cdot (B_2 \times \text{id})$ at $t = \varepsilon$: $e_0 \cdot A_2 = e_0 \cdot B_2$, i.e. $f_0 \circ (\rho \cdot A_2) = f_0 \circ (\rho \cdot B_2)$, and since the face inclusion is monic, $\rho(u, u) = \rho(u, 1)$ as cells of $\square^{m-1}$. Evaluating at $u = 0$ gives the vertex equation $\rho(0, 0) = \rho(0, 1)$. A level-1 cell of a representable cube whose two vertices agree is constant (its components lie in $D(1)$ and are determined by their endpoint values). Hence the left edge $\rho \cdot (0, v)$ of ρ

is a constant cell, and applying A ,

$$\iota_{\wedge V} \cdot (0, v) = A \circ \rho \cdot (0, v) = A \circ (\text{constant}) = \text{a constant edge of } Y_{<\beta},$$

contradicting Lemma 23.1(iii): the left edge of $W_{\wedge V}$ is not constant. So no fresh cylinder has an end of class $\iota_{\wedge V}$, and the theorem follows by transfinite induction over the filling presentation. $\square$

Remark 23.3. The proof uses only the diagonal-fold condition to pin the cylinder and the swap fold to make the base facts true; both folds are needed — each single-fold quotient is strictly contractible (artifact). The theorem covers boxes of arbitrary dimension: no dimension bound is used. What it does not cover is the uniform replacement itself, whose shapes include the shell boxes — and there a genuine contraction exists (Section 24.2).

24. THE PROVENANCE CALCULUS OF THE ALGEBRAIC REPLACEMENT

Theorem 23.2 is a statement about free open-box fillings, while the fibrant replacement R is not free: its filler values satisfy the attachment relations (i) and the exchange relations (ii) of Lemma A.2, over box shapes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$ with arbitrary subpresheaves S . This section develops the exact calculus of those identifications — the cartesian generating category, an intrinsic freshness criterion, and a complete normal form (Theorem 24.2), so that provenance in R is exactly as computable as in the free case — and then uses the calculus to settle the type side, in the direction *opposite* to the one the cellular theorem suggests. A first example (Example 24.3) shows that strong isolation fails in R ; the systematic search for what replaces it terminates in the *shell contraction* (Theorem 24.5): fresh fillers over shell boxes, whose fold tracks are entirely prescribed by the box, contract the identity of $W_{\wedge V}$ in every fibrant object.

24.1. The cartesian generating category and the based normal form. Recall from Remark A.3 that the morphisms of the generating category $\mathbb{J}$ are the *cartesian* squares: $\sigma: [m] \rightarrow [n]$ with $S' = \sigma^{-1}(S)$ exactly, and ε fixed, acting as $\hat{\sigma} = \sigma \times \text{id}_{\square^1}$. Cartesianness is not a convenience but a consistency requirement, and it is what makes the following two results true.

Lemma 24.1 (Freshness is intrinsic). *For a cartesian $\hat{\sigma}: j' \rightarrow j$ one has $\hat{\sigma}^{-1}(\text{dom } j) = \text{dom } j'$: membership of the value cell in the box domain is invariant along every exchange. Consequently a filler value $\text{val}(b, c)$ is old if and only if $c \in \text{dom}(j)$, in which case it equals $b(c)$; fresh classes contain no attached pair, and equality between fresh values is generated by the exchange relations alone.*

Proof. For a cell $c = (c', c_s)$ of $\square^m \times \square^1$: $\hat{\sigma}c = (\sigma c', c_s)$, and $\sigma c' \in S \iff c' \in \sigma^{-1}(S) = S'$ (cartesianness), while the $\square^1$ -component is untouched, so $\hat{\sigma}c \in S \times \square^1 \cup \square^n \times \{\varepsilon\} \iff c \in S' \times \square^1 \cup \square^m \times \{\varepsilon\}$. The attachment relations (i) are available exactly at domain cells, and on them the two sides

agree along exchanges: $(b \circ \hat{\sigma})(c) = b(\hat{\sigma}c)$. A class containing no domain pair therefore never meets an old cell, and by Lemma A.2(2) all remaining equality is exchange-generated. $\square$

Theorem 24.2 (Based normal form). *Call a pair based if it has the form $(\beta, (\text{id}_{[k]}, s))$ with β a box of shape $(T \hookrightarrow \square^k) \hat{\otimes} \delta^\varepsilon$ and $s \in D(k)$, the cell being the graph cell $(u_1, \dots, u_k, s)$. Every pair (b, x) with $x = (x', x_s)$ at level k is exchange-equal to its base form*

$$N(b, x) := (b \circ (x' \hat{\times} \text{id}), (\text{id}_{[k]}, x_s)), \quad T = x'^{-1}(S),$$

and the base form is invariant under every exchange move. Hence base forms are a complete invariant: two fresh values are equal iff their base forms coincide, and the fresh cells of each stage at level k correspond bijectively to the based pairs (β, s) with $\text{id}_{[k]} \notin T$ and $s \neq \text{const}_\varepsilon$. Restriction is computed on based pairs by

$$\text{val}(\beta, s) \cdot f = \text{val}(\beta \circ (f \hat{\times} \text{id}), s \circ f),$$

and $\text{val}(\beta, s) \cdot f$ is old iff $f \in T$ or $s \circ f = \text{const}_\varepsilon$.

Proof. The base move is legal: $x': [k] \rightarrow [n]$ is a cube map, $T := x'^{-1}(S)$ makes $x' \times \text{id}$ cartesian, its Leibniz square maps the based domain into $\text{dom}(j)$, and $(x' \times \text{id}) \circ (\text{id}, x_s) = (x', x_s) = x$, so $(b, x) \sim N(b, x)$ by one exchange. Invariance: for an exchange $(b \circ \hat{\sigma}, c) \sim (b, \hat{\sigma}c)$ with $c = (c', c_s)$, $N(b \circ \hat{\sigma}, c) = (b \circ \hat{\sigma} \circ (c' \hat{\times} \text{id}), (\text{id}, c_s)) = (b \circ ((\sigma c') \hat{\times} \text{id}), (\text{id}, c_s)) = N(b, \hat{\sigma}c)$, since $\hat{\sigma} \circ (c' \hat{\times} \text{id}) = (\sigma \circ c') \hat{\times} \text{id}$ and $(\hat{\sigma}c)_s = c_s$. A based pair is its own base form, so two based pairs are exchange-equal iff equal; combined with Lemma 24.1 (fresh classes are pure-exchange) this gives the bijection. The restriction formula is strict naturality (Lemma A.2(3)) followed by rebasing; the oldness criterion is Lemma 24.1 applied to the based pair. $\square$

Theorem 24.2 restores, for the honest algebraic replacement, everything the free provenance Lemma 22.1 provided — with the based box and the coordinate $s \in D(k)$ as the exact bookkeeping data. What it does *not* restore is the conclusion of Theorem 23.2:

Example 24.3 (Free translates: strong isolation fails in R). Take the shape $(\emptyset \hookrightarrow \square^3) \hat{\otimes} \delta^0$ — an empty-sided box, i.e. a free cylinder attachment, a shape which representable open-box calculi do not contain — with box $B := \iota_{\wedge\vee} \circ \text{pr}: \square^3 \times \{0\} \rightarrow R$ (the degenerate fold-cylinder on $\iota_{\wedge\vee}$) and coordinate $s = w$. Then $H := \text{val}(B, w)$ satisfies both fold conditions: by Theorem 24.2 the two tracks of fold i are the values at the based boxes pulled back along the track maps, which coincide because $\iota_{\wedge\vee} \cdot A_i = \iota_{\wedge\vee} \cdot B_i$ ($\iota_{\wedge\vee}$ is a fold cell), and their coordinates agree ($w = w$). Its 0-end is the old cell $B(n_0, 0) = \iota_{\wedge\vee}$; its 1-end is *fresh* (the empty side contains nothing and $1 \neq 0$), hence not equal to $\iota_{\wedge\vee}$. So $\iota_{\wedge\vee}$ is fold-cylinder-adjacent to a fresh cell: in R the identity class is not isolated in the strong sense — as it must be, in a fibrant object, where every cell acquires free homotopy translates.

More is true: Section 24.2 shows that with a larger shell the translates step *outside* the strict component of the identity.

24.2. The shell contraction. The based calculus directs attention to fold-cylinders whose four track values are all *old* — prescribed by the box rather than created by the filler, so that the fold conditions become finitely many congruence identities of the boundary data. An exhaustive search over such configurations at the first filling stage (artifact, `dedekind_pcs0.py`) produced the following data.

Let $T \subseteq \square^3$ (coordinates u, v, w) be the union of the six squares

$$\begin{aligned} a_2 &= (u, u, w), & b_2 &= (u, 1, w), & a_1 &= (0, u, w), \\ b_1 &= (u, 0, w), & n_0 &= (u, v, 0), & n_1 &= (u, v, 1), \end{aligned}$$

and define Φ on $T \times \square^1 \cup \square^3 \times \{0\}$, with prism coordinates (p, q, r) on each square, by the constant v_{00} on $\square^3 \times \{0\}$ and by

$$\begin{aligned} a_2 &\mapsto (p \wedge r, p \wedge r), & b_2 &\mapsto (p \wedge r, r), & a_1, b_1 &\mapsto (0, p \wedge r), \\ n_0 &\mapsto (p \wedge r, q \wedge r), & n_1 &\mapsto (p \wedge q \wedge r, r \wedge (p \vee q)). \end{aligned}$$

Lemma 24.4. *Φ is a well-defined map to $W_{\wedge\vee}$: the six prisms restrict to the constant on their 0-slices and agree pairwise on all shared cells. (The pairwise intersections of the six squares are generated by shared edges; on these the displayed formulas agree, in two cases up to the fold congruence — on $b_1 \cap n_0$ one compares $(x \wedge y, 0)$ with $(0, x \wedge y)$, identified by the swap fold, and similarly on $b_1 \cap n_1$. Machine-checked exhaustively; artifact, `dedekind_shell.py`.)*

Theorem 24.5 (Shell contraction). *The identity class of $W_{\wedge\vee}$ is type-homotopic to the strict endomorphism $h = (u \wedge v, u \vee v)$, which is strictly null-homotopic. Hence $W_{\wedge\vee}$ is type-contractible.*

Proof. Let Y be fibrant and $u: W_{\wedge\vee} \rightarrow Y$ any map. The shell box $(T \hookrightarrow \square^3) \otimes \delta^0$ is a pushout product of a cofibration with an endpoint inclusion, i.e. a generating trivial cofibration, so the box datum $u \circ \Phi$ extends to a filler $L: \square^3 \times \square^1 \rightarrow Y$. Let $K := L \circ (\text{id}_{[3]}, 1)$ be the graph cell at the far end. All four track cells $(a_i, 1), (b_i, 1)$ lie in the shell $T \times \square^1$, where $L = u \circ \Phi$ is prescribed, so

$$\begin{aligned} K \cdot (A_2 \times \text{id}) &= u(p, p), & K \cdot (B_2 \times \text{id}) &= u(p, 1), \\ K \cdot (A_1 \times \text{id}) &= K \cdot (B_1 \times \text{id}) &= u(0, p), \end{aligned}$$

and $(p, p) \sim (p, 1)$ is a generator of the diagonal fold: K satisfies both fold conditions, hence is a cylinder $W_{\wedge\vee} \times \square^1 \rightarrow Y$ by Lemma 22.2. Its ends are the 1-slices of the two end prisms: n_0 gives (p, q) — the generic cell, i.e. $u \circ \text{id}$ — and n_1 gives $u \circ h$. Finally, the single strict cylinder

$$H = (w \vee (u \wedge v), u \vee v \vee w) \in W_{\wedge\vee}([3])$$

satisfies both fold conditions and has ends h and the constant v_{11} . So $u \circ \text{id} \simeq u \circ h \simeq \text{const}$ in every fibrant Y ; taking $Y = R(W_{\wedge\vee})$ and u the unit,

Lemma 22.3 shows $W_{\wedge\vee}$ is type-contractible. (Every finite ingredient — the gluing of Φ , the fold conditions, the two ends, the strict null-homotopy of h , and, for contrast, the strictly isolated identity component — is verified in `dedekind_shell.py`.) $\square$

Remark 24.6 (Why cellular calculi cannot see it). Every filler value over a representable open box has a free track — Theorem 23.2 exploits exactly this through the diagonal-track lemma — while a shell prescribes all four tracks, and representable-box calculi contain no shell shapes. The two theorems together exhibit a sharp phenomenon: on the Dedekind site, *uniform fibrancy is strictly more connective than cellular box-filling*. Note also that the identity’s *strict* homotopy component is the singleton $\{\iota_{\wedge\vee}\}$ (artifact), so the bridge from $\iota_{\wedge\vee}$ to h genuinely requires the fresh shell interior. Every future finite separation candidate on this site must be tested against shell fillers; for a graph cell over a shell the fold conditions reduce to finitely many congruence identities of the candidate’s boundary data, so the test is a finite search.

24.3. Shell moves, the census, and the new candidate. The contraction of Theorem 24.5 is an instance of a general, replacement-free *shell move*. For a cell x of $\square^n$ at level j write $\text{cone}_{\wedge}(x) := (x_1 \wedge r, \dots, x_n \wedge r)$ — the level- $(j+1)$ connection cone contracting x to the bottom vertex, r the new last variable — and dually $\text{cone}_{\vee}$.

Theorem 24.7 (Shell moves). *Let $Q = \square^n / (A_i \sim B_i)_{i \in I}$ be a finite congruence quotient on the Dedekind site and let g_0, g_1 be endomorphism cells of Q admitting raw representatives $\hat{g}_0, \hat{g}_1$ with*

$$[\text{cone}_{\wedge}(\hat{g}_0 \circ T)] = [\text{cone}_{\wedge}(\hat{g}_1 \circ T)] \quad \text{in } Q$$

for every track $T \in \{A_i, B_i : i \in I\}$. Then $u \circ g_0 \simeq u \circ g_1$ for every map $u: Q \rightarrow Y$ with Y fibrant; dually with $\text{cone}_{\vee}$. Consequently the type homotopy relation on endomorphism classes contains the closure of strict homotopy under these shell moves.

Proof. Form the shell $T_{\text{sh}} \subseteq \square^{n+1}$ (w the cylinder coordinate) as the union of the track squares $T \times \text{id}_w$ and the two end cubes $n_{\varepsilon'} = (\text{id}, \varepsilon')$, and define Φ on $T_{\text{sh}} \times \square_r^1 \cup \square^{n+1} \times \{0\}$ into Q by: the constant $[\vec{0}]$ on the 0-end; the end prisms $[\text{cone}_{\wedge}(\hat{g}_{\varepsilon'})]$; and the track prisms $[\text{cone}_{\wedge}(\hat{g}_0 \circ T)] \circ \text{pr}_w$. The pieces glue: among themselves and with the 0-end by functoriality of the cone in x , and with the end prisms because the T -edge of $\text{cone}_{\wedge}(\hat{g}_{\varepsilon'})$ is $\text{cone}_{\wedge}(\hat{g}_{\varepsilon'} \circ T)$, equal in Q to the track prism by hypothesis. For fibrant Y the box $u \circ \Phi$ extends to a filler L , and the graph cell $K = L \circ (\text{id}, 1)$ has all cylinder tracks $(A_i \times \text{id}, 1)$, $(B_i \times \text{id}, 1)$ inside the shell, with values $u[\hat{g}_0 \circ A_i] = u[\hat{g}_0 \circ B_i]$ (equal because g_0 is an endomorphism cell): K is a cylinder $Q \times \square^1 \rightarrow Y$ with ends $u \circ g_0$ and $u \circ g_1$. $\square$

Theorem 24.5 is the instance $g_0 = \text{id}$, $g_1 = (u \wedge v, u \vee v)$ on $W_{\wedge\vee}$: the two share their A_2 -, B_2 -, A_1 -tracks on the nose, and at the B_1 -track the cone

classes agree by the swap fold, $(p \wedge r, 0) \sim (0, p \wedge r)$. Shell moves are not the end of the calculus: shells also *compose*, through their fresh ends.

Theorem 24.8 (Tent moves). *Work in the algebraic replacement $R(Q)$, for any finite congruence quotient Q of a cube, and consider two shell boxes over a common shape (a common track set and end-square configuration, one orientation ε) whose far-end data agree in Q : equal $w=1$ -slices of the bases and of the corresponding track prisms. Then the two graph cylinders have literally the same fresh end — by the completeness of the based normal form (Theorem 24.2) a fresh cell is its based box — and therefore compose: the two old ends are type-homotopic.*

Proof. The fresh end of each graph cylinder at $w = 1$ is the filler value of the based pair $(B \circ (n_1 \hat{\times} \text{id}), \text{const}_{1-\varepsilon})$, whose box consists of the $w=1$ -slice of the base on $\square^2 \times \{\varepsilon\}$ and the $w=1$ -slices of the track prisms on the side; equal data give equal based pairs, hence equal cells (Theorem 24.2). The two cylinders then form a chain from one old end to the other through the common fresh apex. $\square$

We have run the resulting calculus — strict homotopies, cone shell moves, and tents — as a census over all 1,533 one- and two-relation quotients of the square (artifact, `dedekind_shellreach.py`, `dedekind_freshchain.py`). Call a quotient $\mathbb{F}_2$ -acyclic if its reduced $\mathbb{F}_2$ -homology vanishes — the finite criterion the census computes ($\mathbb{F}_2$ -simplicial chains of the triangulated realization through level 3, exact for these two-dimensional complexes; artifact, `dedekind_triangulate.py`). Test-contractibility implies $\mathbb{F}_2$ -acyclicity; the census proves the converse on this landscape — every $\mathbb{F}_2$ -acyclic quotient is shown *type*-contractible below, and type-contractible implies test-contractible by (U) — so the $\mathbb{F}_2$ -acyclic, test-contractible, and type-contractible quotients coincide here. Of the 781 $\mathbb{F}_2$ -acyclic quotients, 753 are strictly contractible and 22 more are contracted by cone shell moves ($W_{\wedge\vee}$ among them); exactly six resist both, and they generate, up to the $u \leftrightarrow v$ mirror, a single congruence:

Definition 24.9 (The monotone triple fold). $Q^* := \square^2 / ((0, u) \sim (u, u) \sim (u, 1))$: the left edge, the diagonal, and the top edge identified.

Theorem 24.10 (Q^* contracts by a tent; the move hierarchy is strict). Q^* is simply connected, $\mathbb{Z}$ -acyclic, and test-contractible. Its identity class is contracted by no strict homotopy, by no cone shell move, and by no single shell over any base — the exhaustive first-stage search over all shells with arbitrary old data (`dedekind_fullshell.py`) finds seven full-shell edges among its six strict endomorphism classes, none reaching a constant — but it is contracted by a single tent: the two shells with bases

$$G_1 = ((u \wedge v) \vee (v \wedge w), v), \quad G_2 = (0, v \wedge w),$$

track prisms (coordinates (z, w, r))

$$\begin{aligned} P_1[\ell] &= (z \wedge w, z \vee (w \wedge r)), & P_1[\delta] &= (z \wedge w \wedge r, z \vee (w \wedge r)), \\ P_1[\tau] &= (z \wedge w \wedge r, z \vee w), & P_2[\ell] &= (z \wedge w, (z \wedge w) \vee (w \wedge r)), \\ P_2[\delta] &= (z \wedge w \wedge r, (z \wedge w) \vee (w \wedge r)), & P_2[\tau] &= (z \wedge w \wedge r, w), \end{aligned}$$

and end prisms $C^1 = ((u \wedge r) \vee (u \wedge v), v)$, $C^2 = (0, 0)$, whose far-end data agree (the base slices (v, v) and $(0, v)$ are identified by the left-diagonal fold; the track-prism slices agree on the nose) and whose old ends are the identity and the constant v_{00} . All thirty-two finite compatibilities are machine-verified (artifact, `dedekind-qstar.py`).

Theorem 24.11 (The census). *Every $\mathbb{F}_2$ -acyclic one- or two-relation quotient of the square on the Dedekind site is type-contractible: 753 strictly, 22 more by cone shell moves, and the six copies of Q^* by tents. No quotient with nontrivial $\mathbb{F}_2$ -homology is contracted by any move, consistently with $W_{\text{type}} \subseteq W_{\text{test}}$.*

Remark 24.12 (The expected direction of the comparison reverses). Where the precursor of this Part expected a separation, the moves of the uniform replacement have now contracted every candidate in the small-quotient landscape — including the two dunce-hat-like congruences $W_{\wedge\vee}$ and Q^* that block every cellular contraction and, in the case of Q^* , every single shell. Beyond contractibility, the closure identifies *all* strict endomorphism classes of both quotients in a single component (seven for $W_{\wedge\vee}$, six for Q^*), exactly as the test theory predicts for contractible objects. We therefore now expect the opposite resolution and pose it as the question: *do Dedekind cubical sets present spaces — is $W_{\text{type}} = W_{\text{test}}$ on the Dedekind site?* The shell-tent calculus suggests the mechanism: the moves implement connection cones over test-contraction data, composed through fresh apexes. The precise reduction is carried out in Section 24.4: the comparison holds iff the type localizer is regular (Theorem 24.19), and the question becomes shape-by-shape — with the tracts settled and the frontier localized in Section 24.5.

24.4. Toward the positive resolution. We now make the reversed expectation precise. This subsection gives two criteria — a model-categorical and an ∞ -categorical one — then the identification with the minimal Cisinski structure, and finally the regularity reduction; the next subsection settles large tracts of the resulting shape-by-shape question.

Proposition 24.13 (Fibrant-object criterion). *On any site of the paper, $W_{\text{type}} = W_{\text{test}}$ if and only if every type-fibrant object is test-fibrant.*

Proof. $(\Rightarrow)$ is trivial (equal structures). $(\Leftarrow)$: test-fibrant objects are always type-fibrant ($W_{\text{type}} \subseteq W_{\text{test}}$ gives $\text{TC}_{\text{type}} \subseteq \text{TC}_{\text{test}}$, hence the reverse inclusion of fibration classes), so the hypothesis makes the two fibrant classes coincide. Now $X \times \square^1$ is a *common* cylinder object for both structures: its end inclusions are monomorphisms, and the projection to X is inverted by both localizers: the end inclusion $i_0 = (\emptyset \hookrightarrow X) \hat{\otimes} \delta^0$ is a generating

trivial cofibration of the type structure, hence lies in $W_{\text{type}} \subseteq W_{\text{test}}$, and two-out-of-three applied to $p \circ i_0 = \text{id}$ inverts p . For cofibrant X (all objects are) and Z fibrant in either structure, the homotopy relation on $\text{Hom}(X, Z)$ computed through this one cylinder is therefore the same in both structures, and $[X, Z]$ is unambiguous. A map $f: X \rightarrow Y$ is a weak equivalence in a model structure iff $[Y, Z] \rightarrow [X, Z]$ is bijective for every fibrant Z ; with the fibrant classes and the homotopy relations equal, the two weak-equivalence classes coincide. $\square$

So the question “ $W_{\text{type}} = W_{\text{test}}$?” on the Dedekind site is exactly the lifting question: *does every uniform Kan complex lift against every test-trivial cofibration?* The shell–tent calculus is the beginning of precisely such a lifting theory: a shell is a lift constructed from prescribed old data, a tent composes two of them, and Theorem 24.11 is the statement that these lifts already suffice for every contraction problem in the small-quotient landscape.

Proposition 24.14 (Mapping-space criterion). *Let $\mathcal{C}$ denote the underlying ∞ -category of the cubical-type structure on Dedekind cubical sets and $F: \mathcal{S} \rightarrow \mathcal{C}$ the unique colimit-preserving functor with $F(*) = \square^0$. Then the type structure presents spaces if and only if*

- (1) *F is fully faithful: $\text{Map}_{\mathcal{C}}(\square^0, F(T)) \simeq T$ naturally in $T \in \mathcal{S}$; and*
- (2) *F is essentially surjective: every presheaf is, in $\mathcal{C}$, a colimit of a diagram of points.*

Proof. If the type structure presents spaces then, composing an equivalence $e: \mathcal{C} \simeq \mathcal{S}$ with F and using $\text{Fun}^L(\mathcal{S}, \mathcal{S}) \simeq \mathcal{S}$ [15, Thm. 5.1.5.6] together with $eF(*) \simeq *$ (the presheaf $\square^0$ is terminal, and equivalences preserve terminal objects), $e \circ F \simeq \text{id}$, so F is an equivalence. The converse is immediate. $\square$

The census verifies condition (1) at its lowest level: beyond contractibility, the moves identify *all* strict endomorphism classes of $W_{\wedge \vee}$ and Q^* in a single component, exactly as the mapping-space prediction requires.

The lifting question has a classical face. Recall Cisinski’s theory [10]: any interval on a presheaf category generates a smallest *class of anodyne extensions* [10, 1.3.13]. Explicitly, fix a *cellular model* M — a set of monomorphisms whose closure under pushout, transfinite composition, and retract is the class of all monomorphisms; one exists with all sources and targets subobjects of quotients of representables [10, 1.2.26, 1.2.27] — and set [10, 1.3.12]

$$\begin{aligned} \Lambda^0 &= \{ \delta^\varepsilon \hat{\otimes} m : m \in M, \varepsilon \in \{0, 1\} \}, \\ \Lambda^{n+1} &= \{ (\partial \square^1 \hookrightarrow \square^1) \hat{\otimes} f : f \in \Lambda^n \}, \quad \Lambda = \bigcup_{n \geq 0} \Lambda^n. \end{aligned}$$

The smallest anodyne class is the saturation of Λ under pushout, transfinite composition, and retract, independently of the choice of M [10, 1.3.13], and it generates a model structure with cofibrations the monomorphisms — the *minimal Cisinski structure* of the interval — whose fibrant objects are exactly the objects with the right lifting property against Λ .

Lemma 24.15 (Fibrant objects of the minimal interval structure). *On any site of the paper, the fibrant objects of the minimal Cisinski structure of the interval $\square^1$ are exactly the uniform Kan complexes.*

Proof. Fibrancy on both sides is a right lifting property against a set of monomorphisms: type-fibrancy against the open boxes $m \hat{\otimes} \delta^\varepsilon$ with m a monomorphism [14, 20], and minimal-Cisinski fibrancy against the completion Λ recalled above. We compare the two lifting problems in both directions, through the explicit form of Λ .

Every element of Λ is an open box. By induction on the completion stage. An element of Λ^0 is $\delta^\varepsilon \hat{\otimes} m$ with $m \in M$, isomorphic to $m \hat{\otimes} \delta^\varepsilon$: the product of presheaves is symmetric, and the Leibniz construction respects the symmetry. An element of Λ^{n+1} is $(\partial\square^1 \hookrightarrow \square^1) \hat{\otimes} f$ with $f \in \Lambda^n$; inductively $f \cong m' \hat{\otimes} \delta^\varepsilon$ with m' a monomorphism, so associativity and symmetry of the Leibniz product give

$$(\partial\square^1 \hookrightarrow \square^1) \hat{\otimes} f \cong ((\partial\square^1 \hookrightarrow \square^1) \hat{\otimes} m') \hat{\otimes} \delta^\varepsilon.$$

The Leibniz product of two monomorphisms of presheaves is a monomorphism — levelwise it is the inclusion of the union of two subobjects of the product into the product — so the left factor is again monic, and every element of Λ is an open box $m \hat{\otimes} \delta^\varepsilon$ with m a monomorphism. A type-fibrant object therefore lifts against Λ , and is Cisinski-fibrant.

Every open box is a Λ^0 -cell complex. Conversely, let m be an arbitrary monomorphism. The skeletal induction writes m as a transfinite composition of pushouts of elements of the canonical cellular model: attach the nondegenerate cells of the target absent from the source level by level, each attachment a pushout of $S \hookrightarrow Q$ with Q the quotient of a representable recording the cell's degeneracies and identifications and $S \subseteq Q$ the part already attached [10, 1.2.26, 1.2.27]. The functor $(-) \hat{\otimes} \delta^\varepsilon$ preserves pushouts and transfinite compositions (the Leibniz product is cocontinuous in each variable), so $m \hat{\otimes} \delta^\varepsilon$ is a transfinite composition of pushouts of the maps $(S \hookrightarrow Q) \hat{\otimes} \delta^\varepsilon$ — elements of Λ^0 for M the canonical cellular model, a choice we are free to make since the anodyne completion does not depend on it [10, 1.3.13]. Hence $m \hat{\otimes} \delta^\varepsilon$ lies in the saturation of Λ^0 ; a Cisinski-fibrant object lifts against the saturation of Λ , hence against every open box, and is type-fibrant. $\square$

Theorem 24.16 (The type structure is the minimal interval structure). *Classically, on every site of the paper the cubical-type model structure coincides with the minimal Cisinski structure of the interval $\square^1$.*

Proof. The two structures have the same cofibrations and, by Lemma 24.15, the same fibrant objects; the end inclusion i_0 is a trivial cofibration of both (Λ contains $(\emptyset \hookrightarrow X) \hat{\otimes} \delta^0$ in its saturation), so $X \times \square^1$ is a common cylinder and the determination argument of Proposition 24.13 applies verbatim. $\square$

Corollary 24.17 (The phase diagram as a statement of Cisinski's theory). *On the standard cube sites, “the cubical-type structure presents spaces” is*

equivalent to “the minimal Cisinski structure of the interval is the test structure.” Thus: on the cartesian site the minimal interval structure is not the test structure ([1]); on the one-connection site it is ([8]); on the De Morgan, Kleene, and Boolean sites it is not, at every dimension (Theorem 17.2); and the Dedekind question becomes: is the minimal interval structure on Dedekind cubical sets a model for spaces?

Remark 24.18. The reformulation places the program squarely in the classical literature: for the box-only cube categories with connections the analogous statement is Cisinski’s, and the one-connection cartesian case is the relative-elegance theorem of [8] — a method shown there to be unavailable on the Dedekind site. The shell–tent calculus of this Part is the candidate replacement: shells and tents are computations *inside* the anodyne completion Λ , made exact by the provenance calculus.

The reformulation can be pushed one step further, to a single property of the type localizer, using Cisinski’s regularity theory [10]. Recall: an A -localizer W is *regular* if every presheaf is canonically the W -homotopy colimit of the diagram of its representable cells [10, 3.4.13]; a *test A -localizer* is a regular A -localizer in which every representable is aspherical [10, 4.2.11]; and every presheaf category has a *cellular model* — a set of monomorphisms generating all monomorphisms under pushout, transfinite composition, and retract — whose targets are quotients of representables [10, 1.2.26, 1.2.27].

Theorem 24.19 (The regularity reduction). *On each of the Dedekind, De Morgan, Kleene, and Boolean sites, the following are equivalent:*

- (1) $W_{\text{type}} = W_{\text{test}}$;
- (2) *the type localizer is regular*;
- (3) *every source and target of a cellular model — every subobject of every quotient of a representable — is W_{type} -regular, i.e. is the type-homotopy colimit of its own cells.*

Proof. First, W_{type} is an A -localizer: two-out-of-three holds, the trivial fibrations (defined by lifting against the common cofibrations) lie in W_{type} , and W_{type} -trivial cofibrations are closed under pushout and transfinite composition, all because W_{type} carries a model structure with cofibrations the monomorphisms. Second, representables are type-aspherical: the connection cone $(\vec{x}, t) \mapsto \vec{x} \wedge t$ exhibits a $\square^1$ -homotopy equivalence $\square^n \rightarrow \square^0$, and cylinder homotopy equivalences are weak equivalences.

(2) $\Rightarrow$ (1): if W_{type} is regular then, by the above, it is a test A -localizer, so it contains the minimal test A -localizer. The latter is generated, as an A -localizer, by the minimal *regular* localizer together with the asphericity maps of the representables [10, proof of 4.3.19], valid for every small category; the minimal regular localizer is contained in every regular localizer, in particular in W_{type} , and the asphericity maps lie in W_{type} as just shown. Finally the minimal test A -localizer is W_{test} : the correspondence $W \mapsto W_{\text{Cat}}$ of [10, 4.2.15] is an inclusion-preserving bijection between test A -localizers and

fundamental localizers modelable by A , the minimal fundamental localizer is the class of nerve weak equivalences [10, 4.2.19] and is modelable by our sites (they are test categories, Section 2), and an increasing bijection matches least elements — so the minimal test A -localizer corresponds to the minimal fundamental localizer, i.e. equals the class of test equivalences. Hence $W_{\text{test}} \subseteq W_{\text{type}}$, and (U) gives equality.

(1) $\Rightarrow$ (2): W_{test} is regular (it is a test A -localizer by the correspondence just cited).

(2) $\Leftrightarrow$ (3): regular presheaves form a class saturated by monomorphisms [10, 3.4.20–3.4.22], and a localizer is regular iff the sources and targets of some cellular model are regular [10, 3.4.23] — a statement valid for every A -localizer, with no accessibility hypothesis; representables themselves are always regular [10, 3.4.10]. $\square$

Corollary 24.20. *On the De Morgan, Kleene, and Boolean sites the type localizer is not regular: some quotient of a representable is not the type-homotopy colimit of its cells. Indeed the separating quotients of Theorem 17.2 are quotients of representables, and their failure of regularity is exactly what the separations exploit.*

Remark 24.21 (The final form of the program). On the Dedekind site the question is now shape-by-shape: *show that every subobject of every quotient of a representable is the type-homotopy colimit of its cells.* Three observations organize the attack. (i) For a *type-contractible* shape U the statement reduces to the type-contractibility of the homotopy colimit of its cell diagram — a map between type-contractible objects is a weak equivalence by two-out-of-three — so Theorem 24.11 settles the map side for the two-relation quotients of the square once the corresponding cell-colimits are contracted; these colimits are realizations of categories of elements, i.e. objects of the same codiscrete, nerve-like kind as the instances $E(T)$ of Part 3. (ii) The shapes' symmetries on a reversal-free site are all of reflection type (they fix cells), the regime that Part 1 proves harmless — the mixed symmetries that break regularity on the reversal sites do not exist here. (iii) The provenance calculus makes each individual shape a finite computation.

24.5. Mono-cellular shapes: the first regular atoms. The reduction bites immediately, because regularity has strong closure properties valid for *every* localizer.

Proposition 24.22 (Mono-cellular presheaves are regular). *Call a presheaf mono-cellular if it lies in the smallest class containing the representables and closed under small coproducts, pushouts along a monomorphism (that is: if one leg of a span is a monomorphism, the parallel leg is arbitrary, and all three objects of the span lie in the class, then so does the pushout), transfinite compositions of monomorphisms, and retracts. On any site, every mono-cellular presheaf is W -regular for every A -localizer W .*

Proof. Representables are regular [10, 3.4.10]. Diagrams of the three listed colimit shapes — pushouts whose one leg is a monomorphism, well-ordered sequences of monomorphisms, and discrete diagrams — are W -perfect for every W [10, 3.4.9, 3.4.19], and regular presheaves are closed under perfect colimits and under retracts [10, 3.4.20, 3.4.21]. $\square$

Remark 24.23 (The classical sites in one stroke). If a site admits a cellular model whose targets are *representable* (the simplex category; the plain cube category; the Brown–Higgins cube categories with connections — the Eilenberg–Zilber regime), then every presheaf is mono-cellular by skeletal induction, so *every* localizer is regular, and the argument of Theorem 24.19 shows that the minimal Cisinski structure of any aspherical interval coincides with the test structure. This recovers the classical comparisons at one stroke and locates the entire difficulty of the Lawvere-style sites in their non-representable cell shapes [10, 1.2.29], which on our sites are forced by the symmetric and diagonal operations (cf. the proof of Lemma A.1).

On the Dedekind site the closure nevertheless reaches far beyond the representables.

Lemma 24.24 (Graphs are mono-cellular). *Every presheaf generated by cells of level ≤ 1 , and every quotient of such a presheaf, is mono-cellular.*

Proof. Such a presheaf is built from a coproduct of vertices $\square^0$ by attaching, for each generating edge, a copy of $\square^1$ along the monomorphism $\partial\square^1 \hookrightarrow \square^1$ — a pushout whose other leg (the attaching map of the two endpoints) is arbitrary; degenerate or identified configurations (loops, collapsed edges) are covered precisely because the second leg need not be monic. A quotient of a presheaf generated in levels ≤ 1 is again generated in levels ≤ 1 . $\square$

Theorem 24.25 (The first regular atoms). *On the Dedekind site, every quotient of a representable by identifications of cells of level ≤ 1 is mono-cellular, hence W_{type} -regular. In particular $W_{\wedge\vee}$, the triple fold Q^* , and all 1,533 quotients of the census are W_{type} -regular.*

Proof. We prove mono-cellularity by induction on the number of identifications; regularity then follows by Proposition 24.22. The base, a representable, is mono-cellular. Let X be mono-cellular and let a further identification $A \sim B$ of cells of level ≤ 1 be given. Write $U \subseteq X$ for the subobject generated by the (images of the) two cells and $U' := U/(A \sim B)$. Then $X/(A \sim B) = X \cup_U U'$, the pushout of the span $X \leftarrow U \rightarrow U'$ along the monomorphism $U \hookrightarrow X$; U is generated by cells of level ≤ 1 and U' is a quotient of such, so both are mono-cellular (Lemma 24.24), all three objects of the span are mono-cellular, and so is the pushout. For $W_{\wedge\vee}$ the two steps are exactly the homotopy-pushout presentation of Proposition 25.2 (wedges of edges, folded); for Q^* a single step suffices: with $U_3 = \ell \cup \delta \cup \tau \hookrightarrow \square^2$ the union of the three identified edges and $S = \square^1/(v_0 \sim v_1)$ the circle,

$$Q^* = \square^2 \cup_{U_3} S,$$

the fold $U_3 \rightarrow S$ carrying all three edges to the loop (machine check that the two congruences coincide: artifact, `dedekind_monocell.py`). $\square$

Remark 24.26 (The frontier: cyclic atoms as lattice spectra). Every presheaf is assembled from the *cyclic* subobjects $\langle z \rangle \subseteq \square^m$ (images of single cells) by unions along monomorphisms and quotients, so by Proposition 24.22 and the mechanism of Theorem 24.25 the regularity program reduces to the cyclic atoms of level ≥ 2 . These have a lattice-theoretic description: the cells of $\langle z \rangle$ embed into the lattice maps out of the distributive sublattice generated by the components of z — bijectively when the embedding is retractive, and as a *proper* subobject otherwise (Remark 24.29) — so the atoms are subobjects of spectra of finitely generated bounded distributive lattices, Birkhoff-dual to finite posets. In particular an atom is *not* determined by its marked generated lattice: the retractive five-chain spectrum properly contains square subatoms with the same marked lattice (artifact, `referee_measure2.py`). The first atom beyond the graph machinery is the *order triangle* $O = \langle (u \wedge v, u \vee v) \rangle \subseteq \square^2$, the subobject of ordered pairs $\alpha \leq \beta$ (the spectrum of the four-element chain lattice; it has three vertices, so it is not representable). Whether every such spectrum is monocular, or at least W_{type} -regular, is the question the rest of this subsection addresses — O and all two-dimensional atoms are settled affirmatively just below, and the final frontier is isolated in Remarks 24.37 and 24.43. On the reversal sites the answer is negative (Corollary 24.20), the obstruction being a mixed symmetry — which the reversal-free Dedekind atoms do not possess.

Proposition 24.27 (Retractive atoms are regular). *Let $z: [k] \rightarrow [m]$ be a cell with component sublattice $A = \langle z_1, \dots, z_m \rangle \subseteq D(k)$, and suppose A is a retract of $D(k)$: there are $c_1, \dots, c_k \in A$ with $z_j(c_1, \dots, c_k) = z_j$ for all j . Then the cyclic atom $\langle z \rangle$ is a retract of the representable $\square^k$, hence W -regular for every localizer W .*

Proof. Write $\langle z \rangle = \square^k / \ker(z)$ with quotient map p , and let $\sigma_c: [k] \rightarrow [k]$ be the substitution with components c_i . Define $s[w] := \sigma_c \circ w$. This is well defined: each c_i lies in A , i.e. is a lattice term $c_i = s_i(z_1, \dots, z_m)$ in the components of z , so the components of $\sigma_c \circ w$ are $s_i((z \circ w)_1, \dots, (z \circ w)_m)$ and depend only on $z \circ w$. Moreover $z \circ \sigma_c = z$, since its components are $z_j(c_1, \dots, c_k) = z_j$; hence $z \circ \sigma_c \circ w = z \circ w$ and $p \circ s = \text{id}$. $\square$

Corollary 24.28 (The order triangle). *The order triangle $O = \langle (u \wedge v, u \vee v) \rangle$ is a retract of $\square^2$ — the retraction is the median substitution $\sigma = (u \wedge v, u \vee v)$ itself, since $(u \wedge v) \wedge (u \vee v) = u \wedge v$ and $(u \wedge v) \vee (u \vee v) = u \vee v$ — and is therefore W -regular for every localizer.*

Remark 24.29 (The lattice-theoretic frontier). Whether a given atom is retractive is a finite condition on the embedded sublattice $A \subseteq D(k)$, decided by searching $c \in A^k$. An exhaustive sweep (artifact, `dedekind_projtest.py`) finds: every *two-generated* sublattice of $D(3)$

(190 cases, complete) and of $D(4)$ (4,000 sampled cases) is a retract — a fact promoted to a theorem for all k just below (Theorem 24.31) — while 222 of the 1,140 three-generated sublattices of $D(3)$ are *not* retracts along their given embeddings. The smallest failures are five-chains such as $\{0, u \wedge v \wedge w, u \wedge v, u \wedge (v \vee w), 1\}$: in a chain the three displayed generators force $c_w \leq c_v \leq c_u$ and the system is inconsistent. The failure is a property of the *embedding*, not of the lattice: the same five-chain embedded as $\{0, u \wedge v \wedge w, u \wedge v, u, 1\}$ is a retract (take $c = (u, u \wedge v, u \wedge v \wedge w)$), so the abstract lattice is projective. For a non-retractive embedding the atom is a *proper* subobject of the abstract spectrum — not every lattice map $A \rightarrow D(j)$ extends along the embedding (e.g. the five-chain point $0 \leq u \leq 1$ of $D(1)$ does not). Non-retractivity of the given embedding is not, however, the end of the story: the five-chain atom itself turns out to be isomorphic to $\square^2$ along a different coordinatization (Proposition 24.32 below), and the sharp invariant is not the embedded sublattice but the *fiber poset* of the vertex map (Theorem 24.34).

The two-generated case of the sweep is not an accident: it is a theorem, by a splitting lemma for monotone maps onto subposets of the square.

Lemma 24.30 (Monotone sections over the square). *Let $\varphi: \{0,1\}^k \rightarrow \{0,1\}^2$ be a monotone map with image P . Then φ admits a monotone section over its image: a monotone $s: P \rightarrow \{0,1\}^k$ with $\varphi \circ s = \text{id}_P$. The statement is sharp: it fails for images inside $\{0,1\}^3$ containing a four-element chain.*

Proof. All cases use two remarks: for x, y in the cube, $\varphi(x \wedge y)$ lies in P and below both $\varphi(x), \varphi(y)$, and dually for joins. If $|P| = 1$ any preimage is a section. If $P = \{p, q\}$ is an antichain, any preimages form a section. If $P = \{p < q\}$, pick $x \in \varphi^{-1}(p)$, $y \in \varphi^{-1}(q)$ and set $s(p) = x \wedge y$, $s(q) = y$: $\varphi(x \wedge y)$ lies in P below p , hence equals p . A three-element P is a chain $c_0 < c_1 < c_2$, a wedge $\{c_0 < p, c_0 < q\}$, or a co-wedge: for the chain pick $x_1 \in \varphi^{-1}(c_1)$ and y_0, y_2 over the ends and set $s = (x_1 \wedge y_0, x_1, x_1 \vee y_2)$ — the middle is fixed first, and the corrections stay in place because c_0, c_2 are extreme in P ; for the wedge set $s(p) = y_p$, $s(q) = y_q$, $s(c_0) = y_p \wedge y_q$ (the only element of P below both p and q is c_0), and dually for the co-wedge. If $|P| = 4$ then P is the whole square: pick y_{10}, y_{01} over $(1, 0), (0, 1)$ and set $s(0, 0) = y_{10} \wedge y_{01}$, $s(1, 0) = y_{10}$, $s(0, 1) = y_{01}$, $s(1, 1) = y_{10} \vee y_{01}$. In every case s is monotone and a section. What makes the square special is that its chains have at most one interior element; with two interior elements $c_1 < c_2$ the correction $x_1 \vee x_2$ may overshoot into the top fiber, and the sweep's five-chain examples realize exactly this failure. $\square$

Theorem 24.31 (All two-dimensional atoms are regular). *For every k and every cell $z: [k] \rightarrow [2]$ on the Dedekind site, the cyclic atom $\langle z \rangle$ is a retract of $\square^k$, hence W -regular for every localizer W .*

Proof. Write $z = (a, b)$ with $a, b \in D(k)$ and $A = \langle a, b \rangle$; by Proposition 24.27 it suffices to produce $c_1, \dots, c_k \in A$ with $a(\vec{c}) = a$ and $b(\vec{c}) = b$. Elements of A are exactly the lattice polynomials $g(a, b)$ with $g \in D(2)$, and $D(2)$ consists of *all* monotone maps $\{0, 1\}^2 \rightarrow \{0, 1\}$. Let $\varphi = (a, b): \{0, 1\}^k \rightarrow \{0, 1\}^2$, with image P , and let $s: P \rightarrow \{0, 1\}^k$ be a monotone section (Lemma 24.30), extended to a monotone $S: \{0, 1\}^2 \rightarrow \{0, 1\}^k$ by $S(y) := \bigvee_{p \in P, p \leq y} s(p)$. Set $c_i := S_i(a, b) \in A$. Then for every point $\vec{x}$ of the cube,

$$a(\vec{c})(\vec{x}) = a(S(\varphi(\vec{x}))) = (\varphi s \varphi(\vec{x}))_1 = \varphi(\vec{x})_1 = a(\vec{x}),$$

and likewise for b . $\square$

The expected first shape beyond these theorems was the *five-chain atom* $T = \langle (u \wedge v \wedge w, u \wedge v, u \wedge (v \vee w)) \rangle \subseteq \square^3$, whose component sublattice is a five-element chain embedded non-retractively. It is nothing of the sort.

Proposition 24.32 (The five-chain atom is a square). *On the Dedekind site the assignment $(p, q) \mapsto (p \wedge q, p, p \vee q)$ defines an isomorphism $\square^2 \xrightarrow{\cong} T$. In particular T is W -regular for every localizer W .*

Proof. Every cell of T has the displayed form: with $p := \sigma_1 \wedge \sigma_2$ and $q := \sigma_1 \wedge \sigma_3$, distributivity gives

$$z(\vec{\sigma}) = (\sigma_1 \sigma_2 \sigma_3, \sigma_1 \sigma_2, \sigma_1 \wedge (\sigma_2 \vee \sigma_3)) = (p \wedge q, p, p \vee q),$$

and conversely every pair (p, q) is realized, by $\vec{\sigma} = (p \vee q, p, q)$. The assignment commutes with substitution, and it is injective by the cancellation law of distributive lattices: if $p \wedge q = p \wedge q'$ and $p \vee q = p \vee q'$ then

$$q = q \wedge (p \vee q) = q \wedge (p \vee q') = (q \wedge p) \vee (q \wedge q') = (q' \wedge p) \vee (q' \wedge q) = q' \wedge (p \vee q) = q'.$$

Machine check at levels ≤ 4 : the two cell sets coincide and the map is bijective, with $|T(k)| = |D(k)|^2 = 4, 9, 36, 400, 28,224$ (artifact, `dedekind.squaretrick.py`). $\square$

The mechanism behind the square — an unexpected extra coordinate — generalizes to a criterion. For a cell $z \in D(n)^j$ write $\hat{z}: \{0, 1\}^n \rightarrow \{0, 1\}^j$ for its vertex map, and order the set of fibers of $\hat{z}$ by the transitive closure of *realized comparability* ($F \leq F'$ if $a \leq b$ for some $a \in F, b \in F'$) — antisymmetric, hence a partial order, because $\hat{z}$ is monotone.

Lemma 24.33 (Canonical coordinates). *$\text{Hom}(\langle z \rangle, \square^1)$ is naturally isomorphic to the lattice of monotone $\{0, 1\}$ -valued functions on the fiber poset of $\hat{z}$.*

Proof. A map $\langle z \rangle \rightarrow \square^1$ is a $g \in D(n)$ with $z(\vec{\sigma}) = z(\vec{\tau}) \Rightarrow g(\vec{\sigma}) = g(\vec{\tau})$ at every level. The condition is a quasi-identity of bounded distributive lattices, and each $D(k)$ embeds into a power of the two-element chain by evaluation at the points of $\{0, 1\}^k$, where equality is computed pointwise; so the condition holds at every level iff it holds at $\{0, 1\}$ -points, i.e. iff g is constant on the fibers of $\hat{z}$. Monotonicity of the induced function on

fibers with respect to realized comparability is exactly monotonicity of g , and passage to the transitive closure is free. $\square$

Theorem 24.34 (The section criterion). *If the fiber poset of $\hat{z}$ admits a monotone transversal — a monotone map s from the fiber poset to $\{0, 1\}^n$ with $s(F) \in F$ for every fiber F — then $\langle z \rangle$ is isomorphic to the image of the idempotent substitution $e := s \circ \hat{z} \in D(n)^n$, hence to a retract of $\square^n$, and $\langle z \rangle$ is W -regular for every localizer W .*

Proof. The four assertions “ e is idempotent,” “ $z \circ e = z$,” and the fiber collapse “ $z(\vec{a}) = z(\vec{b}) \Rightarrow e(\vec{a}) = e(\vec{b})$ ” are identities or quasi-identities that hold at $\{0, 1\}$ -points by construction of e , hence at every level by the pointwise embedding of the proof of Lemma 24.33; monotonicity of each component of e (i.e. $e \in D(n)^n$) is monotonicity of s composed with $\hat{z}$. The substitution endomorphism $\vec{\sigma} \mapsto e(\vec{\sigma})$ of $\square^n$ is then idempotent with image $\langle e \rangle$, a retract of $\square^n$. Postcomposition with z carries $\langle e \rangle$ onto $\langle z \rangle$ (by $z \circ e = z$), and injectively: if $z(e\vec{\sigma}) = z(e\vec{\tau})$, the fiber collapse gives $e(e\vec{\sigma}) = e(e\vec{\tau})$, i.e. $e(\vec{\sigma}) = e(\vec{\tau})$. So $\langle z \rangle \cong \langle e \rangle$, and retracts of representables are regular for every localizer (Proposition 24.22). $\square$

Remark 24.35. The criterion must be applied to the *fiber poset*, not to the image poset of the given presentation. For T the image is a four-element chain, over which no monotone section exists; but two of the fibers are incomparable in the realized order — the fiber poset is a diamond, the canonical coordinates include the unexpected $u \wedge w$, and the transversal $(100, 101, 110, 111)$ exists, recovering Proposition 24.32 with $e = (1, u \wedge v, u \wedge w)$. Theorem 24.31 is the case $j = 2$, where Lemma 24.30 provides a section of the vertex map itself. Since the criterion depends only on the fiber partition, it decides the retract property along *some* embedding — the sharp invariant that the sublattice sweep of Remark 24.29 could not see.

Corollary 24.36 (Two-variable atoms). *Every cell kernel on $\{0, 1\}^2$ admits a monotone transversal: there are eleven, and the only incomparable merge $\{01, 10\}$ has only 00 below it and 11 above it, so any transversal value works (machine sweep: artifact, `dedekind_spectower.py`). Hence the atom of every two-variable cell $z \in D(2)^j$, for every j , is a retract of $\square^2$ and is W -regular for every localizer W — the transpose of Theorem 24.31, which treats $j = 2$ and arbitrary arity.*

Remark 24.37 (The frontier after the section criterion). Together with Theorem 24.25, Theorem 24.34 settles all graph quotients, all two-dimensional cyclic atoms, and every atom whose fiber poset admits a monotone transversal. The genuine frontier is the atoms with *transversal-less* fiber posets. These exist: of the 880 cell kernels on $\{0, 1\}^3$ (the fiber partitions of monotone maps, characterized by antisymmetry of realized comparability), exactly 236 — fifty up to coordinate symmetry — admit no monotone transversal (artifact, `dedekind_squaretrick.py`). The smallest is the *incomparable-pair merge* C : the spectral quotient identifying the two

vertices $a = (1, 0, 1)$ and $b = (0, 1, 1)$ of $\square^3$ and nothing else — a transversal value at the merged fiber would have to dominate both $(1, 0, 0)$ and $(0, 1, 0)$, and neither vertex does. For C the obstruction is absolute, not an artifact of coordinates: its fiber poset is *not a lattice* (the images of $(1, 1, 0)$ and of the merged pair have two maximal common lower bounds), while the vertex poset of any retract of a cube is a poset retract of $\{0, 1\}^m$, hence a lattice ($r(s(x) \vee s(y))$ is a join). So C is a retract of *no* cube, along any embedding. The remaining assembly mechanisms fail as well, provably: the graph quotient $G = \square^3/(a \sim b)$ is regular (Theorem 24.25) and test-equivalent to a circle, and the spectral surjection $G \twoheadrightarrow C$ fills the loop; but its fold locus contains generators from level 4 on (732 folded classes with sectioned representatives; artifact, `dedekind_cmerge.py`), so the fold-locus pushout of Remark 24.38 degenerates, and the spectral congruence is not generated in bounded level (its level- ≤ 2 pairs give 7,108 classes at level 3 against the true 6,117; artifact, `dedekind_spectower.py`), so no finite tower of cell identifications reaches C either. Every retract and assembly mechanism thus provably fails at C — and yet C is regular: the realization theory of Section 24.6 settles it, along with every other atom (Corollary 24.48).

Remark 24.38 (No assembly for atoms). A companion mechanism explains why mono-cellular assembly cannot reach the remaining atoms. For any surjection $q: X \twoheadrightarrow Y$ and any subobject $S \subseteq X$ containing every cell with a distinct q -partner, the square $Y = q(S) \cup_S X$ is a pushout along the monomorphism $S \hookrightarrow X$ (levelwise: the pushout identifies exactly the S -cells with equal image, and every nontrivially identified cell lies in S); so Y is regular whenever X , S , and $q(S)$ are — the *fold-locus pushout*, of which the proof of Theorem 24.25 is the case of graph-supported folds. For a cyclic atom, however, the fold locus is everything: generating cells acquire partners (for the five-chain presentation the stabilizer already has 27 elements at its own level, and in general a dummy variable creates partners for any expansion of the generator; artifact, `dedekind_foldlocus.py`), and any congruence-generating set must contain a pair of generating cells, since a rewriting chain out of the generator must begin with an instance of a generator. Atoms are irreducible for mono-cellular assembly; the section criterion and the routes of Remark 24.43 are the ways around.

Two structural facts frame the attack on the remaining atoms.

Lemma 24.39 (Cone contractibility of atoms). *If the components of z are lattice polynomials not involving the constant 1, then $z(\vec{\sigma} \wedge t) = z(\vec{\sigma}) \wedge t$, so the $\wedge$ -cone of the cube restricts to the atom: $\langle z \rangle$ is strictly contractible. Dually with 0 and $\vee$. In particular the five-chain atom T is strictly contractible.*

Proof. Induction on the polynomial: variables are clear, $(p \wedge q)(\vec{x} \wedge t) = p(\vec{x}) \wedge q(\vec{x}) \wedge t$, and $(p \vee q)(\vec{x} \wedge t) = (p(\vec{x}) \wedge t) \vee (q(\vec{x}) \wedge t) = (p \vee q)(\vec{x}) \wedge t$ by distributivity; the constant 0 satisfies $0 = 0 \wedge t$, while $1 \wedge t = t \neq 1$ is the one

failure. The cone $(\text{cell}, t) \mapsto \text{cell} \wedge t$ is then a strict cylinder $\langle z \rangle \times \square^1 \rightarrow \langle z \rangle$ from the constant at $\vec{0}$ to the identity. $\square$

Lemma 24.40 (Presentation commas are homotopy-discrete). *Let $p: \square^k \twoheadrightarrow \langle z \rangle$ present an atom and $J_z: A/\square^k \rightarrow A/\langle z \rangle$ the induced functor on categories of elements. For every cell τ of $\langle z \rangle$, at level j with underlying value μ , the comma category $(\tau \downarrow J_z)$ deformation-retracts onto the discrete set L_μ of direct lifts of μ : objects of the comma are pairs (σ, f) with σ a cell of the cube and f a map with $z \circ \sigma \circ f = \mu$; the assignment $r(\sigma, f) := \sigma \circ f$ lands in the lifts, sends every morphism of the comma to an identity (a morphism h satisfies $\sigma' \circ h = \sigma$ and $f' = h \circ f$, so $\sigma' \circ f' = \sigma \circ f$), and the maps $f: (\sigma \circ f, \text{id}) \rightarrow (\sigma, f)$ form a natural transformation from $\text{incl} \circ r$ to the identity. Hence $N(\tau \downarrow J_z)$ is homotopy equivalent to the discrete set L_μ .*

Corollary 24.41 (Naive cofinality is unavailable). *For a non-injective presentation some lift set has more than one element, so J_z is not W -coaspherical, for any localizer W : the cell colimit of an atom can never be computed by naive homotopy cofinality along its presentation.*

Proposition 24.42 (The simplicial criterion). *The type localizer is regular if and only if its simplicial completion contains the argument-wise simplicial weak equivalences [10, 3.4.36].*

Remark 24.43 (The corrected last mile). Two concrete routes remain for the transversal-less atoms. (i) *Fibered analysis*: by Lemma 24.40 the cell colimit of $\square^k$ decomposes over $A/\langle z \rangle$ into the *lift-multiplicity cover*, and the regularity of $\langle z \rangle$ measures exactly the homotopy type of this local system — a Theorem-B-style computation, finite at each level by the provenance calculus. (ii) *Realization*: by Proposition 24.42 it suffices to exhibit the argument-wise simplicial equivalences inside the simplicial completion of W_{type} — a cubical-against-simplicial realization invariance for which the shell and tent constructions are the natural tools: they build precisely the cylinders and cones that a realization argument consumes. A cautionary machine fact frames both routes: the kernel of the five-chain atom is not generated in levels ≤ 2 — the level- ≤ 2 pairs generate 4,288 classes at level 3 against the true 400 (artifact, `dedekind_kernelgen.py`) — and yet the atom is a *square* (Proposition 24.32): a presentation can be irreducibly three-dimensional while the presheaf it presents is not. Presentation-level invariants do not decide regularity; the fiber poset decides the retract property, and for the transversal-less atoms the presentation-free route is forced. Route (ii) is carried out in Section 24.6: the triangulation resolution proves that *every* test-contractible shape has contractible cell-diagram homotopy colimit, so its regularity is exactly its type-contractibility (Theorem 24.47) — settling all atoms, the incomparable-pair merge included, and reducing the remaining comparison to the two tasks of Remark 24.66.

24.6. The triangulation resolution and the contractible tract. The realization route of Remark 24.43(ii) can be carried out. The key is a

canonical realization of the simplex category by *cubical-simplex retracts* in Dedekind cubical sets: a cosimplicial object whose two codegeneracies are the two connections. The order-statistics operators themselves exist on every site with connections (and the contractibility computation below is valid there, cf. Corollary 24.49); the resolution becomes decisive for the Dedekind comparison, and it uses both connections.

Lemma 24.44 (The order-statistics triangulation). *For $1 \leq k \leq m$ let $o_k \in D(m)$ denote the k -th order statistic $o_k(x_1, \dots, x_m) = \bigvee_{|S|=k} \bigwedge_{i \in S} x_i$, with $o_0 := 1$ and $o_k := 0$ for $k > m$. For an order-preserving map $f: \{0 < \dots < m\} \rightarrow \{0 < \dots < n\}$ set*

$$T(f) := (o_{m(f,1)}, \dots, o_{m(f,n)}) \in D(m)^n, \quad m(f,i) := \min\{k : f(k) \geq i\}.$$

Then $T(f \circ g) = T(f) \circ T(g)$, and $T(\text{id})$ is the sort idempotent $\vec{x} \mapsto (o_1, \dots, o_n)$, whose image is the cubical simplex $\Delta_{\square}^n := \langle (x_1, x_1 \wedge x_2, \dots, x_1 \wedge \dots \wedge x_n) \rangle \subseteq \square^n$ of decreasing tuples — the spectrum of the $(n+2)$ -chain, a retract of $\square^n$. Hence $[n] \mapsto \Delta_{\square}^n$ is a cosimplicial Dedekind cubical set $D^\bullet$. Its two codegeneracies $\Delta_{\square}^2 \rightarrow \Delta_{\square}^1$ are induced by $x_1 \wedge x_2$ and $x_1 \vee x_2$: the two connections.

Proof. All assertions are identities between tuples of lattice polynomials, so they hold at every level as soon as they hold at $\{0, 1\}$ -points (the pointwise embedding of Lemma 24.33); there they are elementary statements about counting ones, verified by machine through all composable sizes ≤ 4 and in the structure cases (artifact, `dedekind-triangfunctor.py`): on a decreasing tuple o_k reads off the k -th coordinate, and $T(f)$ applied to a decreasing tuple with k ones returns the decreasing tuple with $f(k)$ ones. Functoriality on the images of the sort idempotent makes T an honest functor with $T(\text{id}) = \text{id}$ there. $\square$

Proposition 24.45 (The triangulation is a resolution). *$D^\bullet$ is a normalized contractile $\mathbf{W}_{\text{type}}$ -cosimplicial resolution in the sense of [10, 2.3.12]: $D^0 = \square^0$; the two vertices of $D^1 = \square^1$ are disjoint (RC1); and each $\Delta_{\square}^n$ is $\square^1$ -contractible, the $\wedge$ -cone of the cube preserving decreasing tuples. Consequently [10, 2.3.25, 2.3.27]: the realization functor $\text{Real}_D: \widehat{\Delta} \rightarrow \widehat{\square_{\wedge \vee}}$ preserves monomorphisms and sends simplicial weak equivalences to type equivalences, and Real_D and pr_1^* induce quasi-inverse equivalences between $\text{Ho}(\mathbf{W}_{\text{type}})$ and the homotopy category of the simplicial completion $(\mathbf{W}_{\text{type}})_\Delta$.*

Proof. RC1 and normalization are immediate; contractibility as stated, so D is a contractile resolution [10, 2.3.12], and the listed consequences are [10, 2.3.10, 2.3.25, 2.3.27], the last two using that $\mathbf{W}_{\text{type}}$ is accessible (it is generated by a set: the open boxes over a cellular model) and hence so is its simplicial completion [10, 2.3.24]. $\square$

Remark 24.46. The realized simplicial horns admit explicit contractions. Under Real_D the boundary $\partial\Delta^n$ becomes the union of the walls $\{x_1 = 1\}$,

$\{x_i = x_{i+1}\}$ ($1 \leq i \leq n-1$), and $\{x_n = 0\}$ in $\square^n$ — outer faces and *diagonal walls* — and the horn Λ_k^n omits the k -th. The partial join-cone $g_t(\vec{x}) = (x_1 \vee t, \dots, x_k \vee t, x_{k+1}, \dots, x_n)$ preserves every wall except the omitted one and ends in the face $\{x_1 = \dots = x_k = 1\} \cong \square^{n-k}$, so every horn — inner horns included — is strictly contractible, and the realized horn inclusions are type-trivial cofibrations. Both connections are needed: the one-connection site contracts only the outer horns.

Theorem 24.47 (The contractible tract). *Let X be a test-contractible Dedekind cubical set. Then the homotopy colimit of its cell diagram is contractible in the simplicial completion, and consequently*

$$X \text{ is } \mathbf{W}_{\text{type-regular}} \iff X \text{ is type-contractible.}$$

Proof. Write $W := (\mathbf{W}_{\text{type}})_\Delta$ for the simplicial completion, an accessible $A \times \Delta$ -localizer containing the Δ^1 -projections. By [10, 3.4.35], X is $\mathbf{W}_{\text{type-regular}}$ iff X , viewed as a constant simplicial presheaf, is W -regular, and by [10, 3.4.27] this holds iff the canonical map $\mathbf{L} \varinjlim \varphi_X \rightarrow X$ is an isomorphism in $\mathbf{Ho}_W$, where φ_X is the cell diagram over A/X . Each cell is a representable, type-contractible by the connection cone, so φ_X is levelwise W -equivalent to the constant diagram at the terminal object e and $\mathbf{L} \varinjlim \varphi_X \simeq \mathbf{L} \varinjlim_{A/X} e$. For any W -fibrant Y , writing $K := \mathbf{Hom}(e, Y)$ (a Kan complex, [10, 3.4.30]),

$$\begin{aligned} \mathbf{RHom}(\mathbf{L} \varinjlim_{A/X} e, Y) &\simeq \mathbf{R} \varprojlim_{(A/X)^{\text{op}}} \mathbf{RHom}(e, Y) \\ &\simeq \mathbf{Map}(N(A/X), K) \simeq K, \end{aligned}$$

the first isomorphism by [10, 3.4.33], the second the homotopy limit of a constant diagram [4], and the third because X is test-contractible and the site is aspherical, so $N(A/X)$ is a contractible simplicial set. By [10, 3.4.31] and the Yoneda lemma in $\mathbf{Ho}_W$, $\mathbf{L} \varinjlim \varphi_X \simeq e$. Finally $X \simeq e$ in $\mathbf{Ho}_W$ iff $X \rightarrow e \in \mathbf{W}_{\text{type}}$ [10, 2.3.27], i.e. iff X is type-contractible; when it is, the canonical map is a map between objects isomorphic to e and is an isomorphism, and when it is not, no map from $\mathbf{L} \varinjlim \varphi_X \simeq e$ to X is one. $\square$

Corollary 24.48 (All atoms are regular; C is regular). *Every cyclic atom on the Dedekind site is strictly contractible, hence $\mathbf{W}_{\text{type-regular}}$. In particular every quotient of a representable by a cell-kernel (spectral) congruence is regular — the incomparable-pair merge C of Remark 24.37 among them — and so is every quotient by a congruence closed under the $\wedge$ -cone.*

Proof. A non-constant monotone polynomial vanishes at $\vec{0}$, so its minimal disjunctive form involves no constants; constant components of a cell can be dropped up to isomorphism of the atom; so Lemma 24.39 applies to every atom. For a spectral congruence the $\wedge$ -cone descends pointwise: at each $\{0, 1\}$ -point the cone multiplies both members of a fiber by the same bit, and equality of cells is pointwise. Test-contractibility follows from strict contractibility, and Theorem 24.47 applies. $\square$

Corollary 24.49 (The reversal separations as regularity failures, revisited). *On the De Morgan, Kleene, and Boolean sites the very same computation — valid on any site with connections — shows that the separating quotients of Theorem 17.2, which are test-contractible but not type-contractible, have contractible cell-diagram homotopy colimits; so they are not regular, recovering Corollary 24.20 conceptually: a test-contractible shape is regular iff it is type-contractible, on every site of the paper with connections.*

The resolution also packages the entire remaining question into a single adjunction.

Proposition 24.50 (The singular adjunction). *The realization functor has a right adjoint, the singular functor $\text{Sing}_D(X)_q := \text{Hom}(\Delta_{\square}^q, X)$, and $\text{Real}_D \dashv \text{Sing}_D$ is a Quillen adjunction from the Kan–Quillen structure to the cubical-type structure. Moreover:*

- (1) *the derived realization is conservative: Real_D is also left Quillen into the test structure, where its derived functor preserves homotopy colimits and the point, hence is an equivalence by $\text{Fun}^L(\mathcal{S}, \mathcal{S}) \simeq \mathcal{S}$ [15, Thm. 5.1.5.6]; so $\text{Real}_D(f) \in \mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$ forces f to be a simplicial weak equivalence;*
- (2) *the unit is an equivalence on simplices: Sing_D carries strict cylinder contractions to simplicial homotopies (through $\text{Sing}(X) \times \Delta^1 \rightarrow \text{Sing}(X \times \square^1) \rightarrow \text{Sing}(X)$), so $\text{Sing}_D(\Delta_{\square}^q)$ is contractible.*

Proof. The adjunction is the standard coend–cotensor adjunction of a cosimplicial object; left Quillen-ness is Proposition 24.45 (monomorphisms are preserved, and simplicial weak equivalences land in $\mathcal{W}_{\text{type}}$, so trivial cofibrations go to trivial cofibrations). For (1), the derived functor into the test structure is colimit-preserving with $\text{Real}_D(\Delta^0) = \square^0$, and the test homotopy category is that of spaces; conservativity follows since $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$. (2) is the displayed factorization applied to the connection cone of $\Delta_{\square}^q$. $\square$

Theorem 24.51 (The comparison as cubical approximation). *The Dedekind cubical-type structure presents spaces — equivalently $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$ — if and only if the counit*

$$\varepsilon_X : \text{Real}_D \text{Sing}_D X \longrightarrow X$$

is a type equivalence for every uniform Kan complex X ; equivalently, iff Sing_D is homotopically fully faithful on fibrant objects.

Proof. By [12, Thm. 1.3.16], a Quillen adjunction with all simplicial sets cofibrant is a Quillen equivalence iff the left adjoint reflects weak equivalences between cofibrant objects — Proposition 24.50(1) — and the counit at every fibrant object is a weak equivalence. A Quillen equivalence with the Kan–Quillen structure means the type structure presents spaces, which by Proposition 2.2 is $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$; conversely if $\mathcal{W}_{\text{type}} = \mathcal{W}_{\text{test}}$ the adjunction is the known test-structure equivalence and the counit is an equivalence. The reformulation through $[X, Z] \rightarrow [\text{Sing } X, \text{Sing } Z]$ is the adjunction applied to homotopy classes into fibrant Z . $\square$

The mechanisms that make the comparison approachable — the sorted-cell description of the singular functor, the sort exchange and absorption calculus, the fundamental-groupoid comparison, the chain filling lemma and the extension engine, and the assembly analysis with its calibrations — are developed in Appendix C. They identify the *chain stratum* $\text{Ch}(X) \subseteq X$, the subpresheaf of instances of the sorted cells (the simplices of $\text{Sing}_D X$), as the image of the counit, and they reduce the entire comparison to a single statement.

Conjecture 24.52 (Chain approximation). *For every uniform Kan complex X on the Dedekind site, the inclusion of the chain stratum $\text{Ch}(X) \hookrightarrow X$ is a type equivalence.*

The first tract of the conjecture is settled by a pushout argument that needs no fibrancy at all.

Theorem 24.53 (Chain approximation for chain-cell quotients). *Let $X = \square^n / \sim$ be a quotient by a presheaf congruence generated by pairs of chain cells. Then $\text{Ch}(X) \hookrightarrow X$ is a pushout of the trivial cofibration $\text{Ch}_n \hookrightarrow \square^n$ of Proposition C.6, hence a type-trivial cofibration: X is chain approximated. In particular this covers the monotone dunce hat and every quotient of the square by identifications of cells of level ≤ 1 — all 1,533 quotients of the census.*

Proof. The chain stratum is a subpresheaf, so the instance closure of the generating pairs stays inside $\text{Ch}(\square^n)$: every identification of the congruence happens between chain cells. Levelwise, the cells of X are therefore the non-chain cells of $\square^n$ together with the classes of chain cells, i.e. the square

$$\begin{array}{ccc} \text{Ch}_n & \longrightarrow & \text{Ch}(X) \\ \downarrow & & \downarrow \\ \square^n & \longrightarrow & X \end{array}$$

is a pushout, where $\text{Ch}(X)$ is the image of Ch_n (sorted cells and their chain instances pass to sorted cells and chain instances, and every sorted cell of X arises this way). Trivial cofibrations are closed under pushout. For $W_{\wedge \vee}$ the three ingredients — every member of every non-singleton class is a comparable pair, the levelwise cell count $|X| = |\square^2| - |\text{Ch}_2| + |\text{Ch}(X)|$ ($26 = 36 - 34 + 24$ at level 2, $362 = 400 - 316 + 278$ at level 3), and $\text{Ch}(X) = p(\text{Ch}_2)$ — are verified by machine (artifact, `dedekind_chainquot.py`); cells of level ≤ 1 of the square are chain cells (each edge has comparable components), which gives the census case. $\square$

The second tract is the cyclic atoms, where three naturality observations close the question completely.

Lemma 24.54 (Chain naturality and retracts). *Every map of presheaves $f: X \rightarrow Y$ carries $\text{Ch}(X)$ into $\text{Ch}(Y)$: f commutes with the sort action and with instances, so sorted cells map to sorted cells and chain instances*

to chain instances. Consequently the class of presheaves satisfying chain approximation is closed under retracts; in particular it contains every retract of a representable, since $\text{Ch}(A) \hookrightarrow A$ is then a retract of the trivial cofibration $\text{Ch}_k \hookrightarrow \square^k$ of Proposition C.6.

Theorem 24.55 (Chain approximation for cyclic atoms). *For every cyclic atom $\langle z \rangle$ on the Dedekind site, the inclusion $\text{Ch}(\langle z \rangle) \hookrightarrow \langle z \rangle$ is a type equivalence.*

Proof. Three steps. (i) *The singular complex of a strictly contractible presheaf is contractible.* If $H: X \times \square^1 \rightarrow X$ is a strict cylinder contraction, then by Lemma C.9 $\text{Sing } H$ is a map $\text{Sing } X \times \Delta^1 \rightarrow \text{Sing } X$, i.e. a simplicial homotopy from a constant to the identity. Every atom is isomorphic to a constant-free atom, which is strictly contractible (Lemma 24.39), so $\text{Sing } \langle z \rangle$ is weakly contractible.

(ii) *The counit is an isomorphism onto the chain stratum for every subobject of a representable.* The presentation-connectivity proof of Theorem C.17 is instance-internal: the sorted parents $s \cdot (\rho \diamond \text{sort})$ and all hub tuples occurring in the junk-hub moves are instances of cells already present, so for $S \subseteq \square^k$ the same T -moves connect the presentations inside S , and $\text{Real}_D \text{Sing}_D S \cong \text{Ch}(S)$.

(iii) *Conclusion.* By (i), (ii) and Proposition 24.45, $\text{Ch}(\langle z \rangle) \cong \text{Real}_D \text{Sing}_D \langle z \rangle$ is the realization of a weakly contractible simplicial set, hence type-contractible; $\langle z \rangle$ itself is type-contractible by Corollary 24.48; and a monomorphism between type-contractible objects is a trivial cofibration by two-out-of-three. $\square$

The atoms and the chain stratum combine into a closure statement that will drive the extension machinery: every subobject anchored on the chain stratum is aspherical.

Conjecture 24.56 (Chain anchoring). *Let $A \subseteq \square^n$ be a cyclic atom or $\square^n$ itself, and let $W \subseteq A$ be any subobject containing the ambient chain part $A \cap \text{Ch}_n$. Then W is type-contractible, and its inclusion into A is a type-trivial cofibration.*

Conjecture 24.57 (Equivariant chain anchoring). *The same, for subobjects of A/H , $H \leq \Sigma_n$ a group of coordinate permutations preserving A , containing the image of $A \cap \text{Ch}_n$.*

Remark 24.58 (Status of the anchoring conjectures). Two ingredients of a proof are in place, and one is refuted. *Sound:* the base — ambient chain parts of constant-free atoms are strictly contractible, the $\wedge$ -cone preserving both the atom (Lemma 24.39) and pairwise comparability — and the Mayer–Vietoris mechanics, mono-pushouts of type-contractible objects over type-contractible ones being homotopy pushouts in a left proper structure. *Refuted:* the well-founded measure that was to drive the induction. A proper inclusion of atoms need *not* decrease the marked lattice:

inside the retractive five-chain spectrum $\langle (u \wedge v \wedge w, u \wedge v, u) \rangle$ the cell $(x \wedge y, x, x \vee y)$ generates a proper subatom isomorphic to $\square^2$ with the *same* five-element marked lattice (artifact, `referee_measure2.py`). Nor do low-level cell counts repair it: the *median substitution* $u = (x_2, x_1, \text{med})$ carries the atom $\langle (x_1 \vee x_2, x_1 \wedge x_3, x_2 \wedge x_3) \rangle \subseteq \square^3$ onto a proper subatom with *identical* cells at levels ≤ 2 (level-3 counts 1786 versus 1780; artifacts, `dedekind_nudetermined.py`, `dedekind_flatmech.py`), so no measure computed from cells of bounded level can drive the induction. Whether the subatom order itself is well-founded — equivalently, in the light of the level-wise finiteness of cell sets, whether strict descent can recur at unboundedly high levels — is open; the naive 5-ary majority lift of the median pair fails to descend (the original generator is recovered by a substitution; artifact, `dedekind_medtower.py`), and the question is whether *every* atom of $\square^n$ is generated at some level bounded in n , which would make the atoms finite in number and the order well-founded. Machine evidence for the conjectures (forty-two chain-anchored subobjects with mod-2 Betti numbers 1, 0, 0; artifacts, `dedekind_usub.py`, `dedekind_usubH.py`) is test-level only and, for the atom-relative half of the sweep, vacuous: every cell of the five-chain atom has pairwise-comparable components.

Proposition 24.59 (The anchoring conjectures hold over the square). *In ambient dimension two, Conjectures 24.56 and 24.57 are theorems: every subobject of $\square^2$ (or of an atom of $\square^2$, or of a permutation quotient) containing the corresponding ambient chain part is type-contractible.*

Proof. The missing ingredient was a terminating induction, and over the square the atom poset is *finite*. An atom $\langle z \rangle \subseteq \square^2$ is determined by two data: its marked two-generated lattice $F(2) \twoheadrightarrow L$, and the class of maps $L \rightarrow F(k)$ extendable along the embedding $L \hookrightarrow F(m)$ — for a cell w lies in $\langle z \rangle$ exactly when w^* kills the kernel and the induced map on L extends. By Lemma 24.30 every embedding of a two-generated sublattice admits a monotone section (the mechanism of Theorem 24.31), so every embedding is retractive, every extendability class is full, and the atom is determined by the finite datum L alone: the atoms of the square are finitely many (twelve; machine confirmation through generator level four, artifact, `dedekind_subatomchains.py`), and each is a retract of a cube, hence strictly contractible. The Mayer–Vietoris induction of the conjectures now runs by induction on the finite atom poset, with the cone-descent base as before; the permutation-equivariant case descends along the symmetric cone as in the general scheme. $\square$

Theorem 24.60 (Cone descent). *Let K be a presheaf congruence on $\square^n$ generated by parallel pairs all of whose components are expressible without the constant 1 — equivalently, vanish at the bottom vertex, a monotone function with $p(\vec{0}) = 1$ being identically 1. Then the $\wedge$ -cone descends to $Q = \square^n/K$: the assignment $([c], d) \mapsto [c \wedge d]$ is a well-defined map $Q \times \square^1 \rightarrow Q$, and it is a strict contraction of Q onto its bottom vertex. Dually for the*

constant 0 and the $\vee$ -cone. The cone is symmetric, so the statement persists in the presence of any coordinate-permutation identifications.

Proof. The generated congruence consists of the instance pairs $(A \diamond v, B \diamond v)$ of the generators. For a 1-free component A_i the cone identity of Lemma 24.39 gives $(A_i \diamond v) \wedge d = A_i \diamond (\vec{v} \wedge d)$, so the cone image of an instance pair is the instance of the same generator along $\vec{v} \wedge d$ — again in K . Hence the cone map is compatible with K levelwise and descends. Its 1-end is the identity and its 0-end the constant at the bottom vertex, since $c \wedge 0$ is the bottom constant for every cell c . (Machine checks: the key identity on 300 random 1-free instances; the descent, with the level-4 strictification conditions, and the two ends on both M -fold quotients; and the negative control that the cone does *not* descend along the dunce-hat fold $(u, u) \sim (u, 1)$, whose second component involves the constant 1 — matching the strict rigidity of $W_{\wedge \vee}$ in the census; artifact, `dedekind.conedescent.py`.) $\square$

Corollary 24.61 (Split-pair fold quotients are strictly contractible). *The components of a split epimorphism are constant-free: $t \diamond w = \text{id}$ forces $t_i(\vec{0}) = 0$ and $t_i(\vec{1}) = 1$ by monotone sandwiching through the section. Hence every quotient of $\square^k$ by identified pairs of split epimorphisms — at any level m , together with any permutation identifications — satisfies the hypothesis of Theorem 24.60 for both cones and is strictly contractible. (Whether the chain-anchored Mayer–Vietoris induction of Conjecture 24.56 runs inside such quotients is part of that conjecture’s status, Remark 24.58: the pieces of the induction are atom images under the congruence, and neither the restriction of the descended cone to them nor a well-founded measure has been established.)*

Remark 24.62 (The extension route). Conjecture 24.56 is the intended shape input for extending maps along $\text{Ch}(X) \hookrightarrow X$: filtering the complement of the chain stratum by nondegenerate cells in level order, each attaching subobject $S_x \subseteq \square^n$ contains Ch_n — chain instances of *any* cell are chain cells, by the sort-exchange factorization — so each attachment is a pushout of the trivial cofibration $S_x \hookrightarrow \square^n$, modulo the identifications among the instances of x . Against a fibrant target the fillers exist, and along stabilizer symmetries they would be chosen equivariantly by lifting in the permutation quotient (Conjecture 24.57; invertible substitutions on the Dedekind site are coordinate permutations). The remaining coherence — descending the fillers along the non-generator trivial cofibrations $S_x \hookrightarrow \square^n$ with their algebraic filling structure — is the sharpened form of the type-level upgrade. Its content can be located exactly: after reducing to a minimal presentation (splitting non-invertible self-stabilizers through their idempotent retract atoms), the descent conditions on a filler split into the prescriptions through S_x (automatic), the stabilizer symmetries (Conjecture 24.57), the *proper* fold coincidences — pairs whose members generate proper atoms, which are absorbed into the chain-anchored subobject T and

hence would be filled compatibly for free by Conjecture 24.56, the tracks being prescribed — and the *split-epi fold coincidences*: identified pairs of split epimorphisms with a common section, whose atoms are the whole cube and which no track can carry. The last class is nonempty: at $(k, m) = (2, 3)$ exactly twelve identified pairs survive minimality, forming two shapes up to symmetry, both of the form $(m_i, m_j) \sim (m_i \vee m_l, m_j \vee m_l)$ for the three pairwise meets $m_1 = x \wedge y$, $m_2 = x \wedge z$, $m_3 = y \wedge z$ — a diagonal join-correction by the third meet, whose instance closure merges the diagonal vertices (artifacts, `dedekind_splitpairs.py`, `dedekind_foldquot.py`). Both these fold quotients are *strictly contractible*, by one and the same single cylinder $H = (m_1 \vee m_2, m_1 \vee m_3)$ with cylinder coordinate x_3 : its 1-end is the identity, its 0-end is the diagonal cell $(x \wedge y, x \wedge y)$, which the fold congruence collapses to a constant, and the strictification descent condition at level 4 is a machine-verified lattice identity (artifacts, `dedekind_foldquot2.py`, `dedekind_foldwitness.py`); so at $(k, m) = (2, 3)$ the split-pair residue dissolves without any shell machinery — and by Corollary 24.61 the same holds at *every* (k, m) : the single cylinder is an avatar of the descended cone. An adversarial audit of a first assembly of this route (Remark 24.63) has, however, refuted its two load-bearing bookkeeping devices, so the route remains a program, its sound and its open parts now sharply separated. The type-level upgrade is thus precisely the *relative shell-tent statement for the split-pair fold quotients*: that in a fibrant target the top cell can be filled over its chain-anchored boundary compatibly with these identifications. The corollary above shows the chain stratum itself never obstructs, and the census verifies the statement for the 781 test-contractible small quotients of the square; the general case is the remaining mathematical content of the Dedekind entry.

Remark 24.63 (The audit of the first assembly). A first assembly of the extension route into a chain extension theorem — every map to a fibrant target defined on a subobject containing the chain stratum extends, whence universal chain approximation and the comparison itself — was subjected to an independent adversarial audit, which refuted it at three points while confirming the rest. *Refuted*: (a) the well-founded measure $\mu = |L|$ on minimal presentations — a proper subatom can carry the same marked lattice (Remark 24.58), so the transfinite well-order and the emptiness of the proper-fresh family collapse; (b) the chain-anchoring of the attachment subobject — for a minimal *atom* presentation the ambient chain cells of the atom need not have chain values: the atom $\langle (x \wedge y, x, x \vee y) \rangle \cong \square^2$ lies entirely inside Ch_3 while its generator is not a chain cell of the atom (artifacts, `referee_counterex.py`, `referee_chainreal.py`); the correct anchor is the *intrinsic* chain stratum, whose image lies in $\text{Ch}(X)$ by Lemma 24.54, and the corresponding intrinsic-anchored contractibility statement is open; (c) the Mayer–Vietoris induction inside the fold quotients (Corollary 24.61). *Confirmed sound*: the cone descent theorem and its negative control, the

chain dictionary, the four-family analysis of a minimal attachment (the fresh/prescribed dichotomy, the impossibility of mixing generator-reaching with proper members, and the constant-freeness of generator-reaching members at every level), the reduction of the comparison to the extension statement through the dictionary and Theorem 24.51, and the absence of circularity throughout. The comparison problem thus stands reduced to: a well-founded measure for subatom chains (or a proof that the subatom order is well-founded), and the intrinsic-chain-anchored asphericity of minimal atoms and their fold and stabilizer quotients.

Remark 24.64 (The shape-by-shape program). Theorems 24.53 and 24.55 start an induction parallel to the regularity program: chain approximation for all cellular shapes, propagated by gluing and box attachment. The bookkeeping identities are themselves machine-calibrated: the union identity $\text{Ch}(A \cup B) = \text{Ch}(A) \cup \text{Ch}(B)$ is automatic, the intersection identity $\text{Ch}(A) \cap \text{Ch}(B) = \text{Ch}(A \cap B)$ and the box identity $\text{Ch}(\square^{n+1}) \cap P = \text{Ch}(P)$ hold on all tested wall shapes and atom pairs of the square, while the *ambient* identity $S \cap \text{Ch}(\square^n) = \text{Ch}(S)$ fails for seven of twenty-six sampled atoms of $\square^3$ — the five-chain atom included — through permuted order-statistic cells with no sorted parent inside the atom; the identities must therefore be managed per shape, not assumed (artifacts, `dedekind_shapes1.py`, `dedekind_chintersect.py`, `dedekind_chambient.py`). For quotients, the counit statement (A) reduces to the generation of the restricted congruence by its identified *sorted* pairs — sort idempotency makes every sort-fixed class contain a sort-fixed representative — and this generation condition passes an adversarial machine sweep (eighteen quotients of the square, including cross-incomparable and multi-pair folds; artifact, `dedekind_sgen.py`) but *fails* at the M -fold quotients of Remark 24.62, where the chain cells $(x \wedge y, x)$ and $(y, x \vee y)$ are identified through no sorted pair — statement (A) is genuinely restrictive, and by Lemma C.18 it is, fortunately, not load-bearing for the comparison criterion; the compatibility of the induction with the coequalizer identifications of Lemma A.2, and the intersection identities beyond the square are the open items of the program. On the *test* side the program is complete: by Theorem C.13 the singular complex computes the test homotopy type of every presheaf and the counit is always a test equivalence, so the entire remaining content of the conjecture is the type-level upgrade at fibrant objects.

Remark 24.65 (Calibration of the conjecture, corrected). Machine computations calibrate the conjecture. It holds for representables (both sides are strictly contractible) and, by Theorem 24.53, at every chain-cell quotient — fibrancy is *not* the operative hypothesis there. An earlier calibration claimed the opposite: the T -face complex on *all* cells of $\text{Ch}(W_{\wedge \vee})$ has $\mathbb{F}_2$ -Betti numbers 1, 0, 14, which we read as a failure of chain approximation without fibrancy. That complex is not a homotopy invariant: it counts each

chain instance of a sorted cell (the same simplex reparametrized) as an independent generator. The homotopically correct model is the simplicial chain complex of $\text{Sing}_D W_{\wedge V}$ — by Proposition 24.45 and Remark 24.46 the realization is left Quillen, so $\text{Ch}(W_{\wedge V}) \cong \text{Real}_D \text{Sing}_D W_{\wedge V}$ has the homotopy type of the simplicial set $\text{Sing}_D W_{\wedge V}$ — and its Betti numbers are 1, 0, 0, 0 through degree 3, with trivial fundamental group by Proposition C.5: no gap (artifacts, `dedekind_singWbetti.py`, `dedekind_chainquot.py`; the superseded computation is retained in `dedekind_chW.py` as a methodological caution, together with its companion 1, 0, 152 for $\text{Ch}_3 \cup P_3$ in `dedekind_An.py`, which is withdrawn for the same reason). Consistently with this, an exhaustive sweep of stage-one replacement boxes — all data over $W_{\wedge V}$ for the coordinate boundary and horn shapes, the shell box with its contraction certificate, the realized-horn shapes with diagonal walls, and the wad foldings along the cartesian swap and sort endomorphisms of Lemma A.2(ii) — confirms that box fillers never change the sorted-complex homology: free attachments are homologically inert, as they must be if nothing is missing (artifacts, `dedekind_decay3.py–decay9.py`). Finally, the transfer routes — conservativity of the singular functor on fibrant objects, and descent along the one-connection restriction through the theorem of Cavallo–Sattler — are each equivalent to the same rectification problem, of upgrading partially natural data to fully natural data; the uniform fillers of Lemma A.2, which are fully natural, are the designated tool. By Theorem 24.51 and the reductions above, the conjecture (with the cell-injectivity of the counit, verified for the dunce hat) is equivalent to the Dedekind phase-diagram entry: *Dedekind cubical sets present spaces if and only if fibrant objects are chain approximated.*

Remark 24.66 (What remains for the phase diagram). By Theorem 24.19 the Dedekind comparison is the regularity of every cellular shape. Theorem 24.47 settles the contractible tract: for test-contractible shapes, regularity *is* the type-contractibility question — the general form of the census program, verified for all 1,533 small quotients of the square by the shell–tent calculus, and now including every cyclic atom. Two tasks remain. (a) *Contraction*: every test-contractible cellular shape is type-contractible — the shell–tent conjecture at full generality. (b) *Assembly*: regularity of the non-contractible sources of the cellular model. Here the comparison map $u_X: \text{Real}_D(N(A/X)) \rightarrow X$ is natural, its source and target carry mono-pushouts to homotopy pushouts (the structures are left proper, with all objects cofibrant, and the nerve of the cell category sends mono-pushouts of presheaves to homotopy pushouts because the *test* localizer is regular), so the class of shapes where u_X is an equivalence is saturated by monomorphisms. A natural finiteness that would ground the assembly — *rank-collapse*: every atom generated at level equal to its number of components, making the subatom family of $\square^n$ finite — holds in the small cases ($\square^1$ has three subatoms, $\square^2$ twelve, stable at level = dimension; no exceptions at

$(l, n) = (2, 1), (3, 2))$ but *fails* at $(3 + 1, 3)$: level-4 cells generate new subatoms of $\square^3$ beyond the 168 visible at level 3 (40 novel subatoms among 305 samples; artifact, `dedekind_subatoms.py`, `dedekind_subatoms3.py`). The shape-by-shape assembly is therefore not the right endgame. It is also not needed: by Theorem 24.51 the entire comparison — and with it, a posteriori, the regularity of every shape — is equivalent to the single cubical approximation statement for the counit ε_X , a uniform filling problem on fibrant objects that the shell calculus of this Part is built to address.

24.7. Immunity of the group quotients. The shell mechanism raises an immediate question: does it not also threaten the separations of Parts I–IV, whose witnesses live over the quotients $\square^n/H$? It does not, and the reason is structural. (Theorem 24.2 and its proof are site-independent: the exchange calculus never uses the interval theory.)

Proposition 24.67 (Equivariant shells are strict). *Let H act on $\square^n$ by automorphisms of a site with connections, let Y be any object, and let $K = \text{val}(B, s)$ be a fresh filler value in $R(Y)$ at level $n + 1$, invariant in the source coordinates — $K \circ (\sigma \times \text{id}) = K$ for all $\sigma \in H$, i.e. a cylinder $(\square^n/H) \times \square^1 \rightarrow R(Y)$. Then:*

- (1) *the based data of K are equivariant: $s \circ (\sigma \times \text{id}) = s$, the shell T is $(\sigma \times \text{id})$ -invariant, and $B \circ ((\sigma \times \text{id}) \hat{\times} \text{id}) = B$;*
- (2) *consequently the ε -end $G := B|_{\square^{n+1} \times \{\varepsilon\}}$ and the end-square prisms $C_{\varepsilon'} := B \circ (n_{\varepsilon'} \times \text{id})$ (when $n_{\varepsilon'} \in T$) are invariant cells of the previous stage — strict cylinders over $\square^n/H$ at that stage;*
- (3) *every old end of K — a G -slice, or a section $C_{\varepsilon'} \cdot (\cdot, s_{\varepsilon'})$ at the invariant parameter $s_{\varepsilon'}$ — is connected to the slice $G \cdot n_{\varepsilon'}$ by invariant cylinders of the previous stage (the connection reparametrizations $C_{\varepsilon'} \cdot (\cdot, s_{\varepsilon'} \vee r)$ and $C_{\varepsilon'} \cdot (\cdot, s_{\varepsilon'} \wedge r)$), and G itself connects its two slices.*

Hence a fresh invariant cylinder with old ends never relates two cells that were not already related at the previous stage: over group quotients the shell mechanism does not extend the strict homotopy relation. The same argument applies generator-wise to maps from the instances $E(T)/N$.

Proof. (1) $K \circ (\sigma \times \text{id})$ is the value of the pair $(B, ((\sigma \times \text{id}), s \circ (\sigma \times \text{id})))$, whose base form is $(B \circ ((\sigma \times \text{id}) \hat{\times} \text{id}), s \circ (\sigma \times \text{id}))$ over the pulled-back shell $(\sigma \times \text{id})^{-1}(T)$. Both values are fresh, so by the completeness of the based normal form (Theorem 24.2) their equality forces the based data to coincide. (2) The displayed cells are restrictions of the equivariant B along invariant squares. (3) The parameters $s_{\varepsilon'} = s(\cdot, \varepsilon')$ are invariant by (1), so the reparametrized cells are invariant; they are cylinders with the stated ends by the connection identities, exactly as in Theorem 24.5's target site, and connections exist on all sites of Parts II–IV. $\square$

Remark 24.68 (The boundary of the shell mechanism). The contrast with Theorem 24.5 is now sharp. Over the congruence quotient $W_{\wedge\vee}$ the conditions for being a cylinder are *fold* conditions — equalities of restrictions along non-invertible maps, which lower dimension and therefore land in the shell, where the box prescribes them; nothing forces the shell data to be strict. Over a group quotient the conditions are *equivariance* conditions — indexed by automorphisms, which preserve dimension and therefore act on the based data themselves, forcing the entire shell to consist of strict cylinders. Fold conditions are shell-absorbable; equivariance is not. This is why the reversal-site theory of Parts I–IV, whose homotopy data live over group quotients, is untouched by the phenomenon that reverses the Dedekind expectation. The audit is corroborated by an exhaustive census over the Boolean Klein quotient (artifact, `boolean_shell_audit.py`): 12 strict endomorphism classes and 128 invariant strict cylinders, with the identity’s component the singleton $\{\text{id}\}$ and the constants elsewhere; all invariant-cylinder slice pairs and all invariant sections are strictly pre-connected, as the proposition predicts.

25. TEST-CONTRACTIBILITY

We prove test-contractibility by exhibiting $W_{\wedge\vee}$ as an iterated homotopy pushout of contractible presheaves. Recall that in the test structure every object is cofibrant (cofibrations are the monomorphisms) and representables are contractible, so a pushout along a monomorphism is a homotopy pushout, and a wedge of two representable intervals at a vertex is contractible.

Lemma 25.1 (Wedge monomorphisms). *(i) The wedge of the left and bottom edges at their common vertex v_{00} maps monomorphically into $\square^2$. (ii) Let $W_1 := \square^2 / ((0, u) \sim (u, 0))$. The images $d, t: \square^1 \rightarrow W_1$ of the diagonal and the top edge are levelwise injective, and their images meet exactly in the degenerate cells of the common vertex v_{11} ; hence the wedge $d \vee_{v_{11}} t \rightarrow W_1$ is a monomorphism.*

Proof. (i) is immediate: distinct cells of a representable. (ii) The congruence of W_1 is generated levelwise by the pairs $(0, w) \sim (w, 0)$, $w \in D(k)$; since no cell of the form $(0, w)$ or $(w, 0)$ equals another such cell except as forced, every congruence class is either a singleton or one swap pair $\{(0, w), (w, 0)\}$. A diagonal cell (u, u) lies in a swap pair only if $u = 0$, i.e. only the degenerate cells of v_{00} ; a top cell $(u, 1)$ only if $u = 1$ hits nothing (the pair $(0, w)$ requires a zero first component, $w = 1$ gives $(0, 1) \neq (u, 1)$ unless $u = 0$, and $(w, 0)$ has second component $0 \neq 1$); injectivity of d and t and the computation of $d \cap t$ follow by inspection of these classes: $(u_1, u_1) \sim (u_2, u_2)$ forces $u_1 = u_2$, $(u_1, 1) \sim (u_2, 1)$ forces $u_1 = u_2$, and $(u, u) \sim (w, 1)$ forces $u = w = 1$. (These facts are also machine-verified through level 3; artifact.) $\square$

Proposition 25.2. $W_{\wedge\vee}$ is test-contractible.

Proof. Identifying two parallel cells $A \sim B$ that share their source vertex is the pushout of the fold map along the wedge inclusion:

$$\square^2/(A \sim B) = \square^2 \cup_{A \vee B} \square^1,$$

where $A \vee B$ is the wedge at the shared vertex, included by (A, B) , and the right leg folds both intervals onto one. For the swap fold, Lemma 25.1(i) makes the wedge inclusion a monomorphism, so the square is a homotopy pushout of test-contractible objects ($\square^2$, the wedge, and $\square^1$ are contractible), and $W_1 \simeq_{\text{test}} *$. In W_1 the diagonal and top edges share their target vertex v_{11} (and, after the first fold, $d(0) = v_{00} \neq v_{01} = t(0)$), and Lemma 25.1(ii) makes the second wedge inclusion a monomorphism, so

$$W_{\wedge \vee} = W_1 \cup_{d \vee t} \square^1$$

is again a homotopy pushout of contractibles, and $W_{\wedge \vee} \simeq_{\text{test}} *$. $\square$

Proof of Theorem 21.5. Test-contractibility is Proposition 25.2; type-contractibility is Theorem 24.5; and the cellular rigidity is Theorem 23.2. In particular $W_{\wedge \vee} \rightarrow *$ lies in $\mathbf{W}_{\text{type}} \cap \mathbf{W}_{\text{test}}$ and separates nothing. $\square$

Corollary 25.3 (The phase diagram; Theorem 1.1). *Among the standard cube categories — cartesian, one connection, Dedekind, De Morgan, Kleene, Boolean — the cubical-type model structure presents spaces if and only if $\mathbf{W}_{\text{type}} = \mathbf{W}_{\text{test}}$; on all sites but the Dedekind site the comparison is settled, and the type structure presents spaces exactly on the one-connection site:*

site	type = test?	reference
cartesian	no	[1]
one connection	yes	[8]
Dedekind	open	witness fails (Thm. 21.5)
De Morgan, Kleene, Boolean	no, at every n	Theorem 17.2

Proof. “Presents spaces $\iff \mathbf{W}_{\text{type}} = \mathbf{W}_{\text{test}}$ ” is Corollary 2.3; the entries are as cited. $\square$

Part 6. Machine verification

26. THE VERIFICATION SUITE

Every finite combinatorial claim of this paper is verified by computation — exhaustive except where a sampling is explicitly noted — in the scripts of the artifact repository of the series,

<https://github.com/r-shibusawa/cubical-strict-j>

releases **v1.16.0–v1.21.0** (version-pinned, with Zenodo DOIs on the release pages). The releases accompanying this paper are **v1.20.0** (Parts I–III), DOI 10.5281/zenodo.22012992, and **v1.21.0** (Parts IV–V), DOI 10.5281/zenodo.22038185. The main entries:

- `kleene_site.py`, `boolean_site.py`: the Kleene (unmixed-poset) and Boolean machineries; parity lemmas at the unmixed witness points; stage-zero censuses of the Klein machines (essential classes 0/18/2 across DM/KL/BA); the collage factorization $F(d, a, t) \oplus F(d, \neg a, t) = t$.
- `boolean_delta.py`, `kleene_delta.py`, `boolean_certificates3.py`, `kleene_certificates3.py`: the exact certificate systems (Boolean: parity-exact; Kleene: with the monotone-realizability filter) for the two-stratum family and the six beyond-two-stratum groups of dimension three; survivor statistics of Table 2.
- `boolean_fate.py`: interior realizations of consistent twists, exact slide-null decisions (positives verified by reconstruction), collage images — the data behind the rigidity discriminant, 12/12.
- `discriminant_check.py`: the discriminant ($\varepsilon \circ \delta$) against the fate data, and the reflection-generation characterization on fourteen groups across $n = 2, 3, 4$ — all matching.
- `boolean_certificates4.py`, `degenerate_search.py`: the five dimension-4 primitive groups $V_{\text{even}}(4) \rtimes P$ (all certified) and the five degenerate groups with no one-step equivariant consensus pattern.
- `separation_criterion.py`: the criterion against the family recursion of [11] on all 98 subgroups of B_3 : 78 agreement / 20 separation, zero mismatches; the reflection closure is generated by its own reflections in every case; every reflection fixes a vertex.
- `isotropy_poset.py`, `el_betti.py`, `el_betti_a4.py`, `el_betti_nested.py`: the incidence posets of isotropy loci with setwise stabilizers and residual actions; the exact $\mathbb{F}_2$ -cochain engine for the collage formula (normalized bar complexes, induced cochain maps, homotopy pushout models with tracked cocycle representatives); the computations of Proposition B.3, validated against the closed-form join $B\mathbb{Z}/2 \star B\mathbb{Z}/2$ and against $H^*(BN_{A_4}; \mathbb{F}_2)$.
- `edom_check.py`, `realizability_check.py`: stage-zero verification of Hypothesis 14.4(ii) for the $n \leq 3$ bases (the hypothesis is superseded by the dichotomy in the body; the check documents the historical route), and the unified realizability theorem (25/25 across the three sites).
- `frontier_census.py`, `frontier_census2.py`, `q4_check.py`: the six $n = 4$ classes left undecided by q_* (all later settled by the census) — incidence complexes (trees, S^1 , $(1, 5, 0)$, $(1, 0, 15)$), and q_* -surjectivity through degree 4.
- `tree_factor_check.py`, `nontree_factor_check.py`, `tree_witness_analysis.py`, `twostep_factor_check.py`, `rigidity_census.py`: the sieve-landing dichotomy — one-step witnesses (256 and 32768 at the two reduction classes, none elsewhere), witness structure (single-stratum bijective images for all

witnesses of the first class and 8192 of the second; residuals K and an order-4 mixed block), deck-freedom counts, and the closed-form census of Theorem 15.6 over all 34 classes.

- **reduction.structure.py**: all four claims of Theorem 15.7 and the conclusion of Lemma 15.9, verified on every witness of both reductions ($256 + 32768$).
- **etm_library.py**, **etm_decisive.py**, **etm_decisive2.py**: the twelve M -set computations behind the arrangement dichotomy — the $\mathbb{F}_2$ - and $\mathbb{F}_3$ -acyclicity of S_3 on three points, the decisive $(\mathbb{Z}/2)^2$ on three 2-orbits and $\mathbb{Z}/4$ on a regular orbit plus a 2-orbit, and the fixed-point-free elementary abelian 2-subgroups in all 34 cube classes.

On the Boolean site every decision procedure above is *exact*: truth tables are unconstrained, so a consistent parity assignment is a cell, and positives are re-verified by reconstruction. On the Kleene site exactness is restored by the monotone-realizability filter over the unmixed poset. Several of the paper’s theorems were found as empirical laws in this data and then proved: the rigidity discriminant, the realizability divide, the reflection-generation characterization, and the criterion itself.

For Parts IV–V (release v1.21.0):

- **selfdual.interpolant.py**: the identities of Lemma 18.1 on the literal-cube representation of the free De Morgan algebras.
- **dm_witness_check.py**: exact 2-SAT decision of De Morgan/Kleene witnesses at the two $n = 4$ reduction classes (both admit them, on the same invariant plane as their Boolean witnesses).
- **dedekind.site.py**, **dedekind.sweep.py**,
dedekind.stage1.py/15/2, **cand05_independent.py**,
case4_check.py, **m4_escape.py**, **dedekind.triangulate.py**,
dedekind.pi1.py, **dedekind.zhomology.py**,
dedekind.sections.py, **dedekind.coherence.py**,
dedekind.pcs0.py, **dedekind.shell.py**: the Dedekind engine — congruence quotients of the square, the base facts of the separation by two independent implementations, the diagonal-track computation over all of $D(3)$, the single-fold contractions, the wedge monomorphism facts, exhaustive fresh-cylinder searches through box dimension 4 (462,672 candidates, all empty as the cellular isolation theorem predicts), the stage-one shell-configuration sweep that discovered the contraction, and the complete certificate of the shell contraction: the gluing of Φ , the fold conditions, the two ends, the strict null-homotopy of h , and the strict isolation of the identity component.
- **boolean_shell_audit.py**: the group-quotient immunity audit on the Boolean Klein quotient (Proposition 24.67): census of strict endomorphisms and invariant cylinders, cluster partition, section reparametrizations.

- `dedekind_shellreach.py`, `dedekind_fullshell.py`, `dedekind_freshchain.py`, `dedekind_qstar.py`: the strict-plus-shell reachability census over all 1,533 one- and two-relation quotients of the square, the exhaustive single-shell search over arbitrary bases on the six resistant quotients, the tent-move closure contracting all of them (Theorems 24.11 and 24.10), and the self-contained thirty-two-check certificate of the Q^* tent.
- `dedekind_monocell.py`: the mono-cellular presentation of Q^* (Theorem 24.25): identity of the triple-fold congruence with the triangle-to-circle pushout congruence at all stored levels.
- `dedekind_projtest.py`: the retractivity sweep over embedded sublattices of $D(3)$ and $D(4)$ (Proposition 24.27, Remark 24.29): all two-generated embeddings retractive; the three-generated failures and the five-chain phenomena ($D(4)$ by sampling).
- `dedekind_kernelgen.py`: the five-chain kernel is not generated in levels ≤ 2 (Remark 24.43) — a presentation-level fact; the atom itself is a square (Proposition 24.32).
- `dedekind_squaretrick.py`: the square isomorphism of the five-chain atom at levels ≤ 4 (cell sets, bijectivity, $|T(k)| = |D(k)|^2$), and the transversal sweep over all 880 cell kernels on $\{0,1\}^3$, 236 without monotone transversal (Proposition 24.32, Theorem 24.34, Remark 24.37).
- `dedekind_foldlocus.py`: the stabilizer and fold data of the five-chain presentation (27 stabilizer elements, fully folded generator), documenting the assembly obstruction (Remark 24.38).
- `dedekind_triangfunctor.py`: the order-statistics triangulation functor — functoriality through all composable sizes ≤ 4 , the sort idempotent, the connection codegeneracies, and the coface formulas (Lemma 24.44).
- `dedekind_chainquot.py`, `dedekind_singwbetti.py`, `dedekind_decay3.py`–`decay9.py`, `dedekind_realsingw2.py`: the corrected calibration of the chain approximation conjecture — the pushout verification for the dunce hat (Theorem 24.53), the simplicial Betti numbers 1,0,0,0 of $\text{Sing}_D W_{\wedge \vee}$, the exhaustive stage-one inertness sweep, and the cell-injectivity of the counit on the dunce hat (2,5,24 cells = values); the superseded cell-basis computations (1,0,14 and 1,0,152) are retained in `dedekind_chw.py` and `dedekind_An.py` as a methodological caution (Remark 24.65).
- `dedekind_shapes1.py`, `dedekind_chintersect.py`, `dedekind_chambient.py`, `dedekind_sgen.py`: the shape-by-shape program — the T -versus-singular Betti sweep over the atoms and wall shapes of the square and the sampled frontier atoms of $\square^3$ (all equal), the intersection and box identities, the failure of the ambient identity at seven $\square^3$ atoms, and the sorted-generation sweep for the counit condition (Theorem 24.55, Remark 24.64).

- `dedekind_normalform.py`: normal presentations of chain cells and the T -move connectivity of their presentation groupoids at $(n, m) = (2, 1), (2, 2), (3, 2)$, sources of level ≤ 3 , no disconnected instance (Lemma C.17).
- `dedekind_protocol.py`: the absorption identities, the level-1 chain fact, and the chain-instance statistics at level 2 (34/400, empty conflict locus) (Lemma C.10).
- `dedekind_sortzigzag.py`: the sort-zigzag endpoint identities, the impossibility of a single homotopy, and the cylinder counts $|\text{Sing}(\square^1 \times \square^1)_q| = (q+2)^2$ (Lemmas C.8, C.9).
- `dedekind_chaincells.py`: the chain-cell subobject — closure, the $\wedge$ -cone, the two-chamber identification of Ch_2 , sorted $\subseteq$ chain, and sort-stability of decreasing rearrangements ($n \leq 3, k \leq 3$; Proposition C.6).
- `dedekind_sortexchange.py`: the sort-exchange factorization ($k, q \leq 3$, no exceptions) and the face identities of the sorted triangles and the connection square (Lemma C.4, Proposition C.5).
- `dedekind_singsort.py`: the sorted-cell description of the singular functor — fixed points equal the sort image, simplicial instances of sorted cells are sorted, and the counts $\text{Sing}(\square^n)_q = (q+2)^n$ and $q+1$ for the cubical circle (Lemma C.1).
- `dedekind_subatoms.py`, `dedekind_subatoms3.py`: the subatom counts of $\square^1, \square^2, \square^3$ (3; 12, stable at level 2; 8/27/66/168 through level 3), the rank-collapse sweeps at $(l, n) = (2, 1), (3, 2)$ (no exceptions), and its refutation at level 4 over $\square^3$ (40 novel subatoms among 305 samples) (Remark 24.66).
- `dedekind_cmerge.py`, `dedekind_specttower.py`: the incomparable-pair merge — level sizes, fold classes of the spectral surjection (732 folded generator classes at level 4), non-generation of the spectral congruence in low levels (7,108 vs 6,117 at level 3), the $n = 2$ transversal sweep (11 kernels, all transversal-ed), and the non-lattice fiber poset (Corollary 24.36, Remark 24.37).

Data availability. All verification scripts and the version-pinned artifacts are archived at releases v1.20.0 (DOI 10.5281/zenodo.22012992) and v1.21.0 (DOI 10.5281/zenodo.22038185).

Use of AI tools. An AI assistant was used in the preparation of this manuscript and its verification scripts; the author takes full responsibility for the content.

Appendix

APPENDIX A. THE REPLACEMENT MACHINERY

This appendix collects, in self-contained form, the machinery of algebraic fibrant replacements used throughout Parts I–V: the uniformity descent

criterion, the coequalizer presentation of the replacement, strictification of maps from quotients, the equivariant descent lemma with its sector calculus, the equivariant filler slide, and the first-exit induction. The statements and proofs are those of the companion manuscript [21], reproduced here at the level of generality at which they are used (an arbitrary reversal site; arbitrary finite group or congruence quotients); the proofs are unchanged except for notation.

Lemma A.1 (Descent via uniformity). *An object equipped with fillers for the boxes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, where S ranges over arbitrary subpresheaves of representables, strictly natural in cartesian substitution squares (Remark A.3), is fibrant.*

Proof. Let W carry the stated filling structure; we show that $W \rightarrow 1$ has the right lifting property against $m \hat{\otimes} \delta^\varepsilon$ for every monomorphism $m : X \hookrightarrow Y$.

Step 1: cellular decomposition of m . Well-order the nondegenerate cells of Y not in X , say $(y_\beta)_{\beta < \lambda}$ with $y_\beta \in Y([n_\beta])$, and let $X_\beta \subseteq Y$ be the subpresheaf generated by X together with $\{y_\gamma : \gamma < \beta\}$; then $X_0 = X$, $X_\lambda = Y$, and $X_{\beta+1} = X_\beta \cup R_\beta$, where $R_\beta \subseteq Y$ is the image of $y_\beta : \square^{n_\beta} \rightarrow Y$. Writing $p_\beta : \square^{n_\beta} \twoheadrightarrow Q_\beta := \square^{n_\beta} / E_\beta$ for the image factorization of y_β — so E_β is the kernel congruence, $p_\beta \sigma = p_\beta \tau \iff y_\beta \cdot \sigma = y_\beta \cdot \tau$ — the square

$$X_{\beta+1} = X_\beta \cup_{P_\beta} Q_\beta, \quad P_\beta := Q_\beta \cap X_\beta,$$

is a pushout (unions of subpresheaves are effective pushouts over the intersection in a presheaf topos). On this non-Reedy site the cell shapes Q_β are genuinely quotients of representables [7], which is why representable *boundaries* alone would not suffice; this is the cellular-model phenomenon of [10, §1.2].

Step 2: reduction to one attachment. A lift against $m \hat{\otimes} \delta^\varepsilon$ is an extension of a given $h : Y \times \{\varepsilon\} \cup_{X \times \{\varepsilon\}} X \times \square^1 \rightarrow W$ to $Y \times \square^1$. We construct compatible extensions $L_\beta : Y \times \{\varepsilon\} \cup X_\beta \times \square^1 \rightarrow W$ by transfinite induction, with $L_0 = h$ and unions at limit stages (the value of L on each cell stabilizes, so the union is well-defined). At a successor stage the required datum is an extension of

$$b_\beta : Q_\beta \times \{\varepsilon\} \cup P_\beta \times \square^1 \longrightarrow W$$

(restriction of L_β and h) to $Q_\beta \times \square^1$: a filling problem for the box $(P_\beta \hookrightarrow Q_\beta) \hat{\otimes} \delta^\varepsilon$.

Step 3: pullback to the representable and descent. Let $S := p_\beta^{-1}(P_\beta) \subseteq \square^{n_\beta}$ and let $b := b_\beta \circ (p_\beta \times \text{id})$ be the pulled-back box for $j_{S, n_\beta, \varepsilon} = (S \hookrightarrow \square^{n_\beta}) \hat{\otimes} \delta^\varepsilon$; the filling structure provides $\text{val}(b) : \square^{n_\beta} \times \square^1 \rightarrow W$ extending b . Now let $\sigma, \tau \in \square_{\mathcal{V}}([m], [n_\beta])$ be any congruence pair, $p_\beta \sigma = p_\beta \tau$. Then $\sigma^{-1}(S) = (p_\beta \sigma)^{-1}(P_\beta) = (p_\beta \tau)^{-1}(P_\beta) = \tau^{-1}(S) =: S'$, and the restricted boxes agree, $b \circ \hat{\sigma} = b \circ \hat{\tau}$ as boxes for $j_{S', m, \varepsilon}$, because b factors through $Q_\beta \times \square^1$. Strict naturality gives

$$\text{val}(b) \circ \hat{\sigma} = \text{val}(b \circ \hat{\sigma}) = \text{val}(b \circ \hat{\tau}) = \text{val}(b) \circ \hat{\tau}.$$

Since Q_β is the coequalizer of its congruence pairs and $(-)\times\Box^1$ preserves this coequalizer (quotients and products of presheaves are computed levelwise), $\text{val}(b)$ descends to $\bar{v} : Q_\beta \times \Box^1 \rightarrow W$. On the box domain, $\bar{v}$ agrees with b_β because $\text{val}(b)$ extends b and $p_\beta \times \text{id}$ maps $S \times \Box^1 \cup \Box^{n_\beta} \times \{\varepsilon\}$ onto $P_\beta \times \Box^1 \cup Q_\beta \times \{\varepsilon\}$. Set $L_{\beta+1} := L_\beta \cup \bar{v}$; this is well-defined precisely because $\bar{v}$ restricts to L_β on the overlap $P_\beta \times \Box^1$.

The induction terminates at λ with the required lift $L_\lambda : Y \times \Box^1 \rightarrow W$. $\square$

Lemma A.2 (Algebraic fibrant replacement). *Every $X \in \widehat{\Box}_V$ admits a functorial trivial cofibration $u : X \rightarrow R$ with R uniformly fibrant, constructed as the colimit of a transfinite sequence $X = R_0 \rightarrow R_1 \rightarrow \dots$ of monomorphisms in which*

$$R_{\beta+1} := \text{coeq}\left(\coprod_{\substack{\sigma:j' \rightarrow j \\ b \text{ box for } j \text{ over } R_\beta}} \text{cod}(j') \rightrightarrows \coprod_{(j,b)} \text{cod}(j) \right) \sqcup_{\coprod_{(j,b)} \text{dom}(j)} R_\beta,$$

the two parallel arrows being the insertion at $(j', b \circ \hat{\sigma})$ and the composite of $\hat{\sigma}$ with the insertion at (j, b) , and the pushout taken along the evaluations b . Consequently:

- (1) every cell of $R_{\beta+1}$ is a cell of R_β or a filler value $\text{val}(b, c)$, $c \in \text{cod}(j)$, and the identifications among filler values are exactly those generated by
 - (i) $c \in \text{dom}(j)$: $\text{val}(b, c) = b(c)$, a cell of R_β (the pushout);
 - (ii) $\text{val}(b \circ \hat{\sigma}, c) = \text{val}(b, \hat{\sigma} \circ c)$ (the coequalizer);
- (2) the stage maps are monomorphisms and no identifications are imposed on old cells: the evaluations agree on the cells identified by the coequalizer, precisely because $b \circ \hat{\sigma}$ is by definition the restricted box; hence equality of cells of R is detected at a stage and is generated by (i)–(ii);
- (3) filler values are strictly natural, $\text{val}(b, c) \cdot f = \text{val}(b, c \cdot f)$, and R carries the tautological uniform filling structure, so R is fibrant by Lemma A.1;
- (4) by the standard property of the algebraically free awfs generated by $\mathbb{J}$, the unit u lies in the left class cofibrantly generated by the box inclusions — Garner’s construction interposes coequalizers, so u is a retract of a transfinite composite of pushouts of the generators rather than literally such a composite — and since the box inclusions are trivial cofibrations of the type structure, whose left class is closed under pushout, transfinite composition and retract, u is a trivial cofibration; the construction is functorial in X .

Proof. This is Garner’s algebraic small object argument [13] for the generating category $\mathbb{J}$ (rather than a mere set), in the free-monad-on-a-pointed-endofunctor form; the displayed one-step construction is its explicit colimit presentation, as in the treatment of uniform fibrations in [14]. The

listed properties are read off from the presentation: (1) is the description of a coequalizer-then-pushout of coproducts of presheaves, whose cell-level relations are precisely the displayed generators; for (2), both relation schemes have a fresh member on each side except (i), whose old member is uniquely determined, and chains through fresh cells terminating in old cells on both ends force equal old cells because restricted boxes evaluate compatibly ($b(\hat{\sigma} \circ c) = (b \circ \hat{\sigma})(c)$ on the box domain); (3) the presheaf structure on a coproduct of representable-products acts on the c -coordinate, and the filling structure assigns to a box its inserted values, uniformity holding by (ii); (4) is the standard comparison of the algebraically free awfs with the cofibrantly generated one [13]. $\square$

Remark A.3 (The generating category is cartesian). The generating category $\mathbb{J}$ has objects the box shapes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, S an arbitrary subpresheaf, and morphisms the *cartesian* squares: cube maps $\sigma: [m] \rightarrow [n]$ with $S' = \sigma^{-1}(S)$ exactly and ε fixed, acting as $\hat{\sigma} = \sigma \times \text{id}_{\square^1}$ — as for the uniform fibrations of [14]. This is forced, not chosen: a non-cartesian shrinking morphism ($\sigma = \text{id}$, $S' \subsetneq S$) would impose $\text{val}(b|, c) = \text{val}(b, c)$ for every box b extending the side data, hence $b(c) = b'(c)$ for any two extensions b, b' and any $c \in S \setminus S'$ — identifying old cells, contradicting property (2). The congruence pairs used in Step 3 of the proof of Lemma A.1 are cartesian, so the descent argument needs nothing more. (Morphisms with $\varepsilon' \neq \varepsilon$ exist only when σ factors through S , in which case every value of the small box is attached; they play no role.)

Lemma A.4 (Strictification). *Let $H \leq \square_{\mathcal{V}}([n], [n])^\times$ be a finite group of invertible substitutions and Y any presheaf. Precomposition with the quotient of the Yoneda embedding induces a bijection*

$$\widehat{\square}_{\mathcal{V}}(\square^n/H, Y) \cong \{s \in Y([n]) : s \cdot h = s \text{ for all } h \in H\},$$

and likewise $\widehat{\square}_{\mathcal{V}}((\square^n/H) \times \square^1, Y)$ is the set of elements of $Y([n+1])$ invariant under H acting on the first n coordinates. More generally, for a quotient by a presheaf congruence generated by pairs of parallel cells $A_i \sim B_i$, maps to Y are the elements s with $s \cdot A_i = s \cdot B_i$, and cylinders the elements one level up with $s \cdot (A_i \times \text{id}) = s \cdot (B_i \times \text{id})$.

Proof. The quotient is a coequalizer of representables and $[a] \times [b] = [a + b]$ in $\square_{\mathcal{V}}$; apply Yoneda and the universal property, levelwise. $\square$

Lemma A.5 (Equivariant descent). *Let a finite group G act on $\tilde{X}$, and let $\tilde{Z} \subseteq \tilde{X}$ be a G -stable subpresheaf with image $Z := \tilde{Z}/G \subseteq X$.*

- (1) Cells. *Stage-zero cells of $R_G(\tilde{X})$ at level $[n]$ are the G -orbits of cells of $\tilde{X}([n])$; every fresh cell is a filler value $\text{val}(b, c)$ of a box datum b over an earlier stage, presented by representable-shaped data via the pullback in the proof of Lemma A.1. Call b supported in Z if all its prescribed cells lie in the (inductively defined) Z -part of its stage.*

- (2) Subreplacements. *The canonical map $R(Z) \rightarrow R(X)$ is injective, and its image — the $\tilde{Z}$ -sector — consists exactly of the old cells of Z and the filler values of Z -supported boxes. Identifications among sector cells computed in $R(X)$ coincide with those of $R(Z)$.*
- (3) Projections. *A strict G -equivariant map $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ descends to $f : X \rightarrow \tilde{Y}/G$ and induces $R(f) : R_G(\tilde{X}) \rightarrow R_G(\tilde{Y})$, which carries the $\tilde{Z}$ -sector into the $\tilde{f}(\tilde{Z})$ -sector. Moreover, for any subgroup $H \leq \square_V([n], [n])^\times$ of invertible substitutions, $R(f)$ sends H -invariant n -cells to H -invariant n -cells, and likewise for cylinder homotopies; here invariance is under the source-coordinate substitution action, the notion produced by Lemma A.4, and is not to be confused with the covering group G , which acts on $\tilde{X}$ but has no residual action on the quotient replacement. In particular, for any quotient $\square^n/H$ (such as the Klein quotient $\square^2/K$ of Part 1), post-composition with $R(f)$ carries $[\square^n/H, R(X)]$ -data to $[\square^n/H, R(\tilde{Y}/H)]$ -data.*

Proof. (1) is the presentation of Lemma A.2 applied to X , combined with the pullback of boxes to representable shapes in the proof of Lemma A.1; quotients are levelwise, so stage zero is the orbit presheaf.

(2) Functoriality of the algebraic construction along the inclusion $Z \hookrightarrow X$ sends stages to stages and box data to box data, giving the canonical map, with the stated image. The substance is injectivity: two cells of $R(Z)$ identified in $R(X)$ must already be identified in $R(Z)$. By Lemma A.2(2), identifications in $R(X)$ are generated by the relations (i) and (ii). A generator of type (i) relates a filler value of a box b to a prescribed cell of the same b ; when b is Z -supported this is a relation of $R(Z)$. A generator of type (ii) can, however, relate two sector cells through a box b that is *not* Z -supported:

$$\text{val}(b \circ \hat{\sigma}_1, c_1) = \text{val}(b, \hat{\sigma}_1 c_1) = \text{val}(b, \hat{\sigma}_2 c_2) = \text{val}(b \circ \hat{\sigma}_2, c_2)$$

with $\hat{\sigma}_1 c_1 = \hat{\sigma}_2 c_2$ and both restricted boxes Z -supported. Such a step normalizes through the common restriction. Write the common cell as a pair of substitutions, $\hat{\sigma}_i c_i = \langle d_0, d_1 \rangle$ (the site has diagonals); then $c_i = \langle c_{i0}, d_1 \rangle$ with $\sigma_i \circ c_{i0} = d_0$, and relation (ii) applied *inside* each restricted box gives

$$\text{val}(b \circ \hat{\sigma}_i, c_i) = \text{val}((b \circ \hat{\sigma}_i) \circ \hat{c}_{i0}, \langle \text{id}, d_1 \rangle),$$

$$(b \circ \hat{\sigma}_1) \circ \hat{c}_{10} = b \circ \hat{d}_0 = (b \circ \hat{\sigma}_2) \circ \hat{c}_{20}$$

(with pulled-back box subpresheaves throughout), and $b \circ \hat{d}_0$, being a restriction of the Z -supported box $b \circ \hat{\sigma}_1$, is itself Z -supported. An arbitrary chain of generating relations between two sector cells is reduced by successively eliminating its maximal excursions through non- Z -supported boxes. Along an excursion, relations of type (ii) preserve the absolute position in the codomain of the ambient box, so an excursion that enters or leaves through a relation of type (i) has old cells at both ends sitting at the same position, and these are equal by the evaluation compatibility of restricted boxes (Lemma A.2(2)) — such an excursion is deleted outright. An excursion that

enters and leaves through relations of type (ii) is replaced by the common-restriction pair displayed above. Each elimination strictly decreases the number of non- Z -supported boxes visited, so the process terminates. Thus every identification between sector cells is generated by relations among Z -supported boxes, which are precisely the relations of $R(Z)$.

(3) Functoriality along the strict map f : $R(f)$ carries a Z -supported box to an $\tilde{f}(\tilde{Z})$ -supported box. For the invariance statement, $R(f)$ is a map of presheaves, hence commutes with the right action of every substitution $h \in H$ on cells; if $x \cdot h = x$ then $R(f)(x) \cdot h = R(f)(x \cdot h) = R(f)(x)$, and the same computation applies one level up for cylinder homotopies. The last statement is Lemma A.4. $\square$

The last two lemmas are stated at the generality at which the body uses them: a *machine* is a presheaf decomposed into two closed sieves — the *sectors* — whose intersection is the *interface*, with a group or congruence action preserving the decomposition; the running examples are the witness machines J_N (body and cylinder sectors) and the quotients $\square^n/N$ themselves (stabilized sieve and free part).

Lemma A.6 (Equivariant filler slide). *Let a cylinder sector $\bar{C} = (\Sigma \times E \times \square^1)/\sim$ (the end $\tau = 1$ collapsed) sit inside a machine over a reversal site, with its group action. The slide $(x, e, \tau) \mapsto (x, e, \tau \vee z)$ is a totally defined, equivariant strict homotopy from the identity of $\bar{C}$ to the projection onto the collapsed end, and it is compatible with the replacement: it extends to the cylinder subreplacement, carrying filler values to filler values.*

Proof. The connection $\vee$ is a site operation, so the slide is a strict map $\bar{C} \times \square^1 \rightarrow \bar{C}$; its ends are the identity ($z = 0$) and the collapse ($z = 1$, using the collapsed end), and it commutes with the action, which does not touch the cylinder coordinate. Compatibility with fillers is relation (ii) of Lemma A.2 applied to the substitution $\tau \mapsto \tau \vee z$: the slide of a filler value is the filler value of the slid box. $\square$

Lemma A.7 (First exit). *In the replacement of a machine, every invariant chain of cells and one-step homotopies beginning at a class supported in one sector either remains in that sector's subreplacement (Lemma A.5(2)) or contains a first step whose box is not supported in the sector; at that step the chain passes through a class supported in the interface. Consequently every class is homotopic to a class supported in one of the two sectors, through interface classes, and the homotopy analysis reduces to the sector subreplacements together with the interface.*

Proof. Grammar. By Lemma A.2, every class of the replacement is presented by old cells and filler values $\text{val}(b, c)$; identifications are generated by the relations (i)–(ii); one-step homotopies are end restrictions of filler values of homotopy boxes, by strict naturality $\text{val}(b, c) \cdot f = \text{val}(b, c \cdot f)$; and the prescribed cells of a box live at strictly earlier stages. For a closed sieve S of

the machine write $P_S(\beta)$ for the S -part of stage β : the old cells lying in S together with the values of S -supported boxes; the parts are closed under the presheaf action (faces of old S -cells stay in the sieve; $\text{val}(b, c) \cdot f = \text{val}(b, c \cdot f)$ stays in the same box), and by Lemma A.5(2) they assemble to the subreplacement $R(S)$. The sectors S_1, S_2 cover the cells of the machine, with interface $I = S_1 \cap S_2$.

Two structural facts. (a) A relation of type (ii) relates two presentations $\text{val}(b \circ \hat{\sigma}, c) = \text{val}(b, \hat{\sigma}c)$ of one class and preserves the *absolute position* $\hat{\sigma}c$ in the codomain of the ambient box b . Consequently the purely fresh presentations of a single class all lie in the restriction system $\{b \circ \hat{\sigma}\}$ of one ambient box at one absolute position e , and two such presentations $\text{val}(b \circ \hat{\sigma}_1, c_1) = \text{val}(b \circ \hat{\sigma}_2, c_2)$, $\hat{\sigma}_1 c_1 = e = \hat{\sigma}_2 c_2$, are merged by the *common restriction*: writing $e = \langle d_0, d_1 \rangle$ as a pair of substitutions (the site has diagonals), so that $c_i = \langle c_{i0}, d_1 \rangle$ with $\sigma_i \circ c_{i0} = d_0$, relation (ii) applied *inside* each restricted box gives

$$\begin{aligned} \text{val}(b \circ \hat{\sigma}_i, c_i) &= \text{val}((b \circ \hat{\sigma}_i) \circ \hat{c}_{i0}, \langle \text{id}, d_1 \rangle), \\ (b \circ \hat{\sigma}_1) \circ \hat{c}_{10} &= b \circ \hat{d}_0 = (b \circ \hat{\sigma}_2) \circ \hat{c}_{20} : \end{aligned}$$

the class is a value of the single restricted box $b \circ \hat{d}_0$, a restriction of *both* entry boxes, and this box is the replacement for the excursion. (b) A relation of type (i) identifies a presentation with an old cell; a chain of generators connecting values of *different* ambient systems must pass through an old cell, since (ii) never leaves a system.

The interface identity $P_{S_1}(\beta) \cap P_{S_2}(\beta) = P_I(\beta)$, *by induction on the stage.* At stage zero this is $S_1 \cap S_2 = I$. Let a class at stage $\beta + 1$ have presentations $\text{val}(b_1, c_1)$ and $\text{val}(b_2, c_2)$ with b_a supported in $P_{S_a}(\beta)$. If the connecting chain of generators passes through an old cell x , then the class is old, and it lies in each sieve (x belongs to the part of S_a iff it lies in S_a), hence in I . Otherwise, by (b) both presentations lie in one ambient system, by (a) at one absolute position, and the common restriction $b \circ \hat{d}_0$ displayed above is a restriction of a $P_{S_1}(\beta)$ -supported box and of a $P_{S_2}(\beta)$ -supported box; its prescribed cells therefore lie in $P_{S_1}(\beta) \cap P_{S_2}(\beta) = P_I(\beta)$ by the stage induction, so $b \circ \hat{d}_0$ is I -supported and the class lies in $P_I(\beta + 1)$. The reverse inclusion is immediate.

Splitting a mixed box. Let b be a box over stage β and $j_a \subseteq \text{dom}(j)$ the subpresheaf of positions whose prescribed cells lie in $P_{S_a}(\beta)$. Since b is a map of presheaves and the parts are closed under the action, j_1 and j_2 are closed subshapes; they cover $\text{dom}(j)$, the sieves covering the machine; and the prescription over $j_1 \cap j_2$ lies in $P_{S_1}(\beta) \cap P_{S_2}(\beta) = P_I(\beta)$. Every box is thus the pasting of the S_1 -supported restriction $b|_{j_1}$ and the S_2 -supported restriction $b|_{j_2}$ along the I -supported common restriction $b|_{j_1 \cap j_2}$, and by relation (ii) the values of each restriction are cells of the corresponding sector, those of $b|_{j_1 \cap j_2}$ cells of the interface subreplacement.

First exit. Consider an invariant chain of cells and one-step homotopies beginning at a class supported in S_1 . If every step's box is S_1 -supported the chain lies in $R(S_1)$, by Lemma A.5(2). Otherwise take the first step whose box b is not S_1 -supported. The chain's positions adjacent to the step are old cells or values in b 's restriction system. A crossing through an old cell lies in $S_1 \cap S_2 = I$ by closedness of the sieves. A crossing between a class of the S_1 -part and a class of the S_2 -part inside the system is merged, by the displayed common restriction, into a value of a box that is a restriction both of an S_1 -supported and of an S_2 -supported box — an I -supported box by the interface identity — so the chain is rewritten to pass through a class of the interface subreplacement at that step, entering through $b|_{j_1}$ and leaving through $b|_{j_2}$ of the splitting. Each such rewriting strictly decreases the number of non-sector-supported boxes the chain visits, so iterating terminates and reduces the chain to sector chains linked through interface classes, which is the statement. $\square$

APPENDIX B. THE SUPERSEDED COHOMOLOGICAL ROUTE

This appendix documents, for the record, the conditional route that preceded the sieve-landing dichotomy — the following theorem and Proposition B.3 are superseded by Theorem 15.10, but the low-dimensional $\mathbb{F}_2$ -computations of Proposition B.3 remain of independent interest for the arrangement dichotomy of the outlook, and the collage-formula evaluations they contain are cited from the body.

Theorem B.1 (Separation at reflection-generated groups; superseded). *Under Hypothesis 14.4, $\bar{\Phi}_N \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$ for every reflection-generated mixed N .*

Proof. Test: Proposition 14.2. Type: if $\bar{\Phi}_N$ were a type weak equivalence, the identity of $\square^n/N$ would lift along it up to homotopy, hence by the trichotomy of Theorem 14.3 the identity would be homotopic to a constant, to a class factoring through Σ/N , or to a strictly invariant endomorphism class. The first two are excluded by Hypothesis 14.4(i) after localizing to the test side (contractibility being the empty-sieve case of domination), and the third by Hypothesis 14.4(ii). $\square$

Lemma B.2 (Degree-one generation). *If Y is a homotopy retract of X , with section s and retraction r , then $s^*: H^*(X; \mathbb{F}_2) \rightarrow H^*(Y; \mathbb{F}_2)$ is a surjective ring homomorphism. In particular, if $H^*(X; \mathbb{F}_2)$ is generated as a ring in degrees ≤ 1 , then so is $H^*(Y; \mathbb{F}_2)$; if moreover $H^1(Y; \mathbb{F}_2) = 0$, then $\tilde{H}^*(Y; \mathbb{F}_2) = 0$.*

Proof. $(r \circ s)^* = s^* \circ r^* = \text{id}$, so s^* is surjective, and it is a ring map because it is induced by a map of spaces. A surjective ring map carries a generating set onto a generating set; if the target has no nonzero elements in degree one, the images of the degree- ≤ 1 generators lie in H^0 , so the reduced part of the target vanishes. $\square$

Proposition B.3 (Hypothesis 14.4 in low dimensions). *Hypothesis 14.4 holds for every base occurring in the $n \leq 3$ classification.*

Proof. The bases are the Klein block groups, the A_4 -type group, and the nested order-24 group. (i) is settled uniformly by Lemma B.2. Every component of the stabilized sieve quotient is a point or a $B\mathbb{Z}/2$: for the Klein group, $\text{el}(\Sigma/K) \simeq B\mathbb{Z}/2 \sqcup B\mathbb{Z}/2$ (two isolated reflection loci); for the A_4 -type group, $\Sigma/N \cong \square^1$, so $\text{el}(\Sigma/N) \simeq *$; for the nested group, the plane stratum contributes $\text{el}(\square^2/R) \simeq B\mathbb{Z}/2$ (the residual R is a free reversal), the line stratum a point, and the gluing is along a point, so $\text{el}(\Sigma/N) \simeq B\mathbb{Z}/2$. In each case $H^*(\cdot; \mathbb{F}_2)$ is generated in degrees ≤ 1 (as $\mathbb{F}_2[t]$, or trivially), and a connected retract of a disjoint union retracts through one component. On the other side, the collage of Theorem 10.1 computes, by exact $\mathbb{F}_2$ -cochain models (normalized bar complexes and iterated homotopy pushouts; artifact):

$$\begin{aligned} H^*(\text{el}(\square^2/K)) &= (1, 0, 0, 1, 2, 3, \dots), & H^*(\text{el}(\square^3/N_{A_4})) &= (1, 0, 1, 2, 1), \\ H^*(\text{el}(\square^3/N_{24})) &= (1, 0, 0, 1, \dots), \end{aligned}$$

the first being the join $B\mathbb{Z}/2 \star B\mathbb{Z}/2$, the second equal to $H^*(BN_{A_4})$ (the odd comma $B(\mathbb{Z}/3)$ is $\mathbb{F}_2$ -acyclic), the third assembled from the graph-of-groups comma $B(\mathbb{Z}/2 \times \mathbb{Z}/2) \leftarrow B(\mathbb{Z}/2) \rightarrow B(S_3)$. In all three, $H^1 = 0$ while $\tilde{H}^* \neq 0$ (in degree 3, 2, 3 respectively), so by Lemma B.2 none is a homotopy retract of its stabilized sieve quotient, nor contractible. (ii) is the stage-zero verification of Remark 14.5 (artifact). $\square$

APPENDIX C. THE APPROXIMATION LADDER

This appendix develops the mechanisms behind Theorem 24.51 and Conjecture 24.52: the combinatorial structure of the singular functor, the filling technology, and the assembly analysis with its machine calibrations.

Lemma C.1 (The singular functor is the sorted part). *For every Dedekind cubical set X there is a natural isomorphism*

$$\text{Sing}_D(X)_q \cong \{x \in X([q]) : x \cdot \text{sort}_q = x\}, \quad \text{sort}_q = (o_1, \dots, o_q),$$

the cells fixed by the sort substitution, with the simplicial operators acting by $x \mapsto x \cdot T(f)$ (instances of sorted cells along the triangulation are sorted). Thus Sing_D is restriction along the triangulation of Lemma 24.44. In particular Sing_D preserves products, $\text{Sing}_D(\square^n) \cong (\Delta^1)^n$ is the standard simplicial cube, and the singular complex of the cubical circle $\square^1/(v_0 \sim v_1)$ is the simplicial circle $\Delta^1/\partial\Delta^1$.

Proof. $\Delta_\square^q$ is the image of the idempotent substitution sort_q (Lemma 24.44), so maps out of it correspond to cells fixed by precomposition with sort_q ; the operator formulas are those of the triangulation, and sortedness is preserved because $T(f) \circ \text{sort}$ -identities hold at $\{0, 1\}$ -points. Products are preserved by any right adjoint; $\text{Sing}_D(\square^1)$ has $q+2$ cells at level q (the sorted elements

of $D(q)$ are the order statistics and the constants) with the operators of Δ^1 . Machine check of the fixed-point description, the closure, and the counts (artifact, `dedekind_singsort.py`). $\square$

Proposition C.2 (The type structure is cartesian). *The Leibniz product of a monomorphism with an open box is an open box: $n \hat{\otimes} (m \hat{\otimes} \delta^\varepsilon) = (n \hat{\otimes} m) \hat{\otimes} \delta^\varepsilon$. Hence the cubical-type model structure is a cartesian closed model structure, and for fibrant X the canonical map $X \rightarrow X^{\Delta_\square^q}$ is a type equivalence.*

Proof. Associativity of the Leibniz construction gives the displayed identity, and Leibniz products of monomorphisms are monomorphisms (proof of Lemma 24.15); the pushout-product axiom follows by saturation, and the unit is the terminal object, which is cofibrant. For the last claim, a vertex $e \hookrightarrow \Delta_\square^q$ is a trivial cofibration between cofibrant objects, so the induced map of cotensors of a fibrant object is a trivial fibration, and it retracts the displayed map. $\square$

Remark C.3 (The resolution form of the counit). In the simplicial completion the counit becomes a roof: with $R(X)_q := X^{\Delta_\square^q}$ the cotensor resolution, the projections give $\mathrm{pr}_2^* \mathrm{Sing}_D X \rightarrow R(X) \leftarrow \mathrm{pr}_1^* X$, the right leg a levelwise type equivalence for fibrant X (Proposition C.2); the counit is a type equivalence iff the left leg lies in the simplicial completion of $\mathbf{W}_{\mathrm{type}}$. The left leg is *not* a levelwise equivalence — at each level it compares the discrete set of sorted cells with the whole cotensor — and the diagonal cancellation across the two directions is precisely what a filling argument must supply: the exact analogue of the classical proof that $|\mathrm{Sing} Y| \rightarrow Y$ is a weak equivalence.

Two combinatorial lemmas begin the filling theory that the counit demands.

Lemma C.4 (Sort exchange). *A lattice polynomial evaluated on a chain of arguments returns an argument or a constant. Consequently, for every substitution $\sigma: [k] \rightarrow [q]$ the sorted instance $\sigma \diamond \mathrm{sort}_k$ has all its components in the chain $\{0, o_1, \dots, o_k, 1\}$, and factors as*

$$\sigma \diamond \mathrm{sort}_k = \rho_\sigma \diamond T(f_\sigma),$$

with $T(f_\sigma)$ a simplicial operator (the decreasing rearrangement of the components) and ρ_σ a variable pattern (each component a variable or a constant). After sorting, every cubical operator is a pattern followed by a simplicial one. (Machine sweep with no exceptions through $k, q \leq 3$; artifact, `dedekind_sortexchange.py`.)

Proof. Meets and joins of two chain elements are chain elements, so the first claim follows by induction on the polynomial; constants are absorbed as $o_0 = 1$ and $o_{k+1} = 0$. The components of $\sigma \diamond \mathrm{sort}_k$ are then chain values; their decreasing rearrangement is a decreasing chain-valued tuple, which is exactly an operator in the image of the triangulation (Lemma 24.44), and the original order is restored by the variable pattern that places each component at its rank. $\square$

Proposition C.5 (The fundamental groupoid lives in the sorted part). *For every Dedekind cubical set X , the groupoid generated by the vertices and edges of X modulo the sorted-triangle relations $[s \cdot (t, t)] = [s \cdot (1, t)] \circ [s \cdot (t, 0)]$, s a sorted 2-cell, coincides with the same groupoid modulo the square relations $[h \cdot (t, 1)] \circ [h \cdot (0, t)] = [h \cdot (1, t)] \circ [h \cdot (t, 0)]$, h any 2-cell. Hence Sing_D induces an isomorphism of fundamental groupoids, naturally in X : the $\pi_{\leq 1}$ -rung of the approximation ladder of Theorem 24.51 holds for all X .*

Proof. Both directions are connection computations, with the face identities machine-checked (artifact, `dedekind_sortexchange.py`). For a 2-cell h , its two sorted triangles $h \cdot (x_1 \vee x_2, x_1 \wedge x_2)$ and $h \cdot (x_1 \wedge x_2, x_1 \vee x_2)$ are sort-fixed with simplicial faces (right, diagonal, bottom) and (top, diagonal, left) respectively, so the two triangle relations compose to the square relation of h . Conversely, for a sorted 2-cell s the *connection square* $h_1 := s \cdot (x_1, x_1 \wedge x_2)$ has faces top = the diagonal of s , bottom = its bottom, right = its right, and left degenerate, so the square relation of h_1 is precisely the triangle relation of s . $\square$

Proposition C.6 (The chain filling lemma). *Let $\text{Ch}_n \subseteq \square^n$ be the set of cells whose components are pairwise comparable. Then Ch_n is a subpresheaf, strictly contractible via the $\wedge$ -cone, and the inclusion $\text{Ch}_n \hookrightarrow \square^n$ is a type-trivial cofibration. Consequently every map $\text{Ch}_n \rightarrow Z$ with Z fibrant extends to $\square^n$: fibrant objects extend chain data to cubes in one lift. For $n = 2$, Ch_2 is the union of the two chamber simplices glued along the diagonal, so two sorted 2-cells sharing their diagonal always extend to a square with the four prescribed sides. Every sorted cell is a chain cell, and every chain cell is a variable-pattern instance of its decreasing rearrangement, which is sort-stable.*

Proof. Comparability of components is preserved by substitution ($p \leq q$ implies $p(\vec{v}) \leq q(\vec{v})$) and by the cone ($p \leq q$ implies $p \wedge t \leq q \wedge t$), so Ch_n is a subpresheaf and the $\wedge$ -cone of the cube restricts to it; both Ch_n and $\square^n$ are then strictly contractible and the monic inclusion is a weak equivalence, i.e. a trivial cofibration. The identification of Ch_2 , the containment of the sorted cells, and the sort-stability of decreasing rearrangements are checked by machine at $n \leq 3$, $k \leq 3$ and hold in general by the pointwise arguments above (artifact, `dedekind_chaincells.py`). $\square$

Remark C.7 (The ladder above π_1). Propositions C.5 and C.6 are the pattern for the approximation ladder of Theorem 24.51. Given a simplicial map $f: \text{Sing } X \rightarrow \text{Sing } Z$ with Z fibrant, a candidate $g: X \rightarrow Z$ is built by induction on cell level: in levels ≤ 1 every cell is sorted and $g := f$; at a cell x of level n the sorted shadow $\{f(x \cdot (\rho \diamond \text{sort}))\}$ is chain data in Z , and the chain filling lemma provides the extension $g(x)$. What the ladder still requires is the *coherence* of these extensions — naturality in x , to be enforced through the uniformity of the Kan structure and the functorial fillers of the algebraic replacement — and the corresponding injectivity statements; the

counit factors through the subobject of chain-instances of sorted cells, and the intermediate goal is the *chain approximation lemma*: for fibrant X the inclusion of that subobject is a type equivalence.

Lemma C.8 (The sort zigzag). *The adjacent merge $s_i = (\dots, x_i \vee x_{i+1}, x_i \wedge x_{i+1}, \dots)$ is joined to the identity by the zigzag of strict homotopies*

$$\begin{aligned} H_1 &= (x_i \vee (x_{i+1} \wedge t), x_{i+1}): \text{id} \Rightarrow (x_i \vee x_{i+1}, x_{i+1}), \\ H_2 &= (x_i \vee x_{i+1}, x_{i+1} \wedge (x_i \vee t)): s_i \Rightarrow (x_i \vee x_{i+1}, x_{i+1}), \end{aligned}$$

and by no single homotopy: at the vertex $(x_i, x_{i+1}) = (0, 1)$ the second component would have to fall while t rises, contradicting monotonicity. Composing along a bubble sort, sort_n is joined to the identity by a natural zigzag of strict homotopies; hence in every X , every cell is joined to its sorted cell by a natural zigzag.

Proof. The endpoint identities are machine-checked (artifact, `dedekind.sortzigzag.py`) and immediate: $x \vee (y \wedge 0) = x$, $x \vee (y \wedge 1) = x \vee y$, $y \wedge (x \vee 0) = x \wedge y$, $y \wedge (x \vee 1) = y$. All formulas are substitution homotopies, so they are natural in every presheaf and every cell. $\square$

Lemma C.9 (Cylinders are compatible). $\text{Sing}_D(X \times \square^1) = \text{Sing}_D X \times \Delta^1$ naturally. Consequently Sing_D carries cubical homotopies to simplicial homotopies, and the injectivity half of the full faithfulness in Theorem 24.51 is an instance of the following relative extension problem, applied to the cofibration $X \sqcup X \hookrightarrow X \times \square^1$:

($\star$) *given a cofibration $A \hookrightarrow X$, a map $a: A \rightarrow Z$ with Z fibrant, and a simplicial map $f: \text{Sing } X \rightarrow \text{Sing } Z$ extending $\text{Sing}(a)$, produce $g: X \rightarrow Z$ extending a with $\text{Sing } g \simeq f$ relative to A .*

Proof. Right adjoints preserve products and $\text{Sing}_D(\square^1) = \Delta^1$ (Lemma C.1). A simplicial homotopy $\text{Sing } g_0 \simeq \text{Sing } g_1$ is a map $\text{Sing}(X \times \square^1) \rightarrow \text{Sing } Z$ restricting to the $\text{Sing } g_i$ at the ends, i.e. an instance of ($\star$) whose solution is a cubical homotopy $g_0 \simeq g_1$. $\square$

Lemma C.10 (Sort absorption and the chain stratum). $\text{sort} \diamond \text{sort}^{\text{op}} = \text{sort}$ and $\text{sort}^{\text{op}} \diamond \text{sort} = \text{sort}^{\text{op}}$: every sorted cell is fixed by every sort idempotent, so both chamber restrictions $x \cdot \text{sort}_2$, $x \cdot \text{sort}_2^{\text{op}}$ of a 2-cell are simplices of the singular complex, with automatic diagonal agreement — the canonical Ch_2 -datum of Proposition C.6 always exists. Moreover instances absorb sorting, $s \cdot u = s \cdot (\text{sort} \diamond u)$ for sorted s , so the subpresheaf of instances of sorted cells — the image of the counit — consists exactly of the chain instances, and a chain instance is itself reached only through chain data: at level 2 of the universal case, 34 of the 400 cells are chain instances, all of them chain cells, and none of the remaining cells admits a forced value (artifact, `dedekind.protocol.py`).

Proof. The identities are instances of the chain-evaluation principle of Lemma C.4 (order statistics of a chain are its members), as is the absorption; the level-2 statistics are the machine sweep. $\square$

Lemma C.11 (The sort retraction). *Write $T(X)$ for the simplicial set with q -simplices $X([q])$ and faces the wall restrictions (the restriction of X along the triangulation, computing the test homotopy type), and give $\text{Sing}_D X$ its simplicial structure by the $T(\delta^i)$. Then*

$$r_X : T(X) \longrightarrow \text{Sing}_D X, \quad c \longmapsto c \cdot \text{sort},$$

is a natural simplicial map restricting to the identity on sorted cells. Indeed $T(\delta^i)$ is the wall coface followed by the sort (order-statistic composition), and precomposing with sort_q is absorbed, $\text{sort}_q \diamond T(\delta^i) = T(\delta^i)$, because the components of $T(\delta^i)$ form a decreasing tuple of order statistics and order statistics of a decreasing tuple are its entries (Lemma C.4); machine confirmation through $q \leq 4$ and on the folded quotient $W_{\wedge\vee}$ (artifact, `dedekind.universal.py`). The singular complex is thus a natural simplicial sort-retract of the total triangulation.

Lemma C.12 (Cylinder pairing for the triangulation). *The assignment $(c, g) \mapsto (c, \hat{g})$, where $\hat{g} := x_1 \wedge \cdots \wedge x_{t(g)}$ is the prefix meet at the threshold of $g \in \Delta_q^1$ (the empty meet being the constant 1 and the all-zero simplex going to the constant 0), is a simplicial map $T(X) \times \Delta^1 \rightarrow T(X \times \square^1)$: the outer walls shift the threshold correctly and the inner walls duplicate a variable, which the meet absorbs (machine check through $q \leq 4$; artifact, `dedekind.singtest.py`). Hence T carries strict cylinder homotopies to simplicial homotopies; in particular $T(\square^n)$ is contractible, as is T of every strictly contractible presheaf.*

Theorem C.13 (The singular complex computes the test type). *For every Dedekind cubical set X the sort retraction $r_X : T(X) \rightarrow \text{Sing}_D X$ is a weak equivalence of simplicial sets. Consequently:*

- (1) $\text{Sing}_D X$ has the test homotopy type of X , with no fibrancy or shape hypothesis;
- (2) a map of Dedekind cubical sets is a test equivalence if and only if its image under Sing_D is a simplicial weak equivalence;
- (3) the counit $\varepsilon_X : \text{Real}_D \text{Sing}_D X \rightarrow X$ is a test equivalence for every X ; wherever it is injective on cells — statement (A) — the chain stratum inclusion $\text{Ch}(X) \hookrightarrow X$ is a test equivalence.

Proof. Fix a simplicial degree q and consider two set-valued diagrams on the category of elements $\int X$: the full diagram $F(n, x) := \square^n([q])$ and the sorted subdiagram $F^s(n, x) := \text{Sing}(\square^n)_q$, with transition along $v : (m, y) \rightarrow (n, x)$ given by $w \mapsto v \diamond w$ (which preserves sortedness, since $(v \diamond w) \diamond \text{sort} = v \diamond (w \diamond \text{sort})$). The colimits are $X([q])$ and $\text{Sing}_D X_q$; for the sorted diagram, surjectivity holds because $s = x \cdot u$ implies $s = s \cdot \text{sort} = x \cdot (u \diamond \text{sort})$ with $u \diamond \text{sort}$ sorted.

The homotopy colimit of a set-valued diagram is the nerve of its category of elements, and the comparison with the discrete colimit is a weak equivalence precisely when each fiber category — the category P_c of presentations of a fixed colimit element — is aspherical. Both fiber categories here are *contractible*, by one and the same device. For $c \in X([q])$ the object $\omega_c := ([q], c, \text{id}_{[q]})$ lies in the full fiber, and for sorted s the object $\omega_s := ([q], s, \text{sort}_q)$ lies in the sorted fiber (sort_q is a sorted cell of $\square^q$ and $s \cdot \text{sort} = s$). For any object $o = (n, x, u)$ of the fiber, the substitution u itself is a morphism $\omega \rightarrow o$: it carries s to x ($x \cdot u = s$) and satisfies $u \diamond \text{id} = u$, respectively $u \diamond \text{sort} = u$ by sortedness. The assignment $\eta_o := u$ is a *natural transformation* from the constant functor at ω to the identity: for a fiber morphism $v: (m, y, w) \rightarrow (n, x, u)$, the required identity $v \circ \eta_{(m, y, w)} = \eta_{(n, x, u)}$ is exactly the defining condition $v \diamond w = u$. A natural transformation from a constant functor contracts the nerve, so both fibers are contractible.

Therefore both vertical comparisons in the square

$$\begin{array}{ccc} \text{hocolim}_{\int X} T(\square^n) & \longrightarrow & \text{hocolim}_{\int X} \text{Sing}(\square^n) \\ \downarrow & & \downarrow \\ T(X) & \xrightarrow{r_X} & \text{Sing}_D X \end{array}$$

are levelwise weak equivalences (assembled over q by the realization lemma for bisimplicial sets), the top map is a weak equivalence because each $r_{\square^n}$ is a map between contractible simplicial sets (Lemma C.12; $\text{Sing}(\square^n) = (\Delta^1)^n$ by Lemma C.1), and the square commutes by naturality of r . Hence r_X is a weak equivalence. For the consequences: (1) follows since $T(X) \simeq \text{hocolim}_{\int X} \text{pt} = N(\int X)$, the test homotopy type; (2) is naturality of r and two-out-of-three; (3) the unit $\eta_K: K \rightarrow \text{Sing Real } K$ is a weak equivalence for every simplicial set K — by (1) applied to $\text{Real } K$, whose test type is K since Real_D is the left half of the test-structure Quillen equivalence — and the triangle identity $\text{Sing}(\varepsilon_X) \circ \eta_{\text{Sing } X} = \text{id}$ then forces $\text{Sing}(\varepsilon_X)$ to be a weak equivalence, which by (2) makes ε_X a test equivalence. $\square$

Corollary C.14 (Chain strata of test-contractible shapes). *Let X be test-contractible and suppose the counit is injective on cells at X (statement (A), e.g. X a subobject of a representable). Then $\text{Ch}(X)$ is type-contractible: by Theorem C.13, $\text{Sing}_D X$ is a contractible simplicial set, and Real_D carries the weak equivalence $\text{Sing}_D X \rightarrow \Delta^0$ to a type equivalence (Proposition 24.45), so $\text{Ch}(X) \cong \text{Real}_D \text{Sing}_D X$ is type-contractible. In particular the chain stratum of every test-contractible cellular shape satisfying (A) is type-contractible — for such shapes the entire contraction question of the phase diagram is carried by the cells outside the chain stratum.*

Remark C.15. Theorem C.13 settles, at one stroke and on the test side, the questions that drove the machine sweeps: the equality of T -Betti and singular Betti numbers observed on every shape and every quotient is now a theorem; the sorted-presentation asphericity behind the colimit comparison

is the contractibility of the fiber categories; and the chain stratum of *any* presheaf satisfying (A) is test-equivalent to it — fibrancy plays no role at the test level. What the theorem does *not* decide is exactly the phase-diagram content: whether the counit at a fibrant object is a *type* equivalence (Conjecture 24.52); the reduction of Remark 24.64 and the box-preservation program remain the designated route for that upgrade.

Proposition C.16 (The extension engine). *Let Z be fibrant and $\varphi: \text{Ch}_n \rightarrow Z$. The pair $(\varphi$ composed with the $\wedge$ -cone on $\text{Ch}_n \times \square^1$; the constant $\varphi(\vec{0})$ on $\square^n \times \{0\})$ is a box for the generating open box $(\text{Ch}_n \hookrightarrow \square^n) \hat{\otimes} \delta^0$, and the top slice of its filler, $E(\varphi) := \text{val}(\text{box}, -)|_{\square^n \times \{1\}}$, extends φ to $\square^n$. The operator E is strictly natural, uniform, and equivariant under coordinate permutations, since the $\wedge$ -cone is a symmetric formula and the filling structure satisfies Lemma A.2(2)–(3). No cellular decomposition of $\text{Ch}_n \hookrightarrow \square^n$ is needed.*

Proof. The two pieces agree on the overlap $(a \wedge 0 = \vec{0})$, the cone preserves Ch_n (Proposition C.6), and the inclusion is a monomorphism, so the datum is a box for a generating trivial cofibration and val applies. At $t = 1$ the cone is the identity, so the top slice restricts to φ on Ch_n . For a permutation π , the π -translate of the box is the box of the π -translated datum (symmetry of $\wedge$), and uniformity (Lemma A.2(1)(ii)) identifies the fillers. $\square$

Theorem C.17 (Normal forms of chain cells; T -connectivity). *In any X , every chain instance $y = s \cdot u$ (s sorted, u chain-valued) has a presentation $y = s'' \cdot d$ with s'' sorted and d a decreasing chain-valued substitution: factor $u = \rho \diamond \text{sort}_r \diamond \vec{c}$ by Lemma C.4 and absorb the pattern, $s'' := s \cdot (\rho \diamond \text{sort})$, sorted by Lemma C.10. The presentations of a fixed cell are permuted by the T -moves $(s \cdot T(h), d) \sim (s, T(h) \diamond d)$, and the presentation groupoid is connected: every presentation reduces to the unique minimal one, whose entries are the distinct non-constant values of the components of y in strictly decreasing order.*

Proof. Reduction to the strict form. Given (r, s, d) , let d^{red} be the strictly decreasing tuple of distinct non-constant values of the entries of d , and let h_d be the ordinal pattern recording, for each position of d , the index of its value (constant entries go to the 1-prefix and 0-suffix of the pattern). Then $d = T(h_d) \diamond d^{\text{red}}$, since an order statistic of a strictly decreasing tuple is its entry, and one T -move lands in $(|d^{\text{red}}|, s \cdot T(h_d), d^{\text{red}})$.

Junk-hub moves. Let $(p, s', \vec{v})$ be strict and suppose the index j is not used by s' (no component of s' is o_j). Consider the position-ignoring pattern T'' with index sequence $(1, \dots, j-1, j-1, j+1, \dots, p)$ — nondecreasing, hence an ordinal operator — and note $s' \cdot T'' = s'$, because T'' agrees with the sort on every index other than j . For *any* entry w with $v_{j-1} \geq w \geq v_{j+1}$, the tuple $\vec{v}^b$ obtained by replacing v_j with the duplicate v_{j-1} factors as $T'' \diamond (\vec{v}$ with w in position j), since T'' never reads position j ; so the single hub presentation $(p, s', \vec{v}^b)$ is T -connected to $(p, s', \vec{v}[w/j])$ for every w —

in particular to the original. The hub itself has a duplicate entry, which a degeneracy pattern removes, deleting the index j altogether.

Termination. Each junk-hub round strictly decreases the length; when every index is used and the tuple is strict and non-constant, the presentation is the minimal one, which is determined by y . Hence any two presentations are connected through the minimal one. (Machine confirmation: all chain cells of $\square^2$, $\square^3$ at target levels ≤ 2 , sources of level ≤ 3 , no disconnected instance; artifact, `dedekind_normalform.py`.) $\square$

Lemma C.18 (The chain dictionary). *For all Dedekind cubical sets X and Z , restriction to sorted cells is a bijection, natural in both variables,*

$$\{\text{maps } \text{Ch}(X) \rightarrow Z\} \cong \{\text{simplicial maps } \text{Sing}_D X \rightarrow \text{Sing}_D Z\},$$

with inverse $f \mapsto (s \cdot d \mapsto f(s) \cdot d)$. The inverse is well defined by the connectivity of the presentation groupoid: for a T -move, $f(s \cdot T(h)) \cdot d = (f(s) \cdot T(h)) \cdot d = f(s) \cdot (T(h) \diamond d)$ since a simplicial map commutes with all simplicial operators (Theorem C.17); it is a presheaf map on the chain stratum because chain substitutions compose with arbitrary ones into chain substitutions. Consequently the comparison criterion of Theorem 24.51 is accessible through $[X, Z] \rightarrow [\text{Ch}(X), Z] \cong [\text{Sing } X, \text{Sing } Z]$ — the cell-injectivity statement (A) is not needed for it, only for the geometric identification of $\text{Ch}(X)$ with the realization.

Remark C.19 (The protocol principle, refined). The surjectivity half of $(\star)$ has a designed route to *strict* naturality. Setting $g := f$ on sorted cells, extending by forced values on the chain stratum, and filling the remaining cells by $g(x) := \text{val}(B(x), \text{top})$ for a box $B(x)$ assembled from the canonical chamber data and the lower cells, three exact mechanisms cooperate: the fillers of the uniform Kan structure are strictly natural and uniform (Lemma A.2(2)–(3)); the sorted shadow is natural in x (Lemma C.4); and the forced stratum is cleanly separated from the filling stratum (Lemma C.10). The residual obstruction is now precisely located: order statistics are *symmetric* functions, so coordinate patterns act invisibly on the simplices of cubes, but on a general X the pattern instances $s \cdot \rho$ of a sorted cell are not sorted and are not seen by Sing_D — the singular complex does not determine the chain stratum, and the missing datum is the *pattern (symmetric) structure*, which the sort zigzag (Lemma C.8) trivializes up to natural homotopy. Decisively, the criterion of Theorem 24.51 needs only the *canonical* instance of the protocol: it suffices to produce a section $\sigma: X \rightarrow \mathbb{R}(\text{Real}_D \text{Sing}_D X)$ inverse to the counit up to homotopy, and there the pattern data exists explicitly, as the presheaf structure of the realization. On the chain stratum σ is the canonical value on normal presentations — well-defined wherever the presentation groupoid is T -connected (Lemma C.17) — and on the remaining cells it is supplied by the extension engine (Proposition C.16), which is strictly natural and permutation-equivariant. What separates the ladder from Theorem 24.51 is now concrete. T -moves preserve the prescription

$\sigma(s \cdot d) := \text{real}(s) \cdot d$ (the realization is T -natural), every cell of $\text{Real}_D \text{Sing}_D X$ is a chain instance of a realized simplex, so $\sigma\varepsilon' = \text{id}$ holds *strictly* on the realization and $\sigma\varepsilon' \simeq \text{id}_Y$ follows automatically from the uniqueness of extensions along the trivial cofibration into the replacement. Decisively, the pattern instances cause no wall: although a single T -move does not relate $\text{real}(s \cdot (\rho \diamond \text{sort}))$ to $\text{real}(s) \cdot (\rho \diamond \text{sort})$, *chains* of T -moves through the hubs of Theorem C.17 do — these are two presentations of one value, and the connectivity theorem joins them. A machine probe on the monotone dunce hat itself — a genuinely folded quotient — confirms it: the cells of $\text{Real}_D \text{Sing}_D W_{\wedge \vee}$ at levels ≤ 2 number 2, 5, 24, in bijection with their counit values, with no split fiber (artifact, `dedekind_realsingw2.py`). The counit is thus injective on cells with image the chain stratum, and the comparison of Theorem 24.51 reduces to two statements: (A) $\text{Real}_D \text{Sing}_D X \cong \text{Ch}(X)$ in general — for cubes this is Theorem C.17, and for quotients it is the statement that accidental identifications between chain instances are generated by sorted ones; and (B) the *chain approximation lemma*: for fibrant X the monomorphism $\text{Ch}(X) \hookrightarrow X$ is a type equivalence. In (B) no fold and no pattern wall remains: a map φ on $\text{Ch}(X)$ is defined on all pattern instances, its chain data is natural, and the extension engine applies; surjectivity and injectivity of $[X, Z] \rightarrow [\text{Ch}(X), Z]$ are the engine and its cylinder version.

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DAIICHI INSTITUTE OF TECHNOLOGY, JAPAN

Email address: r-shibusawa@daiichi-koudai.ac.jp